

Dynamic Market Making with Asymmetric Information and Market Power

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Abstract

We study the dynamics of trading volume and bid-ask spread using a multi-period trading model with oligopolistic market makers. Traders smooth out their trading even though they are not strategic, and thus trading persists after the arrival of information or liquidity shocks. Traders act quickly on their private information while postponing hedging trades until later periods. The market power of market makers enables them to widen bid-ask spreads when trading incentives are heightened. Consequently, both trading volume and bid-ask spread may exhibit U-shaped patterns. Our model calibration aligns with the empirical intraday patterns observed in bid-ask spreads and trading volumes.

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The existing empirical literature has documented U-shaped intraday patterns in both trading volume and bid-ask spread across major stock exchanges. Specifically, studies have shown that the average trading volume tends to peak at the beginning and end of the trading day, with a decline during the middle hours.¹ Similarly, bid-ask spreads are observed to be highest at market opening and closure and lowest during the middle of the trading duration.² The simultaneous occurrence of U-shaped patterns in both trading volume and bid-ask spreads has been puzzling. Existing theories have not been able to explain why these double U-shaped patterns can arise endogenously. In this paper, we present a dynamic trading model based on flows of private information and liquidity shocks with imperfectly competitive liquidity suppliers. The model provides a unified and coherent explanation for the observed U-shaped patterns in both trading volume and bid-ask spread and sheds light on the underlying mechanisms that drive these intriguing intraday patterns in financial markets.

As new information or liquidity shocks arrive, traders engage in speculative trading based on their private information or hedge against the risk from liquidity shocks. Market makers, as liquidity suppliers, may possess market power due to imperfect competition. This market power of market makers leads other traders to smooth out their trading, even though these traders are not strategic. Therefore, the market power of market makers dampens and spreads out trading spikes due to either new information arrivals or liquidity shocks, and thus trading persists even after the arrival of information and liquidity shocks. When both information and liquidity shocks arrive at a constant rate, traders tend to trade more aggressively on their private information during earlier periods and postpone hedging trades to later periods. Consequently, trading volume may exhibit a U-shaped pattern, with higher trading volume observed at the beginning and end of the trading duration and lower volume in the middle. The market power of liquidity suppliers enables them to widen bid-ask spreads when trading incentives are heightened, resulting in wider bid-ask spreads at market opening and closure.³

¹Jain and Joh (1988) and Foster and Viswanathan (1993) find that the average trading volume of NYSE stocks is highest during the first hour of the day and declines monotonically until the fourth hour but increases again towards the close of trading.

²McInish and Wood (1992) and Brock and Kleidon (1992) find that spreads are higher at the beginning and end of the day relative to the interior period for NYSE stocks. Madhavan, Richardson, and Roomans (1997) develop a structural model of intraday price formation and show that the bid-ask spread of NYSE stocks is U-shaped. Abhyankar et al. (1997) also document similar U-shaped patterns in spreads for London Stock Exchange (LSE) stocks. Hua, Kong, and Wang (2023) find that NASDAQ stocks exhibit robust U-shaped patterns in both trading volume and bid-ask spread based on their analysis of TAQ data spanning January 2000 to December 2021.

³Both trading volume and bid-ask spread also exhibit U-shaped patterns when the information arrival rate decreases over time while liquidity shocks increase over time.

Our model considers a market with three types of investors: informed investors, uninformed investors, and market makers (who are also uninformed). Informed and uninformed traders are perfectly competitive, while market makers have imperfect competition. At each trading period $t = 0, 1, \dots, T$, traders can trade a risk-free asset and a risky security to maximize their expected constant absolute risk-averse (CARA) utility from the terminal wealth at period $T + 1$. Informed investors receive private signals about the terminal payoff of the security, motivating them to trade based on their private information. Additionally, informed investors experience non-information-based trading incentives, termed liquidity shocks, which we model as random endowments of nontradable assets. These nontradable risky assets' payoffs are correlated with stock payoffs and can only be liquidated at period $T + 1$. Consequently, informed investors also have trading demand driven by their liquidity needs for hedging.

At each trading period, traders can either sell to market makers at the bid price, buy from them at the ask price, or choose not to trade at all. Market makers, in turn, optimize their price and quantity decisions, determining their bid and ask quantities simultaneously. They take into account the inverse supply function (a function of the market makers' purchasing quantity) of all other participants and the inverse demand function (a function of the market makers' selling quantity) of all other participants. As a result, the bid and ask prices set by market makers are strategically determined to achieve their optimal trading quantities.

In practice, market makers often engage in both agency and principal market making activities. However, allowing market makers to simultaneously perform both types of market making would render our dynamic model highly intractable. To maintain tractability while still capturing essential market dynamics, we have opted to examine two distinct scenarios. In the first scenario, we consider agency trading market makers who offset trades and maximize the expected profit from each trading round without carrying inventory over time. In the second scenario, we analyze principal trading market makers who actively manage inventory over time to maximize their expected total profits.

By examining these simplified scenarios, we can effectively reduce the dimensionality of the model, enabling us to analytically solve for the equilibrium endogenous bid and ask prices and depths. This approach allows us to gain meaningful insights and understand the factors influencing the intraday patterns of trading volume and bid-ask spread. Remarkably, we find that the majority of our main results

remain qualitatively similar in both cases. Traders react promptly to their private information while deferring hedging trades until later periods, leading to potential U-shaped patterns in both trading volume and bid-ask spread.

To better understand the potential U-shaped patterns in trading volume, we examine the distinct effects of information and hedging on their dynamics in two scenarios. We assume that both private information and liquidity shocks are present at market opening to ensure that the equilibrium price is not fully revealing. In the first case, where only information arrives after market opening, as more information arrives over time, both uninformed traders and market makers become less uncertain about the fundamental value of the asset. Consequently, the discrepancy between the reservation prices of informed and uninformed traders diminishes,⁴ leading to a decline in trading volume over time. In contrast, in the second case, where only liquidity shocks occur after market opening, as more liquidity shocks arrive over time, the discrepancy in reservation prices increases, resulting in higher trading volume in later periods.

When both information and liquidity shocks arrive at constant rates, trading volume tends to concentrate at the market opening and closure. Traders are motivated to trade on their private information in earlier periods to capitalize on potential profits, as other traders might also act on similar information. However, they delay their hedging trades to later periods, as the adverse selection effect becomes less severe over time. Holding the competitiveness of market makers constant, the U-shaped pattern of trading volume becomes more pronounced when the volatility of hedging demand increases or when informed traders' private information is noisier. This occurs because high volatility of hedging demand leads uninformed traders to perceive that trades are primarily driven by hedging needs. Consequently, the adverse selection effect becomes less pronounced, allowing informed traders to trade more aggressively on their private information at market opening and shift more trading due to hedging demand at market close. As a result, the U-shaped pattern of trading volume becomes more pronounced.

In the scenario involving new arrivals of liquidity shocks (with or without new arrivals of information), an important distinction arises between agency trading and principal trading market makers. In the case of agency trading market makers, who do not carry inventory and aim to maximize profit in each trading round by offsetting trades with other traders, the trading volume exhibits a hump-shaped pat-

⁴The reservation price is the critical price such that an investor buys (sells) the risky security if and only if the ask (bid) is lower (higher) than this critical price.

tern several periods before market closure. This hump-shaped pattern arises because traders anticipate wider bid-ask spreads towards market closure and, to reduce total trading costs, optimally shift some of their trading to earlier periods when bid-ask spreads are relatively lower.⁵ Conversely, in the case of principal trading market makers, the expected trading volume tends to increase as market closure approaches, without exhibiting a hump-shaped pattern beforehand. This difference is primarily due to the principal market makers' objective to maximize total trading profits over the entire trading duration. They determine their optimal trading quantities (bid depth and ask depth) based on traders' demand schedules at each trading period, which in turn dictate the trading prices. The expected bid-ask spreads set by principal market makers in earlier periods are relatively higher than those set by agency market makers. As a result, traders continue to spread out their trading over time but shift less of their trading to earlier periods. A larger volume of trading occurs toward market closure when market makers can also set wider spreads, as it is the last opportunity to trade. Therefore, the trading volume does not exhibit a hump-shaped pattern.

Our model generates several testable empirical predictions. For example, it predicts that during periods of news events, the variation in trading volume tends to be smaller for stocks with less competitive liquidity suppliers. The U-shaped pattern of the bid-ask spread tends to be more pronounced for stocks with less competitive liquidity suppliers (e.g., small stocks) and stocks with higher order imbalance, allowing liquidity suppliers to have greater market power. Moreover, the U-shaped pattern of the trading volume tends to be more pronounced for stocks with lower information asymmetry or a higher proportion of liquidity traders, holding all other factors constant.

Our main findings are applicable to markets characterized by imperfect competition among liquidity suppliers, where some liquidity providers, such as market makers, have market power. As an illustrative example, we perform model calibration using stocks listed on NASDAQ, which is one of the most active stock trading venues in the US in terms of volume and market capitalization of shares traded. Previous studies (Biais, Bisière, and Spatt, 2010) suggest that competition among market makers on NASDAQ

⁵While the literature typically reports a pronounced U-shaped pattern in trading volume for major US stock exchanges, Abhyankar et al. (1997) uncovered a different trend for the LSE, which operated as a dealers' market during that period. They discovered a two-humped pattern, with trading volume peaking at 09:00 a.m. and 03:00 p.m. This two-humped pattern is also found by Kleidon and Werner (1993). Notably, the market operated from 08:30 a.m. to 04:30 p.m. during the periods examined in these studies. Our model has the capability to generate this two-humped pattern in trading volume. The first hump arises due to information arrival, while the second hump, occurring before market closure, can be attributed to traders shifting their trades or trading spikes due to the emergence of new information.

remains imperfect, even with the introduction of electronic markets.

Specifically, we calibrate the model parameters to fit the empirical intraday U-shaped patterns of bid-ask spreads and trading volume using TAQ data from January 2000 to December 2021. The theoretical expected bid-ask spreads and trading volumes, as predicted by our calibrated model, generally align with the corresponding empirical values at half-hour intervals throughout the trading day. Notably, the calibrated parameter values indicate a higher precision of private information at market opening compared to other periods. Conversely, the uncertainty associated with hedging needs is found to be higher near market closure in our calibration results. This suggests that there tends to be more information arrival at market opening, while a greater demand for hedging emerges near market closure compared to other periods throughout the trading day. The calibration results further validate the relevance and applicability of our model in capturing the intraday patterns observed in real-world trading environments, especially in markets with imperfectly competitive liquidity suppliers.

Applicable Markets and Related Literature. Major financial markets across the globe, including the NYSE, NASDAQ, LSE, and Euronext, operate as hybrid models where both market makers and limit orders coexist. In these markets, designated market makers or dealers play a crucial role in facilitating trading. They actively maintain two-sided markets during trading hours, standing ready to buy and sell at the displayed bids and offers. As a result, they fulfill a fundamental role as essential liquidity providers in these markets.⁶ For specific smaller stocks and periods marked by relatively low liquidity, investors predominantly trade through market makers ([Anand and Venkataraman, 2016](#)). In quote-driven electronic markets, such as the Stock Exchange Automated Quotation (SEAQ) system used for trading small-cap LSE stocks, traders often need to trade through market makers as direct trading among themselves is typically not feasible. This highlights the relevance of our model in financial markets where designated market makers play a crucial role in facilitating trades.⁷

Competition among market makers in many financial markets is far from being perfect. For instance, [Christie and Schultz \(1994\)](#) suggest that NASDAQ dealers might engage in implicit collusion to maintain wide spreads. Although NASDAQ transitioned away from a quote-driven market in 1997, allowing all par-

⁶See, for example, “Designated Market Makers,” 2016, www.nyse.com, and [Venkataraman and Waisburd \(2007\)](#).

⁷Existing inventory-based market making literature ignores information asymmetry ([Garman, 1976](#); [Stoll, 1978](#); [Ho and Stoll, 1981](#)). In contrast to most information-based (rational expectations) models ([Grossman and Stiglitz, 1980](#); [Kyle, 1985](#); [Glosten and Milgrom, 1985](#)), in our model a market maker has significant market power and profits from bid-ask spreads.

participants to submit limit orders to its central limit order book, the competition among liquidity providers remains imperfect. This is evident in the study conducted by [Biais, Bisière, and Spatt \(2010\)](#). They analyze trades and order placements on both NASDAQ and a competing electronic order book, Island. Their findings reveal that even with the introduction of electronic markets, competition among market makers in these markets remains imperfect. They emphasize that according to conversations with market participants, only a limited number of large professional traders actively monitored Island and NASDAQ and strategically undercut NASDAQ quotes to earn trading profits, resulting in imperfect competition. Moreover, they point out “market makers’ quotes were the main source of liquidity on NASDAQ. These quotes were disseminated electronically on the NASDAQ screens.” [Hua, Kong, and Wang \(2023\)](#) show that the magnitude of trading volume spikes at market closure has monotonically increased over time, consistent with the observation that institutional traders’ rebalancing needs at market closure have risen in recent years. Additionally, the decrease in the magnitude of bid-ask spread spikes from 2011-2015 to 2016-2021 suggests that the market power of market makers may have decreased during this period. Nevertheless, the double U-shaped pattern in bid-ask spreads and trading volume (turnover) remains consistent throughout the entire sample period from 2000 to 2021, as documented in their paper. This pattern indicates that competition among liquidity suppliers on NASDAQ is still not perfect, allowing market makers to increase the bid-ask spread in response to heightened trading incentives.

Our model is relevant for markets in which certain liquidity providers, like market makers, possess significant market power. While we do not model dynamic limit order books in the paper, the presence of competing limit orders from customers does not alter our primary intuition, as long as there is imperfect competition among liquidity suppliers and thus some liquidity suppliers possess significant market power. For example, when there is a significant imbalance between the total number of buying orders and the total number of selling orders, indicating a large order imbalance and insufficient competition from customer limit orders that aim to provide liquidity, market makers tend to have greater market power.

In most OTC markets, investors almost always trade with designated dealers (market makers) who typically provide bid and ask prices that adjust according to order sizes.⁸ [Ang, Shtaubert, and Tetlock](#)

⁸For example, [Li and Schürhoff \(2011\)](#) show that dealers intermediate 94% of the trades in the municipal bond market, with most of the intermediated trades representing customer-dealer-customer transactions.

(2013) present evidence indicating that the illiquidity premium in OTC markets is several times higher compared to listed markets, implying that market makers in many OTC markets possess substantial market power. Moreover, the availability of Transaction Reporting and Compliance Engine (TRACE) information and electronic bond trading platforms effectively reduces search costs for both customers and dealers (Vames, 2003; O'Hara and Zhou, 2021; Goldstein and Hotchkiss, 2007) find that the degree of price dispersion decreased after bond transactions were reported through TRACE, highlighting the impact of increased transparency.

These insights suggest that our model could potentially find application in some OTC markets where factors like bilateral negotiation, the need to search for a market maker, and price dispersion are of less critical importance. Examples could include OTCQX and OTCQB stock markets. The OTC Market Group facilitates electronic trading for both OTCQX and OTCQB, providing investors with real-time quoted bid and ask prices, along with comprehensive market data and depth.

In summary, our primary findings are relevant to markets characterized by imperfect competition among liquidity suppliers, particularly where some liquidity providers such as market makers have substantial market power. Our calibration results help validate the applicability of our model in markets featuring imperfectly competitive liquidity suppliers.

Studying trade dynamics is notoriously difficult because it typically requires a non-stationary dynamic model, which is challenging to solve in closed form. Afonso and Lagos (2015) provide a tractable framework for analyzing intraday dynamics in OTC markets with search and bilateral negotiations, specifically focusing on the intraday trade dynamics of the federal funds market. In contrast, our paper develops a tractable framework to analyze intraday trading dynamics for exchanges where order flows are endogenously determined, and liquidity suppliers can range from competitive to imperfectly competitive. This framework can be extended to explore important questions, such as the impact of information dynamics on trading and return dynamics within a non-stationary dynamic model.

Admati and Pfleiderer (1988) generate a U-shaped pattern of trading volume when liquidity traders concentrate their activity within specific time periods. However, their model suggests that trading costs are low during times of higher trading volume. This is in contrast to the findings of Foster and Viswanathan (1993) who note that higher trading costs often correspond with increased trading volume — a predic-

tion consistent with our model. In our model, intraday U-shaped patterns in both trading volume and bid-ask spread can occur simultaneously. [Brock and Kleidon \(1992\)](#) show that the demand at open and close is greater and less elastic than at other times of the trading day due to periodic market closure and that this periodic demand generates high volume and concurrent wide spreads. In their model, buy and sell orders are Poisson distributed in time and the price process is exogenously assumed. In contrast, in our model, traders optimally choose in which period to trade and the optimal trading quantity within that period. The U-shaped patterns in both trading volume and bid-ask spread arise endogenously because traders trade aggressively on their private information in early periods and postpone their trading due to hedging needs until later periods. [Madhavan, Richardson, and Roomans \(1997\)](#) develop and estimate a structural model of price formation. The estimated parameter values indicate that information asymmetry decreases throughout the day, while dealers' execution costs rise. This result helps explain the U-shaped pattern in intraday bid-ask spreads. Different from our model, their structural model assumes a fixed order size and incorporates an exogenous cost for market makers when providing liquidity. As a result, their model cannot explain the concurrent U-shaped patterns in both bid-ask spreads and trading volume.

[Bogouslavsky and Collin-Dufresne \(2023\)](#) develop a continuous-time inventory model with exogenous order flows and explain the positive time-series relationship between the spread and volume for large stocks. [Lee and Wang \(2023\)](#) show that when an OTC market coexists with an exchange, traders who are more likely to be informed prefer to trade on the exchange. As a result, any variation in the relative mass of informed traders would induce a positive correlation between the bid-ask spread and the market share of the exchange. However, neither paper generates the double-U-shaped intraday pattern in both the spread and the volume. Moreover, given that the volume spike toward market closure is driven by liquidity trades, [Lee and Wang \(2023\)](#) would predict a decrease in the bid-ask spread toward market closure.

[Liu and Wang \(2016\)](#) develop a one-period model of market making with asymmetric information and inventory risk, aiming to explain why bid-ask spreads may decrease with information asymmetry. In contrast, our paper focuses on the intraday dynamics of trading volume and bid-ask spread and sheds light on how information and liquidity shocks influence trading volume and bid-ask spread over time.

Budish, Cramton, and Shim (2015) advocate for the use of frequent batch auctions as a solution to counteract the arms races in high-frequency trading. Our model focuses on discrete time trading without engaging in such arms races and traders in our model respond to commonly observed public information simultaneously. Lester et al. (2023) show that reducing trading or search frictions can potentially slow down dealers' learning and can increase the bid-ask spread in the long run. Our paper differs from theirs as we focus on the dynamic patterns of bid-ask spread and trading volume without search frictions.

The remainder of the paper proceeds as follows. In Section 1, we present the model. In Section 2, we derive the equilibrium bid and ask prices, bid and ask depth, and trading volume. In Section 3, we study the dynamics of bid-ask spreads and trading volume. Section 4 presents model calibration. We conclude in Section 5. All proofs are provided in the Appendix and some additional results are presented in the Internet Appendix of the paper.

1. The Model

We consider a multi-period model of trading. There are a continuum of identical informed investors with mass N_I (I investors), a continuum of identical uninformed investors with mass N_U (U investors), and N_M designated market makers (M investors) who are also uninformed. Both informed and uninformed traders are perfectly competitive and market makers are imperfectly competitive. Traders can trade one risk-free asset and one risky security at trading periods $t = 0, 1, \dots, T$. There is zero supply of the risk-free asset, which also serves as the numeraire, and thus the risk-free interest rate is normalized to 0. The total supply of the risky asset is $\bar{\theta}$ and each share of the stock pays a liquidation value of $V \sim \mathcal{N}(\bar{V}, \tau_V^{-1})$ at period $T + 1$, where $\bar{V}$ is a constant, $\tau_V^{-1} > 0$, and $\mathcal{N}$ denotes the normal distribution. No investor is endowed with any risk-free asset. Each type $i \in \{I, U, M\}$ investor is endowed with $\bar{\theta}^i$ shares of risky assets before date 0. Therefore, the total supply of the risky security $\bar{\theta}$ is equal to the total endowment $N_I \bar{\theta}^I + N_U \bar{\theta}^U + N_M \bar{\theta}^M$. At each trading period $t = 0, 1, \dots, T$, informed investors may observe a private signal

$$v_t = V + \varepsilon_t, \tag{1}$$

about the liquidation value V , where $\varepsilon_t \sim \mathcal{N}(0, \tau_{\varepsilon,t}^{-1})$ is an independent noise. To prevent informed investors' private information from being fully revealed in equilibrium, at each period $t = 0, 1, \dots, T$, we assume that each informed investor is also endowed with X_t total number of shares of a nontradable risky asset at period t ,⁹

$$X_t = X_{t-1} + \eta_t, \quad (2)$$

where $X_{-1} = 0$, $\eta_t \sim \mathcal{N}(0, \tau_{\eta,t}^{-1})$, η_t is the number of shares received by each informed trader at period t ($t = 0, 1, \dots, T$) and is independent of V and ε_t . The number of the nontradable risky asset X_t is only observable to informed traders and the nontradable risky asset can only be liquidated at period $T + 1$. Without loss of generality, we assume that each unit of the nontradable risky asset pays a liquidation value of $V - \bar{V}$ at the final period $T + 1$.¹⁰

At periods $t = 0, 1, \dots, T$, I and U investors sell to the designated market makers at the bid price B_t or buy from them at the ask price A_t or do not trade at all. Market makers simultaneously choose how much to buy at bid given the inverse supply function of all other participants (a function of the market makers' purchasing quantity) and how much to sell at ask given the inverse demand function of all other participants (a function of the market makers' selling quantity). The posted bid and ask prices are the required prices to achieve the optimal amount market makers choose to trade.

Define $\underline{Z}_t := (Z_0, Z_1, \dots, Z_t)$ for any stochastic process $\{Z_t\}$, representing the history of Z_t up to t . On each date t , after observing private signals $\underline{v}_t$ and liquidity shocks $\underline{X}_t$, each informed investor chooses a demand schedule $\Theta_t^I(\underline{v}_t, \underline{X}_t; \cdot)$. Each uninformed trader chooses a demand schedule $\Theta_t^U(\cdot)$. The schedules Θ_t^I and Θ_t^U are traders' strategies. Given bid price $\underline{B}_t$ and ask price $\underline{A}_t$, the quantities demanded by informed and uninformed investors can be written as $\theta_t^I = \Theta_t^I(\underline{v}_t, \underline{X}_t, \underline{A}_t, \underline{B}_t)$ and $\theta_t^U = \Theta_t^U(\underline{A}_t, \underline{B}_t)$.

Let $\mathcal{F}_t^i$ denote a type i investor's information set at period t for $i \in \{I, U, M\}$. Investors' information

⁹Similar to Vayanos and Wang (2012) and Goldstein, Li, and Yang (2014), to keep information from being fully revealed in equilibrium, we assume that informed investors have two trading motives. They have liquidity shocks in addition to private information. One can interpret this assumption to mean that there are some pure liquidity traders who trade in the same direction as the informed. The assumption that all informed traders have the same information and the same liquidity shock is for tractability, so that there are only two groups of non-market-makers in the model.

¹⁰In general, we can assume that the nontradable asset has a payoff correlated with V . The correlation between the nontradable asset and the security results in a non-information-based, liquidity demand for the risky asset to hedge the nontradable asset payoff. This generalization would not yield qualitatively different results.

sets can be written as

$$\mathcal{F}_t^I = \{\underline{v}_t, \underline{X}_t, \underline{A}_t, \underline{B}_t\}, \quad \mathcal{F}_t^U = \mathcal{F}_t^M = \{\underline{A}_t, \underline{B}_t\}. \quad (3)$$

Given $\underline{A}_t, \underline{B}_t$, for type $i \in \{I, U\}$, a type i investor's problem is to solve

$$\max_{\theta_t^i} \mathbb{E}[-e^{-\lambda^i W_{T+1}^i} | \mathcal{F}_t^i], \quad (4)$$

where W_{T+1}^i is a type i investor's final wealth, λ^i is the risk aversion coefficient of type i traders, and

$$\begin{aligned} W_{T+1}^I &= W_{-1}^I + \sum_{t=0}^T [(\theta_t^I - \theta_{t-1}^I)^- B_t - (\theta_t^I - \theta_{t-1}^I)^+ A_t] + \theta_T^I V + X_T(V - \bar{V}), \\ W_{T+1}^U &= W_{-1}^U + \sum_{t=0}^T [(\theta_t^U - \theta_{t-1}^U)^- B_t - (\theta_t^U - \theta_{t-1}^U)^+ A_t] + \theta_T^U V, \end{aligned} \quad (5)$$

where $x^+ := \max(0, x)$, $x^- := \max(0, -x)$, and, for $i \in \{I, U\}$, the initial wealth $W_{-1}^i = \bar{\theta}^i \bar{V}$.

Let α_t^j and β_t^j be the number of shares that market maker $j \in \{1, 2, \dots, N_M\}$ sells at ask (i.e., ask depth) and buys at bid (i.e., bid depth), respectively. Given the demand schedules of the informed and the uninformed investors ($\Theta_t^I(A_t, B_t)$ and $\Theta_t^U(A_t, B_t)$), the bid price $B_t(\alpha_t^j, \beta_t^j)$ (i.e., the inverse supply function) and the ask price $A_t(\alpha_t^j, \beta_t^j)$ (i.e., the inverse demand function) can be determined by the following stock market clearing conditions at the bid and ask prices, for any $t = 0, 1, \dots, T$,

$$\begin{aligned} \sum_{j=1}^{N_M} \alpha_t^j &= \sum_{i=I, U} N_i (\Theta_t^i(A_t, B_t) - \Theta_{t-1}^i(A_{t-1}, B_{t-1}))^+, \\ \sum_{j=1}^{N_M} \beta_t^j &= \sum_{i=I, U} N_i (\Theta_t^i(A_t, B_t) - \Theta_{t-1}^i(A_{t-1}, B_{t-1}))^-, \end{aligned} \quad (6)$$

where the left-hand sides represent the total sales and purchases by market makers respectively and the right-hand sides represent the total purchases and sales by other investors respectively.

At t , for $j = 1, 2, \dots, N_M$, the designated market maker j 's problem is to solve

$$\max_{\alpha_t^j, \beta_t^j} \mathbb{E}[-e^{-\lambda^M W_{T+1}^j} | \mathcal{F}_t^M], \quad (7)$$

where market maker j 's terminal wealth, $W_{T+1}^j = W_{-1}^j + \sum_{t=0}^T (\alpha_t^j A_t - \beta_t^j B_t) + \theta_T^j V$, his initial wealth $W_{-1}^j = \bar{\theta}^M \bar{V}$ and his inventory at T , $\theta_T^j = \theta_{T-1}^j + \beta_T^j - \alpha_T^j$, and λ^M is market makers' risk aversion coefficient.

This dynamic model becomes highly intractable for $0 < \lambda^M < \infty$, representing a scenario where market makers engage in both agency and principal market making simultaneously.¹¹ To maintain tractability while still capturing essential market dynamics, we consider two distinct scenarios, one with agency trading market makers and the other with principal trading market makers.

(1) Market makers conduct agency trading. Assume $\lambda^M \rightarrow \infty$ and market makers have zero initial endowment, in this case, market makers conduct agency trading and they do not carry inventory over time, the designated market maker j 's problem (7) is equivalent to solving

$$\max_{\alpha_t^j = \beta_t^j} \mathbb{E} \left[\alpha_t^j A_t - \beta_t^j B_t \mid \mathcal{F}_t^M \right]. \quad (8)$$

This implies that market makers who trade in both the “ask” market and the “bid” market optimally make offsetting trades with other traders to maximize the expected profit from each round of trading conditional on market makers’ information set at period t , $\mathcal{F}_t^M$.¹²

(2) Market makers conduct principal trading. Assume $\lambda^M = \lambda^U = 0$, *i.e.*, market makers and uninformed traders are risk neutral, in this case, market makers conduct principal trading.¹³ The designated market maker j 's problem (7) is then equivalent to solving

$$\max_{\alpha_t^j, \beta_t^j} \mathbb{E} \left[W_{-1}^j + \sum_{t=0}^T (\alpha_t^j A_t - \beta_t^j B_t) + \theta_T^j V \mid \mathcal{F}_t^M \right]. \quad (9)$$

This implies that market makers carry inventory over time and maximize their expected total profits.

This leads to the definition of a perfect Bayesian equilibrium as follows.

Definition 1. An equilibrium $(\Theta_t^I(A_t, B_t), \Theta_t^U(A_t, B_t), A_t, B_t, \alpha_t^j \text{ and } \beta_t^j)$ is such that

¹¹More specifically, at each period, there could be potentially eight cases in equilibrium: both I and U investors buy, both I and U investors sell, I investors buy and U investors sell, I investors sell and U investors buy, I investors buy and U investors do not trade, I investors sell and U investors do not trade, I investors do not trade and U investors buy, and I investors do not trade and U investors sell. As a result, in our dynamic model, at period t , there are potentially 8^t cases, and the dimensionality grows exponentially. The exponential growth in dimensionality poses a significant challenge in solving the model. The expected utility of each trader involves taking integrals of the utility function over eight regions, and finding numerical solutions to the first-order conditions requires handling terms that involve both the cumulative distribution function and the probability density function of the standard normal distribution. This complexity renders the model highly intractable.

¹²We prove formally in Section B.8 of the Internet Appendix that market makers optimally do not carry inventory over time when they are extremely risk averse.

¹³Üslü (2019) demonstrates that market participants with low risk aversion and an advantage in holding inventory naturally take on the role of market makers. In our model, if market makers are risk neutral but uninformed traders are risk averse, the dynamic model again becomes intractable with truncated normal distributions. Please refer to the discussion in footnote 16.

1. given any A_t and B_t , $\Theta_t^i(A_t, B_t)$ solves a type i investor's Problem (4) for $i \in \{I, U\}$, where the information set of the informed is $\mathcal{F}_t^I = \{\underline{v}_t, \underline{X}_t, \underline{A}_t, \underline{B}_t\}$ and the information set of the uninformed is $\mathcal{F}_t^U = \mathcal{F}_t^M = \{\underline{A}_t, \underline{B}_t\}$;
2. given $\Theta_t^I(A_t, B_t)$ and $\Theta_t^U(A_t, B_t)$, α_t^j and β_t^j solve the market makers' Problem (7);
3. $A_t := A_t(\alpha_t^j, \beta_t^j)$ and $B_t := B_t(\alpha_t^j, \beta_t^j)$ clear both the risky security and the risk-free asset markets, where $A_t(\alpha_t^j, \beta_t^j)$ and $B_t(\alpha_t^j, \beta_t^j)$ solve equation (6); and
4. for every realization of the signals $\underline{v}_t$ and liquidity shocks $\underline{X}_t$, the beliefs of all investors are consistent with the joint conditional probability distribution in equilibrium.

2. The Equilibrium

In this section, we solve for the dynamic equilibrium bid and ask prices and trading volume in closed form. We focus on a piecewise linear equilibrium where the equilibrium trading prices can be expressed as a piecewise linear function of the state variables in the model. At period t , both informed investors' private information v_t and liquidity shock η_t affect informed investors' trading and thus the equilibrium prices. From market prices, other investors can only infer the value of the composite signal

$$s_t := v_t - h_t \eta_t = V + \varepsilon_t - h_t \eta_t, \quad (10)$$

where the coefficient h_t will be determined in equation (12) below. Thus both ask prices A_t and bid prices B_t are informationally equivalent to the composite signal s_t , which is a linear combination of informed investors' private signal and the amount of the nontradable risky asset. Therefore, the information set at period t for uninformed investors $\mathcal{F}_t^U = \mathcal{F}_t^M = \{\underline{s}_t\}$. We obtain the following lemma on the dynamic Kalman filtering formulas.

Lemma 1. (Kalman filtering) At t , for $i \in \{I, U\}$, trader i 's estimate of the stock's liquidation value $\hat{V}_t^i := E[V | \mathcal{F}_t^i]$ and trader U 's estimate of the liquidity shocks $\hat{X}_t^U := E[X_t | \mathcal{F}_t^U]$ are

$$\hat{V}_t^I = \hat{V}_{t-1}^I + o_{V,t}^I \tau_{\varepsilon,t} (v_t - \hat{V}_{t-1}^I), \quad \hat{V}_t^U = \hat{V}_{t-1}^U + o_{V,t}^U \tau_t (s_t - \hat{V}_{t-1}^U), \quad \hat{X}_t^U = \hat{X}_{t-1}^U + K_{X,t}^U (s_t - \hat{V}_{t-1}^U), \quad (11)$$

where $\hat{X}_{-1}^U = 0$, $\hat{V}_{-1}^I = \hat{V}_{-1}^U = \bar{V}$, the coefficient $K_{X,t}^U$ is defined in (A-6) in the Appendix, and the conditional variances of V for informed and uninformed traders at t are,

$$o_{V,t}^I := \text{Var}[V | \mathcal{F}_t^I] = (\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s})^{-1}, \quad o_{V,t}^U := \text{Var}[V | \mathcal{F}_t^U] = (\tau_V + \sum_{s=0}^t \tau_s)^{-1},$$

where $\tau_s := (\tau_{\varepsilon,s}^{-1} + h_s^2 \tau_{\eta,s}^{-1})^{-1}$, for $s = 0, 1, \dots, t$. The history of composite signals $\underline{s}_t := (s_0, s_1, \dots, s_t)$ can be summarized by the sufficient statistic $\hat{V}_t^I - \mu_t X_t$, which can be inferred by uninformed traders from prices. The noise-to-signal ratios μ_t and h_t are

$$\mu_t = \lambda^I o_{V,t}^I = \lambda^I (\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s})^{-1}, \quad h_t = \lambda^I / \tau_{\varepsilon,t}. \quad (12)$$

Lemma 1 shows that informed traders' estimate $\hat{V}_t^I$ can be written as a recursive equation of $\hat{V}_{t-1}^I$ with an innovation term $v_t - \hat{V}_{t-1}^I$ and uninformed traders' estimate $(\hat{V}_t^U, \hat{X}_t^U)$ can be expressed as a recursive equation of $(\hat{V}_{t-1}^U, \hat{X}_{t-1}^U)$ with an innovation term $s_t - \hat{V}_{t-1}^U$. The conditional variance and mean of the security's payoff for traders have intuitive expressions. Specifically, the conditional variance of the payoff corresponds to the inverse of the total precision of traders' signals, while the conditional mean of the payoff (or the estimate of V) is determined by the precision-weighted average of signal innovations.

The sufficient statistic $\hat{V}_t^I - \mu_t X_t$ is a composite signal of informed traders, capturing their dual trading motives: speculating based on private information and hedging against risks from liquidity shocks. Since the sufficient statistic $\hat{V}_t^I - \mu_t X_t$ can be inferred by uninformed traders from prices, it follows that $\hat{V}_t^I - \mu_t X_t = \hat{V}_t^U - \mu_t \hat{X}_t^U$. Equation (12) implies that the noise-to-signal ratio for uninformed traders μ_t decreases over time, as prices progressively convey more information about informed traders' estimation of the stock value.

2.1. The Equilibrium with Agency Trading Market Makers

In this subsection, we study the case where market makers only conduct agency trading, they maximize the expected profit from each round of trading and do not carry inventory over time. Specifically,

at period t , given A_t and B_t , the optimal demand schedule of a type i investor ($i \in \{I, U\}$) is

$$\Theta_t^i(A_t, B_t) - \Theta_{t-1}^i(A_{t-1}, B_{t-1}) = \begin{cases} \frac{P_t^{iR} - A_t}{\gamma_t^i} & A_t < P_t^{iR}, \\ 0 & B_t \leq P_t^{iR} \leq A_t, \\ -\frac{B_t - P_t^{iR}}{\gamma_t^i} & B_t > P_t^{iR}, \end{cases} \quad (13)$$

where P_t^{iR} is the reservation price of type i traders at period t and will be presented in equation (14) in Proposition 1. The coefficient γ_t^i is proportional to trader i 's risk aversion and depends on the conditional variance and covariance of state variables ($o_{V,t}^i$, $o_{X,t}^i$, and $o_{VX,t}^i$). Proposition 1 highlights that at period $t = T$, $\gamma_T^i = \lambda^i o_{V,T}^i$. However, at period $t < T$, γ_t^i does not have a simple expression. Instead, it can be determined recursively using equations (A-46) and (A-47) presented in the proof of Proposition 1.

A trader of type i ($i \in \{I, U\}$) does not trade at his reservation price P_t^{iR} . He purchases (or sells) the security only when the ask price is below (or bid price is above) this critical price. Next, we provide a closed-form expression for traders' reservation prices in the following proposition.

Proposition 1. *The reservation prices for traders I and U are given by*

$$\begin{aligned} P_t^{IR} &= \bar{V} + \xi_t^I (\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^I (\hat{V}_t^U - \bar{V}) - f_t^I \bar{\theta} - \gamma_t^I \theta_{t-1}^I, \\ P_t^{UR} &= \bar{V} + \xi_t^U (\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^U (\hat{V}_t^U - \bar{V}) - f_t^U \bar{\theta} - \gamma_t^U \theta_{t-1}^U. \end{aligned} \quad (14)$$

The coefficient μ_t is as defined in equation (12) and the coefficients ξ_t^i , ψ_t^i , f_t^i , and γ_t^i , for $i \in \{I, U\}$ and $t < T$ are all positive and can be computed recursively using equations (A-46) and (A-47). Their values at $t = T$ are $\xi_T^I = 1$, $\xi_T^U = 0$, $\psi_T^I = 0$, $\psi_T^U = 1$, and for $i \in \{I, U\}$, $f_T^i = 0$, $\gamma_T^i = \lambda^i o_{V,T}^i$, implying that the reservation prices at T are

$$P_T^{IR} = \hat{V}_T^I - \mu_T X_T - \lambda^I o_{V,T}^I \theta_{T-1}^I \quad \text{and} \quad P_T^{UR} = \hat{V}_T^U - \lambda^U o_{V,T}^U \theta_{T-1}^U. \quad (15)$$

Equation (15) in Proposition 1 implies that, at period T , an informed trader's reservation price increases with the composite signal $\hat{V}_T^I - \mu_T X_T$ and decreases with his previous inventory θ_{T-1}^I . An uninformed trader's reservation price increases with his estimate of the asset's liquidation value, $\hat{V}_T^U$ and decreases with his preceding inventory θ_{T-1}^U , following the logic of a one-period model. Since the final

payoff V is realized at period $T + 1$, at period T informed traders adjust their positions based on their updated payoff estimates and the realized hedging demand. Uninformed traders adjust based on their updated payoff estimates. Neither group of traders has a dynamic speculative position related to the estimation of the next period's trading prices. Consequently, at period T , each trader's reservation price is independent of the other type of traders' estimates and inventory levels.

In contrast, for $t < T$, both traders' reservation prices increase with the informed traders' composite signal $\hat{V}_t^I - \mu_t X_t$ and the uninformed traders' estimate $\hat{V}_t^U$. This implies that each trader's reservation price is influenced by other traders' estimates. This is because, at $t < T$, each trader's optimal demand consists of a long-term stock position and a short-term speculative position. The former is based on the trader's estimate of the asset's final payoff $\hat{V}_t^i$, while the latter is based on the trader's estimate of the next period's trading prices. Since the next period's trading prices are influenced by both traders' estimates and inventories, each trader's reservation price at period t depends on the informed traders' composite signal $\hat{V}_t^I - \mu_t X_t$, the uninformed traders' estimate $\hat{V}_t^U$, and their previous inventory levels. By applying the market-clearing condition, we show that each trader's reservation price is a function of both traders' estimates, the total supply $\bar{\theta}$, and his own previous inventory, θ_{t-1}^i , as depicted in equation (14).

Given that both the long-term stock position and short-term speculative position of each trader are based on his own estimate of the asset's payoff, and only the short-term speculative position is influenced by the other trader's estimates, the coefficient on $\hat{V}_t^I - \mu_t X_t$ in the informed traders' reservation price is typically greater than that in the uninformed traders' reservation price. This implies that $\xi_t^I > \xi_t^U$ for $t < T$. Similarly, the coefficient on $\hat{V}_t^U$ in the uninformed traders' reservation price tends to be larger than that in the informed traders' reservation price, suggesting $\psi_t^U > \psi_t^I$ for $t < T$.¹⁴ We use Δ_t to denote the difference in the reservation prices of informed and uninformed traders. From Proposition 1, we have

$$\begin{aligned} \Delta_t := P_t^{IR} - P_t^{UR} = & \underbrace{((\xi_t^I - \xi_t^U)(\hat{V}_t^I - \bar{V}) - (\psi_t^U - \psi_t^I)(\hat{V}_t^U - \bar{V}))}_{\text{estimated payoff difference}} - \underbrace{(\xi_t^I - \xi_t^U)\mu_t X_t}_{\text{hedging difference}} \\ & - \underbrace{((f_t^I - f_t^U)\bar{\theta} + (\gamma_t^I \theta_{t-1}^I - \gamma_t^U \theta_{t-1}^U))}_{\text{risk premium difference}}. \end{aligned} \quad (16)$$

¹⁴We conduct an extensive numerical analysis across a wide range of parameters. We find that the parameter values consistently demonstrate $\xi_t^I > \xi_t^U$ and $\psi_t^U > \psi_t^I$ for $t < T$, as illustrated in the Internet Appendix B.3.

The difference in reservation prices Δ_t can be intuitively understood through equation (16). The first term in the equation represents the difference in the estimation of the expected asset payoff between informed and uninformed traders. Since the endogenous parameter values satisfy $\xi_t^I > \xi_t^U$ and $\psi_t^U > \psi_t^I$, the coefficient on $\hat{V}_t^I - \bar{V}$ is positive and the coefficient on $\hat{V}_t^U - \bar{V}$ is negative in equation (16). The second term in equation (16) represents the difference in hedging demand between the two groups. Finally, the third term reflects the difference in the required risk premium for holding the asset between informed and uninformed traders.

At T , as shown in Proposition 1, $\xi_T^I = 1 > \xi_T^U = 0$, $\psi_T^U = 1 > \psi_T^I = 0$, and $f_T^I = f_T^U = 0$. This aligns with our previous discussion, suggesting that at period T , each trader's reservation price is independent of the other type of traders' estimates and inventory levels. It follows that

$$\Delta_T := P_T^{IR} - P_T^{UR} = \underbrace{(\hat{V}_T^I - \hat{V}_T^U)}_{\text{estimated payoff difference}} - \underbrace{\mu_T X_T}_{\text{hedging difference}} - \underbrace{(\lambda^I o_{V,T}^I \theta_{T-1}^I - \lambda^U o_{V,T}^U \theta_{T-1}^U)}_{\text{risk premium difference}}. \quad (17)$$

Similarly, the difference in reservation prices Δ_T can be intuitively understood. The first term in equation (17) captures the difference in the estimation of the expected asset payoff. The second term represents the difference in hedging demand between informed and uninformed traders, while the third term denotes the difference in the required risk premium for holding the asset between informed and uninformed traders. It is worth noting that if the inventory levels at $T - 1$ and risk aversion coefficients were the same for both informed and uninformed traders, the difference in the risk premium would be equal to the difference in the conditional variance of the asset payoff between the two groups.

Define v_t and w_t^I as

$$v_t := \gamma_t^U / \gamma_t^I, \quad w_t^I := \frac{N_M}{N_M + 1} \frac{N_I v_t}{N_I v_t + N_U}, \quad w_t^U := \frac{N_M}{N_M + 1} \frac{N_U}{N_I v_t + N_U}. \quad (18)$$

Note that $N_I v_t + N_U$ represents the aggregate population of non-market-making traders, adjusted for investors' information and risk aversion. Moreover, w_t^I (and correspondingly w_t^U) can be viewed as the adjusted population weight of informed (and uninformed, respectively) traders in the market, taking into account the market power of market makers. As N_M approaches infinity, the ratio $N_M / (N_M + 1)$ tends towards 1.

Assume that the second-order condition (A-37), which involves coefficients expressed recursively, is satisfied. The following proposition provides the equilibrium bid and ask prices and equilibrium quantities at t ($t = 0, 1, 2, \dots, T$) in closed form.

Proposition 2. 1. *The equilibrium ask and bid prices at t are*

$$A_t = w_t^I P_t^{IR} + (1 - w_t^I) P_t^{UR} + \frac{\Delta_t^+}{N_M + 1}, \quad B_t = w_t^I P_t^{IR} + (1 - w_t^I) P_t^{UR} - \frac{\Delta_t^-}{N_M + 1}. \quad (19)$$

2. *The bid-ask spread at t is*

$$A_t - B_t = \frac{|\Delta_t|}{N_M + 1}. \quad (20)$$

3. *The equilibrium quantities demanded at t are*

$$N_I(\theta_t^I - \theta_{t-1}^I) = -N_U(\theta_t^U - \theta_{t-1}^U) = \frac{N_M}{N_M + 1} \frac{\Delta_t}{\gamma_t^I / N_I + \gamma_t^U / N_U}. \quad (21)$$

4. *The equilibrium ask and bid depths, and trading volume at t are*

$$\alpha_t = \beta_t = \frac{1}{N_M + 1} \frac{|\Delta_t|}{\gamma_t^I / N_I + \gamma_t^U / N_U}, \quad Vol_t = \frac{N_M}{N_M + 1} \frac{|\Delta_t|}{\gamma_t^I / N_I + \gamma_t^U / N_U}. \quad (22)$$

Proposition 2 implies that the equilibrium ask (bid) price is a weighted average of traders' reservation prices. Assume $P_t^{IR} > P_t^{UR}$, then I investors buy at ask price and U investors sell at bid price. Equation (19) implies that the ask and bid prices can be written as

$$A_t = \left(w_t^I + \frac{1}{N_M + 1}\right) P_t^{IR} + w_t^U P_t^{UR}, \quad B_t = w_t^I P_t^{IR} + \left(w_t^U + \frac{1}{N_M + 1}\right) P_t^{UR}, \quad (23)$$

where w_t^I and w_t^U are as defined in equation (18) and $w_t^I + w_t^U + \frac{1}{N_M + 1} = 1$. Due to the market power of market makers, the ask price A_t has a weight of $w_t^I + \frac{1}{N_M + 1}$ on the buyers' reservation price (P_t^{IR}) and a weight of w_t^U on the sellers' reservation price (P_t^{UR}). Conversely, for the bid price B_t , the weight on the sellers' reservation price (P_t^{UR}) is $w_t^U + \frac{1}{N_M + 1}$, and the weight on the buyers' reservation price (P_t^{IR}) is w_t^I .

In the absence of market power (e.g., when $N_M \rightarrow \infty$), the equilibrium price becomes $w_t^I P_t^{IR} + w_t^U P_t^{UR}$.

To further help understand the result in Proposition 2, let's assume again that $P_t^{IR} > P_t^{UR}$. The market clearing condition (6) and equation (13) imply that

$$A_t = P_t^{IR} - \frac{\gamma_t^I}{N_I} \sum_{j=1}^{N_M} \alpha_t^j, \quad B_t = P_t^{UR} + \frac{\gamma_t^U}{N_U} \sum_{j=1}^{N_M} \beta_t^j. \quad (24)$$

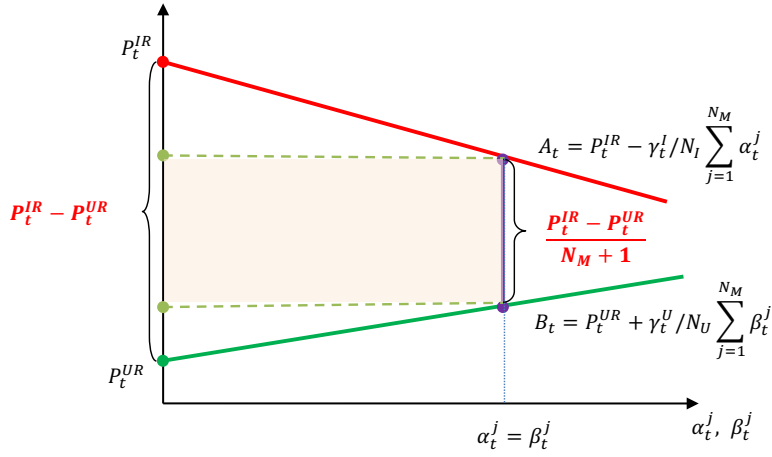


Figure 1. Inverse Demand/Supply Functions and Bid/Ask Spreads

We plot the above inverse demand and supply functions and equilibrium spreads in Figure 1. Figure 1 shows that the higher the bid price B_t , the more a market maker can buy from other investors, and the lower the ask price A_t , the more a market maker can sell to other investors. Facing the demand and supply functions of other investors, an oligopolistic market maker optimally trades off the prices and quantities. Similar to the results of classical models on oligopolistic firms that set a market price to maximize profit, the bid and ask spread is equal to the absolute value of the reservation price difference $|\Delta_t|$ divided by $N_M + 1$. Unlike these oligopolistic firms, however, the market makers are sellers in one market and buyers in another, and make a profit from the bid-ask spread.

In addition, as implied by Proposition 2, the difference between P_t^{IR} (P_t^{UR}) and the ask (bid) price is also proportional to the absolute value of the reservation price difference $|\Delta_t|$. Therefore, the trading quantities of both I and U investors and thus the aggregate trading volume all increase with $|\Delta_t|$. This suggests that bid-ask spreads and trading volume can move in the same direction over time, because

both trading volume and the bid-ask spread increase with $|\Delta_t|$.¹⁵

2.2. The Equilibrium with Principal Trading Market Makers

In this subsection, we study the case where market makers are risk neutral and engage in principal trading. They carry inventory over time and maximize their expected total profits. The market makers' problem is presented in equation (9). For tractability, we assume that uninformed traders are also risk neutral.¹⁶

At period t , given A_t and B_t , the optimal demand schedule of an informed investor is as presented in equation (13), where P_t^{IR} is the reservation price of informed traders at period t and will be presented in equation (26) in Proposition 3 below. The coefficient γ_t^I is proportional to informed traders' risk aversion, and is a function of informed traders' conditional variance and covariance of state variables ($o_{V,t}^I$, $o_{X,t}^I$, and $o_{VX,t}^I$). At period $t = T$, $\gamma_T^I = \lambda^I o_{V,T}^I$; at period $t < T$, γ_t^I can be computed recursively by equation (A-73) in the proof of Proposition 3. When $\lambda^U = 0$, uninformed traders' problem (4) is reduced to solve

$$\max_{\theta_t^U} E \left[W_{-1}^U + \sum_{t=0}^T [(\theta_t^U - \theta_{t-1}^U)^- B_t - (\theta_t^U - \theta_{t-1}^U)^+ A_t] + \theta_T^U V \mid \mathcal{F}_t^U \right]. \quad (25)$$

Let P_t^U denote the quoted price for uninformed traders. In equilibrium, this price aligns with the risk-neutral uninformed investors' valuation, such that $P_t^U = \hat{V}_t^U$. If this weren't the case, and if $P_t^U > \hat{V}_t^U$ (or conversely, $P_t^U < \hat{V}_t^U$), then the demand from uninformed traders would become infinitely negative (or positive, respectively).

Define the reservation price of market makers (P_t^{MR}) as a critical threshold. Market makers buy from (or sell to) informed traders if this threshold is higher (or lower) than the informed traders' reservation price. By establishing this reservation price for market makers, we can present the results for the case with principal market makers in a manner parallel to the case with agency trading market makers.¹⁷

¹⁵Lin, Sanger, and Booth (1995) find that trading volume and effective spreads are positively correlated at the beginning and end of the day. Chordia, Roll, and Subrahmanyam (2001) find a positive relationship between the effective bid-ask spread and trading volume.

¹⁶The case where market makers are risk neutral but uninformed traders are risk averse is not tractable. There are still potentially eight cases in equilibrium, for example, at each period, informed traders may buy and uninformed traders sell, or both informed and uninformed traders sell, etc. The expected utility function at period $t < T$ involves taking expectations of exponential function with truncated normal distribution. There is no closed-form solution for traders' optimal demand.

¹⁷It's worth noting that in the scenario with agency trading market makers, who engage in offsetting trades with other traders, both the bid-ask spread and trading volume are determined by the difference in reservation prices between informed and uninformed traders, rather than the market makers' reservation price.

Next, we provide a closed-form expression for the reservation prices of both traders and market makers in the subsequent proposition.

Proposition 3. *The reservation prices for informed investors, uninformed investors, and market makers are*

$$\begin{aligned} P_t^{IR} &= \bar{V} + \xi_t^I(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^I(\hat{V}_t^U - \bar{V}) - \gamma_t^I \theta_{t-1}^I, \\ P_t^{MR} &= \bar{V} + \xi_t^M(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^M(\hat{V}_t^U - \bar{V}) - \gamma_t^M \theta_{t-1}^I, \\ P_t^{UR} &= \hat{V}_t^U. \end{aligned} \tag{26}$$

The coefficient μ_t is as defined in equation (12) and the coefficients ξ_t^I , ψ_t^I , and γ_t^I for $t < T$ are all positive and can be computed recursively using equation (A-73). The coefficients ξ_t^M , ψ_t^M , and γ_t^M for $t < T$ are all positive and can be computed recursively using equation (A-79). Their values at $t = T$ are $\xi_T^I = 1$, $\xi_T^M = 0$, $\psi_T^I = 0$, $\psi_T^M = 1$, $\gamma_T^I = \lambda^I o_{V,T}^I$, and $\gamma_T^M = 0$.

At T , an informed trader's reservation price is $P_T^{IR} = \hat{V}_T^I - \mu_T X_T - \lambda^I o_{V,T}^I \theta_{T-1}^I$. It increases in the composite signal $\hat{V}_T^I - \mu_T X_T$ and decreases in the previous inventory θ_{T-1}^I . This aligns with the equilibrium presented in Proposition 1 for scenarios involving agency trading market makers. A risk-neutral market maker's reservation price at T is $P_T^{MR} = P_T^{UR} = \hat{V}_T^U$, solely determined by his estimate of the asset's fundamental value, $\hat{V}_T^U$, as in a one-period model.

In contrast, as in Proposition 1, for $t < T$, the reservation price of informed traders increases in the composite signal $\hat{V}_t^I - \mu_t X_t$ and is also influenced by the uninformed market makers' estimation $\hat{V}_t^U$. This is because, at $t < T$, informed investors' optimal demand consists of a long-term stock position and a short-term speculative position. While the long-term stock position is anchored to the informed investors' composite signal $\hat{V}_t^I - \mu_t X_t$, the short-term speculative position is derived from their predictions of trading prices in the upcoming period. Since the uninformed market makers' estimate of the stock payoff affects the next period's trading prices, the informed investors' reservation price is naturally also influenced by market makers' estimate of the fundamental value $\hat{V}_t^U$.

Likewise, the reservation price of risk-neutral market makers at t is influenced by their estimate of the asset's fundamental value, $\hat{V}_t^U$, as well as the composite signal $\hat{V}_t^I - \mu_t X_t$ and the previous inventory of informed traders, θ_{t-1}^I . This is because market makers cater to informed investors, whose demand

consists of both long-term holdings and short-term speculative positions. The latter is anchored to investors' estimates of trading prices in the subsequent period, which are shaped by both groups' stock payoff estimates. It's worth noting that, being risk-neutral, the market makers' reservation price is unaffected by their own inventory levels.

Similar to Proposition 1, each trader tends to assign a higher significance to his own estimates compared to the estimates of other traders. For a wide range of parameter values, as illustrated in the Internet Appendix B.3, the parameter values indicate that $\xi_t^I > \xi_t^M$ and $\psi_t^M > \psi_t^I$ for $t < T$. At T , as shown in Proposition 3, the parameter values $\xi_T^I = 1 > \xi_T^M = 0$ and $\psi_T^M = 1 > \psi_T^I = 0$.

Assume that the second-order conditions (A-71) and (A-76), which involve coefficients expressed recursively, are satisfied.¹⁸ The following proposition provides the equilibrium trading prices and quantities at $t(t = 0, 1, 2, \dots, T)$ in closed form.

Proposition 4. 1. The equilibrium trading prices for informed (P_t^I) and uninformed investors (P_t^U) are

$$P_t^I = (1 - \omega_t^M)P_t^{IR} + \omega_t^M P_t^{MR}, \quad P_t^U = P_t^{UR}, \quad (27)$$

where ω_t^M is positive. It can be computed using equation (A-80) and $\omega_T^M = N_M / (N_M + 1)$.

2. The equilibrium trading quantities are¹⁹

$$\begin{aligned} \theta_t^I - \theta_{t-1}^I &= \frac{\omega_t^M}{\gamma_t^I} (P_t^{IR} - P_t^{MR}), & \theta_t^U - \theta_{t-1}^U &= 0, \\ \alpha_t &= \frac{N_I \omega_t^M}{N_M \gamma_t^I} (P_t^{IR} - P_t^{MR})^+, & \beta_t &= \frac{N_I \omega_t^M}{N_M \gamma_t^I} (P_t^{IR} - P_t^{MR})^-, \end{aligned} \quad (28)$$

where $x^+ := \max(0, x)$ and $x^- := \max(0, -x)$.

3. The trading volume at t is

$$Vol_t = \frac{N_I \omega_t^M}{\gamma_t^I} |P_t^{IR} - P_t^{MR}|. \quad (29)$$

Proposition 4 suggests that the trading price of informed investors (i.e., the ask price when they purchase or the bid price when they sell) is a weighted average of the reservation prices of both informed

¹⁸For both the agency and principal trading scenarios, to satisfy the second-order condition, the variance of liquidity shocks cannot be too large for given parameter values of $\tau_{e,t}$ and τ_V .

¹⁹In equilibrium, uninformed traders are indifferent between trading and not trading, as the quoted price matches the valuation of risk-neutral uninformed traders. However, the trading volume stays the same, as presented in equation (29).

traders and market makers. Consider an instance where $P_t^{IR} > P_t^{MR}$. In this case, informed investors purchase at the ask price, given by $A_t = (1 - \omega_t^M)P_t^{IR} + \omega_t^M P_t^{MR}$, where the weight $\omega_t^M > 0$ for $t \leq T$, and it is determined by factors such as informed traders' risk aversion as well as the conditional variance and covariance of state variables. It can be shown that, at $t = T$, $\omega_T^M = N_M/(N_M + 1)$, and for $t < T$, $\omega_t^M > N_M/(N_M + 1)$. This suggests that the emphasis on market makers' reservation prices is more substantial than on the reservation prices of informed traders when determining the trading prices for informed traders. This is due to the market power of market makers.²⁰

Equation (28) implies that the trading quantity of informed traders is proportional to the difference in reservation prices between informed traders and market makers. Furthermore, the trading volume is proportional to the absolute value of the reservation price difference, as presented in equation (29).

Next, we present the bid-ask spread in one of the equilibria.

Proposition 5. *The bid-ask spread at $t \leq T$ is*

$$A_t - B_t = \begin{cases} |P_t^I - P_t^U| & (P_t^I - P_t^{IR})(P_t^I - P_t^U) \leq 0, \\ \frac{1}{2}|P_t^I - P_t^U| & (P_t^I - P_t^{IR})(P_t^I - P_t^U) > 0. \end{cases} \quad (30)$$

Given that the reservation prices of market makers and uninformed traders coincide at T , $P_T^{MR} = P_T^{UR} = \hat{V}_T^U$ and $\omega_T^M = N_M/(N_M + 1)$, equation (30) implies that the bid-ask spread at T is

$$A_T - B_T = |P_T^I - P_T^U| = \frac{1}{N_M + 1} |P_T^{IR} - P_T^{UR}|. \quad (31)$$

Analogous to the equilibrium in markets with agency trading market makers, the bid-ask spread equals the absolute value of the difference in reservation prices, divided by $N_M + 1$.

In contrast, for $t < T$, the reservation prices of market makers and uninformed traders are distinct (i.e., $P_t^{MR} \neq P_t^{UR}$). This distinction arises because market makers interact with informed investors who engage in both long-term and short-term speculative trades. Consequently, the reservation price P_t^{MR} incorporates added value from market makers' market power and the liquidity they provide to informed

²⁰If $N_M \rightarrow \infty$, the market becomes competitive, then $\omega_t^M \rightarrow 1$, and $P_t^I \rightarrow P_t^{MR}$, which converges to $\hat{V}_t^U$. Trading prices are the same for both informed and uninformed traders and thus the bid-ask spread is zero.

investors. As indicated by equation (27),

$$P_t^I - P_t^U = (1 - \omega_t^M) (P_t^{IR} - P_t^{UR}) + \omega_t^M (P_t^{MR} - P_t^{UR}). \quad (32)$$

This suggests that the difference in trading prices is equal to a weighted average of the reservation price differences. Regarding bid and ask prices, there are four equilibrium scenarios at $t < T$: Informed traders buy or sell while uninformed traders are indifferent between selling and no-trading, and informed traders sell or buy while uninformed traders are indifferent between buying and no-trading.

To elaborate on this result, let's assume $P_t^{IR} \geq P_t^I$. In this case, informed traders buy from market makers. There are two cases: (i) If $P_t^U \leq P_t^I$, then $A_t = P_t^I$ and $B_t = P_t^U$; (ii) If $P_t^U > P_t^I$, then both P_t^U and P_t^I are ask prices, any bid price B_t that satisfies $B_t \leq A_t^I := P_t^I < A_t^U := P_t^U$ can be an equilibrium. Thus, the bid price can be set as $B_t = P_t^I$ and the ask price can be set as the average of the two ask prices, i.e., $A_t = \frac{1}{2} (P_t^U + P_t^I)$. As a result, in this equilibrium, the bid-ask spread can be written as in equation (30).

In the analysis presented in Section 3, we calculate the expected bid-ask spread using both equation (30) and $E[|P_t^I - P_t^U| | (P_t^I - P_t^{IR})(P_t^I - P_t^U) \leq 0]$. The latter represents the expected bid-ask spread under conditions where one equilibrium bid and ask price exists. Notably, the results from both methods exhibit qualitatively similar patterns.

Proposition 5 implies that, for $t < T$, the bid-ask spread depends on the absolute value of a weighted average of two reservation price differences: between informed and uninformed traders ($P_t^{IR} - P_t^{UR}$) and between market makers and uninformed traders ($P_t^{MR} - P_t^{UR}$). Different from the scenario involving agency trading market makers, the bid-ask spread and trading volume in the context of principal trading market makers—where offsetting trades might not always be available—are also influenced by the market makers' reservation price.

3. The Intraday Dynamics of Trading Volume and Bid-Ask Spread

In this section, we study the intraday dynamics of trading volume and bid-ask spread. In our model, both trading volume and bid-ask spread might display U-shaped patterns over the trading duration. Additionally, we examine the impact of factors such as market makers' market power, trading rounds,

information flows, and hedging demands on these intraday dynamics.

3.1. The Impact of Information and Hedging on the Intraday Dynamics

To analyze the dynamics of trading volume and bid-ask spread, we first examine two distinct cases: one where trading is mainly driven by information and another where it is mainly driven by liquidity shocks. This approach allows us to highlight the contrasting impacts of information and hedging on the dynamics of trading volume and bid-ask spread.

For the case with agency trading market makers, trading volume at period t (Vol_t) and bid-ask spread ($A_t - B_t$) are given in equations (22) and (20) in Proposition 2. For the case with principal trading market makers, trading volume and bid-ask spread are given in equations (29) and (30).²¹

Figures 2a and 2b illustrate the dynamics of expected trading volume and bid-ask spread for both principal and agency market-making markets against trading duration in an information-driven market where information arrives every period and liquidity shock only arrives at the beginning to keep information from fully revealing.²² As depicted in Figures 2a and 2b, in an information-driven market, both the expected trading volume and bid-ask spread tend to decrease over time.²³ The intuition is simple. With more information arrivals over time, both uninformed investors and market makers become less uncertain about the fundamental value of the asset in later periods. The difference between the reservation prices of informed and uninformed traders becomes smaller and thus both trading volume and spread also tend to be smaller over time. The expected bid-ask spread and trading volume may experience a slight increase towards market closure (period T) because market makers have market power and period T is the last trading opportunity for traders.

Figures 3a and 3b present the dynamics of expected trading volume and bid-ask spread over the trading duration for both principal and agency market-making markets in a liquidity-driven market.²⁴

²¹The detailed formulas for the expected trading volume and bid-ask spread are presented in the Internet Appendix B.1.

²²Parameters are $N_M = 1$, $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\bar{V} = 1$, $\tau_V = 1$, $\tau_{\varepsilon,t} = 0.5$, and $\tau_{\eta,t} = 10^6$ (i.e., $\tau_{\eta,t}^{-1} \approx 0$) except for $\tau_{\eta,0} = 10$. Moreover, $\lambda^U = 1$ in Figure 2a and $\lambda^U = 0$ in Figure 2b.

²³Across all specifications of information flow and liquidity shocks, the general patterns depicted in the figures of this section remain qualitatively consistent across a wide range of parameter values. Note that the new liquidity shock, denoted as η_t , follows a normal distribution $\mathcal{N}(0, \tau_{\eta,t}^{-1})$. In all figures presented throughout this section, when we set $\tau_{\eta,t} = 10^6$, the value of $\tau_{\eta,t}^{-1}$ is approximately zero. This implies $\eta_t \approx 0$, indicating the absence of a liquidity shock at period t . Similarly, the information noise, denoted as ε_t , follows a normal distribution $\mathcal{N}(0, \tau_{\varepsilon,t}^{-1})$. When we use $\tau_{\varepsilon,t} = 10^{-6} \approx 0$, it indicates that the information is predominately pure noise, implying no information arrival at period t .

²⁴Parameters are $N_M = 1$, $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\bar{V} = 1$, $\tau_V = 1$, $\tau_{\eta,t} = 10$, and $\tau_{\varepsilon,t} = 10^{-6} \approx 0$ except for

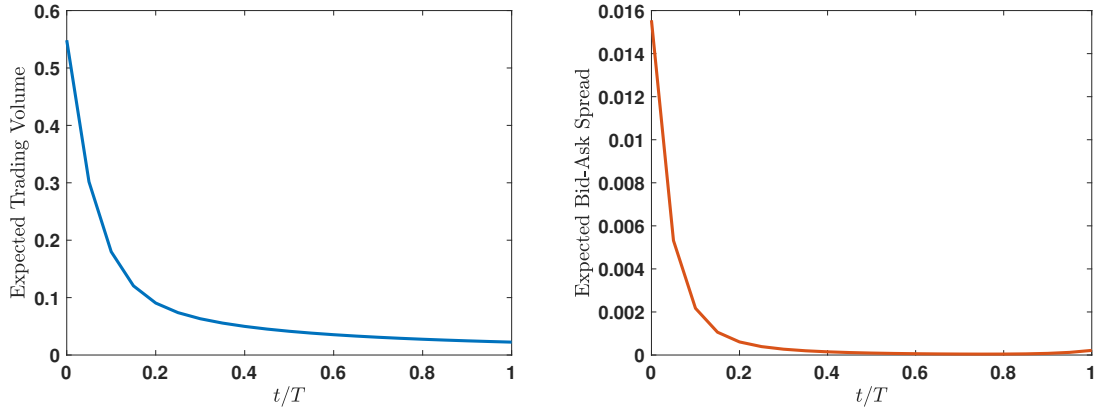


Figure 2a. Agency Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Information-Driven Markets.

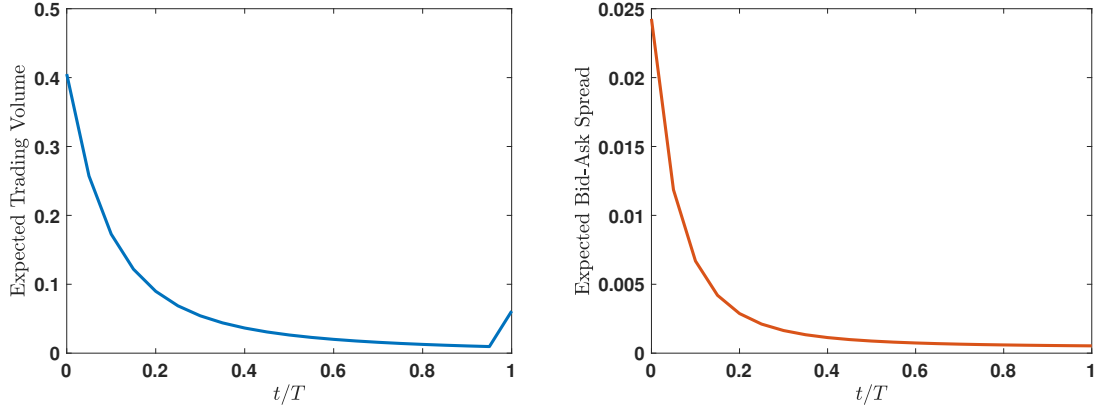


Figure 2b. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Information-Driven Markets.

In this setting, liquidity shocks occur every period, while information arrives only at the beginning. As depicted in the right panel of Figures 3a and 3b, contrary to the case in an information-driven market, in a liquidity-driven market, except for the first several trading periods where trading also responds to the initial information, the bid-ask spread tends to increase over time. This rise in the bid-ask spread is attributed to the widening difference in reservation prices, a consequence of the arrival of liquidity shocks over time.

In scenarios with agency trading market makers—who do not carry inventory and aim to maximize round-by-round profits by offsetting trades with other traders—the trading volume exhibits a hump-

$\tau_{e,0} = 0.5$. Moreover, $\lambda^U = 1$ in Figure 3a and $\lambda^U = 0$ in Figure 3b.

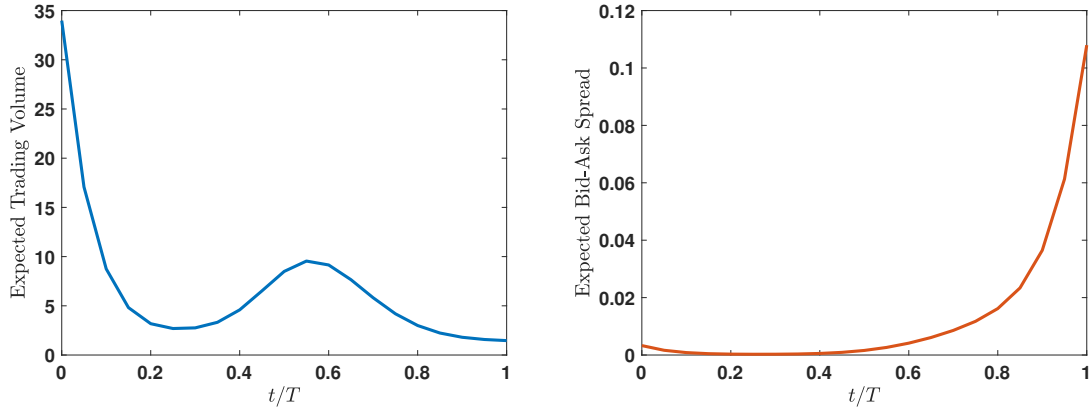


Figure 3a. Agency Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Liquidity-Driven Markets.

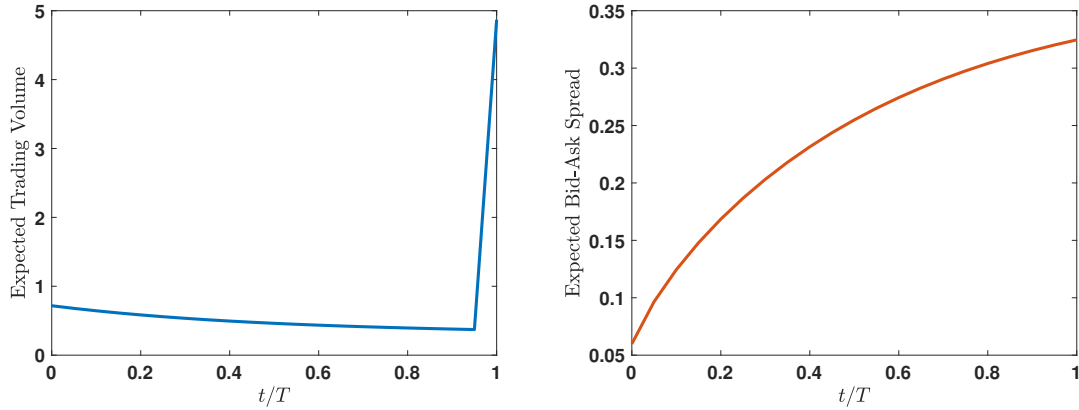


Figure 3b. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Liquidity-Driven Markets.

shaped pattern in the middle, as illustrated in the left panel of Figure 3a. This pattern arises because traders, anticipating a wider bid-ask spread toward market closure, optimally shift more of their trading to earlier periods when bid-ask spreads are relatively lower to reduce total trading costs. Notably, our extensive numerical analyses consistently reveal a hump in the expected trading volume before market closure for cases with agency trading market makers, regardless of the presence or absence of new information arrivals, particularly when new liquidity shocks occur.

In contrast, with principal trading market makers, there is no hump-shaped pattern in the expected trading volume; instead, it tends to increase as market closure approaches.²⁵ This difference is primarily

²⁵When multiple market makers are present (i.e., $N_M \geq 2$), the expected trading volume exhibits a gradual increase over time, due to greater competition among market makers compared to the monopolistic principal trading market maker case.

due to the objective of principal market makers to maximize total trading profits throughout the entire trading duration. They determine their optimal trading quantities (bid depth and ask depth) based on traders' demand schedules at each trading period, which in turn dictate the trading prices. With the arrival of new liquidity shocks, the bid-ask spreads set by principal market makers increase more gradually toward market closure, rather than peaking drastically at market close, as observed with agency market makers. This difference is clearly illustrated when comparing Figure 3a with Figure 3b. Specifically, the expected bid-ask spread with agency trading market makers (Figure 3a) increases tenfold from $t/T = 0.6$ to market close at $t/T = 1$, while the expected bid-ask spread with principal trading market makers (Figure 3b) increases by only approximately 18% from $t/T = 0.6$ to market close at $t/T = 1$. Therefore, the relative levels of the expected bid-ask spreads in earlier periods, compared to those at market close, are much higher in the principal market maker case than in the agency market maker case.

This is because principal market makers, compared to their agency counterparts, optimally buy less from other traders at bid prices (resulting in lower bids) or sell less to other traders at ask prices (resulting in higher asks) in earlier periods. Consequently, the expected bid-ask spreads set by principal market makers in these earlier periods are higher than those in the agency market maker scenario. As a result, traders continue to spread out their trading over time but shift less of their trading to earlier periods. A larger volume of trading occurs toward market closure when market makers can also set wider spreads, as it is the last trading opportunity for traders. Therefore, the trading volume does not exhibit a hump-shaped pattern.

We now explore the dynamics of trading volume and bid-ask spread when informed traders access private information and experience liquidity shocks at constant rates. As illustrated in Figures 4a and 4b, both trading volume and bid-ask spread display U-shaped patterns over the trading duration.²⁶ At the beginning and end of the trading duration, we observe higher trading volume and bid-ask spreads, while in the middle of the trading duration, both trading volume and bid-ask spreads are lower.

The intuition is as follows. Even though traders tend to smooth out their trading due to market mak-

This observation is consistent with the analytical insights provided in Proposition B.2, detailed in Section B.7 of the Internet Appendix. Since the overall patterns remain similar, we proceed to present the principal market maker cases assuming $N_M = 2$ throughout the rest of the paper. For completeness, we also include the Figures with $N_M = 1$ in Section B.6 of the Internet Appendix.

²⁶Parameters are $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\bar{V} = 1$, $\tau_V = 1$, $\tau_{\eta,t} = 10$, and $\tau_{\varepsilon,t} = 0.5$. Moreover, $\lambda^U = 1$, $N_M = 1$ in Figure 4a, and $\lambda^U = 0$ and $N_M = 2$ in Figure 4b.

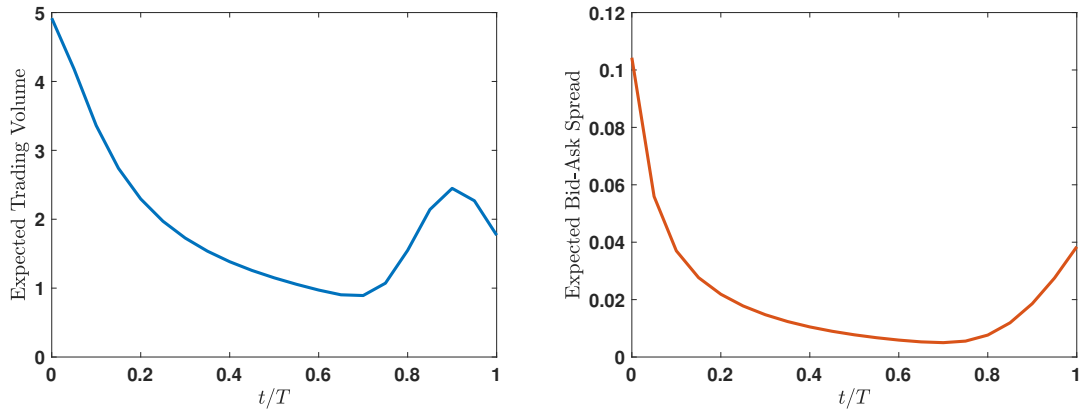


Figure 4a. Agency Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration with Both Information and Liquidity Shocks Arriving at Constant Rates.

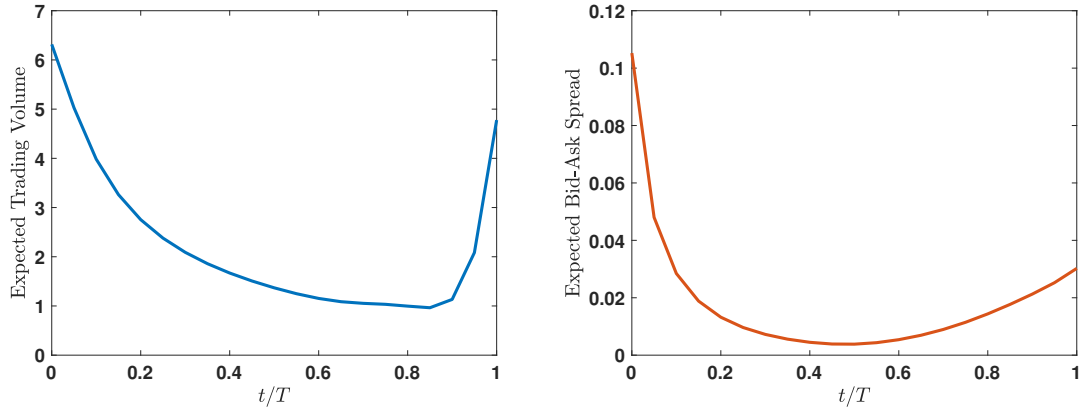


Figure 4b. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration with Both Information and Liquidity Shocks Arriving at Constant Rates.

ers' market power, they are inclined to act quickly on their private information since others might trade on the same information. They tend to postpone their trading due to hedging needs until later periods. This is because, with more information revealed through trading, information asymmetry decreases and thus trading cost due to the adverse selection effect becomes smaller over time. As a result, trading volume tends to be concentrated at market opening and closure.²⁷ Notably, the trading peak near mar-

²⁷While our model features the endogenous dynamic separation of trades based on information and those based on liquidity shocks, achieving a clear separation can be challenging due to the interrelation of these two trading incentives. We cannot separate these components by merely shutting down one of the incentives, or by varying the precision of the signal or the uncertainty of hedging demand. Instead, we focus on analyzing the formula for trading quantity, aiming to decompose it into two distinct components: one related to information and the other to hedging demand. The decomposition aligns with the key insights of this section, indicating that trading volume in later periods is predominantly driven by hedging demand, while trading volume attributed to information gradually diminishes over time. The details are presented in Section B.2 of the Internet Appendix.

ket closure might be reached at a point prior to closure when market makers predominantly engage in agency trading. Since market makers with market power can charge wider bid-ask spreads when trading incentives are heightened, bid-ask spreads are wider at market opening and closure.

3.2. The Effects of Liquidity Shock Uncertainty and Information Precision

Next, we examine the impact of liquidity shock uncertainty and the precision of informed traders' private information on the dynamics of trading volume and bid-ask spread. We assess how variations in these factors shape the magnitude of the U-shaped patterns in trading volume and bid-ask spread.

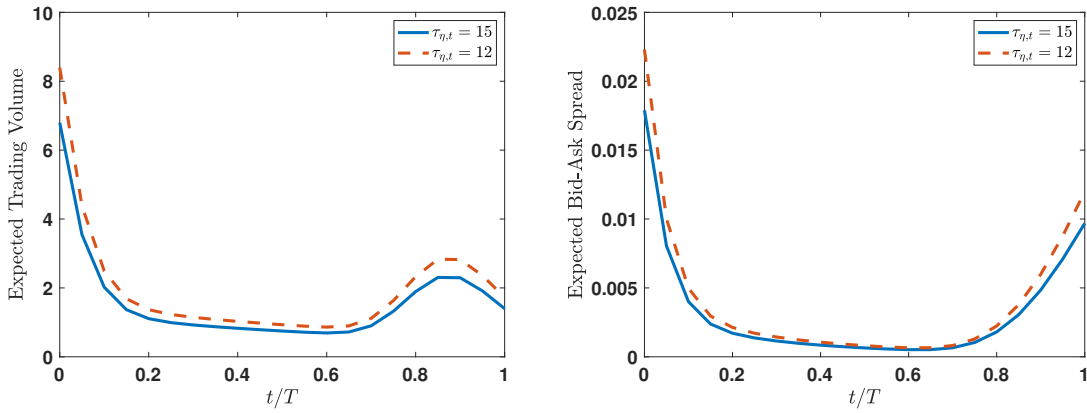


Figure 5a. Agency Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration for Different $\tau_{\eta,t}$.

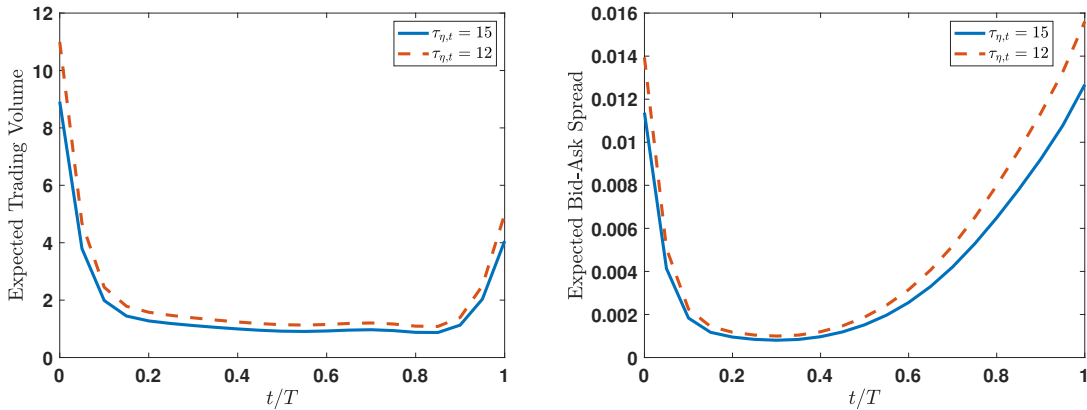


Figure 5b. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration for Different $\tau_{\eta,t}$.

Figures 5a and 5b illustrate the dynamics of expected trading volume and bid-ask spread under two

distinct liquidity shock uncertainties. The solid curve represents the case with an uncertainty of the liquidity shocks at t with $\tau_{\eta,t}^{-1} = 1/15$, while the dashed curve represents the case with $\tau_{\eta,t}^{-1} = 1/12$.²⁸

Figures 6a and 6b show the dynamics of the expected trading volume and bid-ask spread for two different information precisions. The solid curve represents the case with information precision at t with $\tau_{\varepsilon,t} = 0.75$, while the dashed curve represents the case with $\tau_{\varepsilon,t} = 0.5$.²⁹

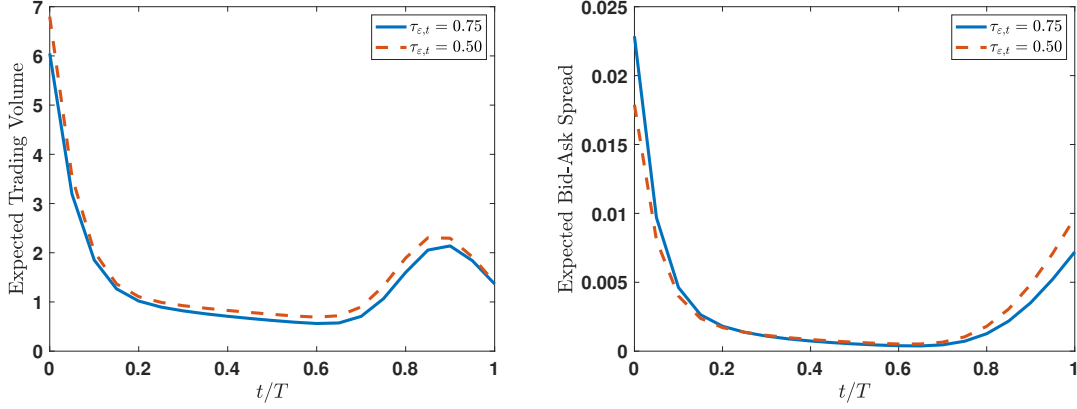


Figure 6a. Agency Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration for Different $\tau_{\varepsilon,t}$.

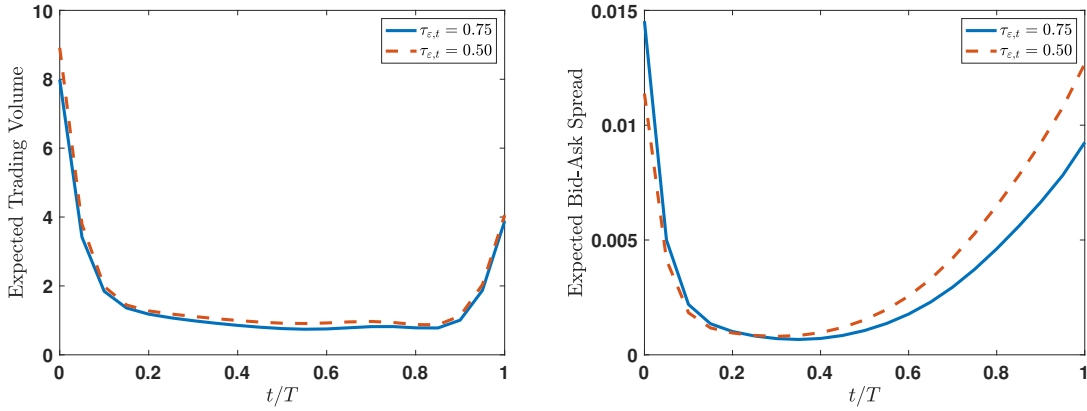


Figure 6b. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration for Different $\tau_{\varepsilon,t}$.

²⁸In Figures 5a and 5b, the total uncertainty of liquidity shocks is $T\tau_{\eta,t}^{-1} = 20/15 = 1.33$ (solid curve) and $T\tau_{\eta,t}^{-1} = 20/12 = 1.67$ (dashed curve). Parameter values in both Figure 5a and Figure 5b are $T = 20$, $N_I = 10$, $N_U = 100$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,0} = 10$, and $\tau_{\varepsilon,0} = 5$. For $t = 1, 2, \dots, T$, $\tau_{\varepsilon,t} = 0.5$ and $\tau_{\eta,t} = 15$ ($\tau_{\eta,t} = 12$) in the solid (dashed) curve. In the agency market making case (Figures 5a and 6a), $N_M = 1$, and $\lambda^U = 1$. In the principal market making case (Figures 5b and 6b), $N_M = 2$ and $\lambda^U = 0$.

²⁹In Figures 6a and 6b, the total precision of informed traders' private information is $T\tau_{\varepsilon,t} = 15$ (solid curve) and $T\tau_{\varepsilon,t} = 10$ (dashed curve). Parameter values in both Figure 6a and Figure 6b are $T = 20$, $N_I = 10$, $N_U = 100$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,0} = 10$, and $\tau_{\varepsilon,0} = 5$. For $t = 1, 2, \dots, T$, $\tau_{\eta,t} = 15$ and $\tau_{\varepsilon,t} = 0.75$ ($\tau_{\varepsilon,t} = 0.5$) in the solid (dashed) curve.

As shown in the left panels of Figures 5a, 5b, 6a, and 6b, the U-shape of trading volume becomes more pronounced when the uncertainty of liquidity shocks increases (i.e., when $\tau_{\eta,t}$ decreases) or when the precision of informed traders' private information is lower (i.e., when $\tau_{\varepsilon,t}$ decreases). This trend arises because uninformed traders may perceive that trades are more likely driven by hedging needs. Consequently, the adverse selection effect becomes less pronounced, allowing informed traders to trade more aggressively on their private information at market opening and shift more trading due to hedging demand at market close. As a result, the U-shaped pattern of trading volume becomes more pronounced. An empirical implication is that the U-shaped pattern of trading volume will be more prominent for stocks with less asymmetric information or a higher proportion of liquidity traders, assuming all other factors remain constant.

Similarly, the U-shaped pattern of the bid-ask spread becomes more pronounced with increased uncertainty of liquidity shocks (when $\tau_{\eta,t}$ is lower), as demonstrated in the right panels of Figures 5a and 5b. When informed traders' private information becomes more precise (keeping the magnitude of liquidity shocks constant), since informed traders tend to act quickly on their private information, the difference in reservation price at market opening becomes larger and it becomes smaller in later trading periods. Therefore, the expected bid-ask spread becomes wider at market opening and narrower during later trading periods when the precision of private information increases (i.e., when $\tau_{\varepsilon,t}$ increases), as illustrated in the right panel of Figures 6a and 6b.

3.3. The Effects of Trading Rounds and Market Makers' Market Power

In this subsection, we explore how variations in the number of trading rounds and the market power of market makers influence bid-ask spreads.

By adjusting T , we modify the number of trading rounds while keeping the time interval fixed at $[0, 1]$. This means traders can execute trades from time 0 to 1 across $T + 1$ rounds in total (spanning trading periods $t = 0, 1, \dots, T$). Additionally, while changing the number of trading rounds, we hold both the total precision of private information and the total uncertainty of liquidity shocks constant.

Figure 7 depicts the dynamics of the expected bid-ask spread against the trading duration t/T for various combinations of N_M (indicating competitiveness among market makers) and the number of trading

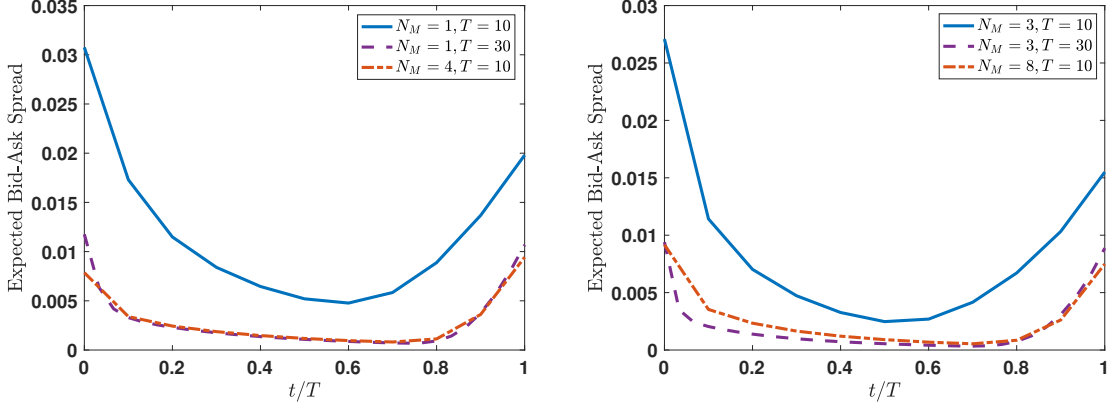


Figure 7. The Dynamics of Expected Bid-Ask Spread against Trading Duration for Different N_M and T for Agency (left) and Principal (right) Trading Market Makers.

rounds T .³⁰ Both the total precision of private information and the total uncertainty of liquidity shocks are held constant. As illustrated in Figure 7, either increasing the trading frequency or the competition among market makers (i.e., increasing N_M) leads to a reduction in the bid-ask spread. The introduction of more trading rounds or additional market makers mitigates the market power of market makers, subsequently narrowing the bid and ask spreads. In Figure 7, it's evident that the U-shaped pattern is more pronounced in less competitive markets (characterized by fewer market makers) or markets with fewer trading rounds. In such environments, market makers possess more market power, which allows them to set wider bid-ask spreads.

An empirical implication of our model is that stocks with fewer liquidity suppliers, such as smaller stocks, and those with a pronounced order imbalance — giving liquidity suppliers market power — tend to exhibit a more significant U-shaped pattern in bid-ask spread.

4. Model Calibration and Implications

Our main findings are applicable to markets characterized by imperfect competition among liquidity suppliers. As an illustrative example, we perform model calibration using stocks listed on NASDAQ.

³⁰Parameters are $N_I = 10$, $N_U = 100$, $\lambda^I = 1$, $\lambda^U = 1$, $\bar{\theta} = 1$, $\bar{V} = 1$, $\tau_V = 1$, $\tau_{\varepsilon,0} = 0.5$, and $\tau_{\eta,0} = 10$. For $t = 1, 2, \dots, T$, $\tau_{\varepsilon,t} = 2/T$ and $\tau_{\eta,t} = 2T$, implying that both the total precision of information and the total uncertainty of liquidity shock are fixed.

4.1. Model Calibration

In this subsection, we calibrate model parameter values to match the empirical intraday dynamics of bid-ask spread and trading volume using TAQ data. Our sample comprises all common stocks listed on NASDAQ from January 2000 to December 2021.³¹ For each stock, denoted by subscript i , we compute both the absolute and relative bid-ask spread at each trading time t , where t represents specific trading times throughout the day. Specifically, we compute:

$$\begin{aligned} \text{Absolute Bid-Ask Spread}_{it} &:= \text{Ask}_{it} - \text{Bid}_{it}, \\ \text{Relative Bid-Ask Spread}_{it} &:= \frac{\text{Ask}_{it} - \text{Bid}_{it}}{\text{Mid-point Price}_{it}}, \end{aligned} \tag{33}$$

at half-hour intervals throughout the trading day, specifically at trading times $t = 9:30, 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$. We then calculate the average bid-ask spreads across all stocks for each of these 14 trading points, and then across dates from January 2000 to December 2021. The resulting mean and standard deviation of the spreads are presented in Table 1.

For each stock i , we also compute the cumulative trading volume during the time interval of $9:30 - 10:00, 10:00 - 10:30, 10:30 - 11:00, \dots, 15:00 - 15:30$, and $15:30 - 16:00$. Specifically, we compute cumulative turnover from 30 minutes prior to time t to time t as,

$$\text{Turnover}_{it} := \frac{\text{Trading Volume}_{it}}{\text{Shares Outstanding}_{it}}, \tag{34}$$

at half-hour intervals throughout the trading day, specifically at trading times $t = 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$. We then calculate the average cumulative turnover across stocks for each of these 13 trading points, and subsequently across dates from January 2000 to December 2021. The resulting averages of turnover are presented in Table 2.

In Table 1, Column (1) presents the averages of absolute bid-ask spreads, while Column (3) presents the averages of relative bid-ask spreads. To calibrate the model, we aim to find parameters that match the theoretical expected absolute bid-ask spreads to the corresponding empirical averages of absolute

³¹Intraday trading data is available from January 1993 onwards for all stocks listed on NASDAQ. Analysis of the data shows that the average intra-day bid-ask spread and trading volume exhibit U-shaped patterns for this period, with the year-by-year U-shape for bid-ask spread becoming more pronounced after 2000. As a result, we have calibrated the model parameters based on the average bid-ask spreads and trading volume from January 2000 to December 2021.

Table 1. Intraday Relative and Absolute Bid-Ask Spreads of NASDAQ Stocks

This table reports the average and standard deviation of bid-ask spreads at half-hour intervals throughout the trading day, specifically at 9:30, 10:00, 10:30, 11:00, ..., 15:00, 15:30, 16:00. At each of these trading points, we identify the bid and ask quotes that were placed nearest to the half-hour mark within a ± 5 minutes interval. For example, the quotes at 9:30 are taken as the quotes that were placed closest to 9:30 within the range of [9:25:00, 9:35:00). We also require that the quotes must be submitted from the listed exchange (e.g., NASDAQ). We trim the relative spreads at 10% and absolute spreads at \$10 to avoid the influence of extreme outliers. We average bid-ask spreads across stocks and then across dates, generating average spreads at these 14 trading points. We also report the number of dates (N) at each time point available from January 2000 to December 2021.

Time	Absolute Spread		Relative Spread		N
	Mean (1)	Std (2)	Mean (3)	Std (4)	
9:30	0.552	0.312	2.59%	0.64%	5535
10:00	0.166	0.058	1.14%	0.35%	5536
10:30	0.131	0.043	0.95%	0.35%	5535
11:00	0.119	0.039	0.88%	0.35%	5536
11:30	0.112	0.037	0.84%	0.35%	5536
12:00	0.108	0.037	0.81%	0.35%	5536
12:30	0.104	0.035	0.79%	0.34%	5535
13:00	0.105	0.043	0.78%	0.35%	5534
13:30	0.112	0.137	0.78%	0.42%	5534
14:00	0.112	0.138	0.78%	0.42%	5535
14:30	0.107	0.112	0.77%	0.39%	5533
15:00	0.106	0.121	0.78%	0.43%	5535
15:30	0.104	0.108	0.78%	0.42%	5533
16:00	0.299	0.183	1.42%	0.52%	5521

bid-ask spreads presented in Column (1) of Table 1³² and the theoretical expected cumulative trading turnover to the empirical averages presented in Table 2.

The expected payoff of the asset ($\bar{V}$) has minimal influence on the expected bid-ask spread and turnover. Therefore, we set $\bar{V}$ to 1. We also set $\tau_V = 1$ because, in our model, what's pivotal are the relative magnitudes of the precision of payoff, information, and liquidity shocks. We normalize the total supply of the asset, $\bar{\theta}$, to 1. As a result, the expected trading volume in our model corresponds to the average turnover, which is empirically used to measure trading volume. To calibrate our model, we aim to

³²Table 1 shows that both the relative bid-ask spreads and absolute bid-ask spreads for NASDAQ stocks follow distinct intraday U-shaped patterns. However, since prices follow a normal distribution in CARA-normal settings, the expected relative bid-ask spreads cannot be expressed in analytically tractable formulas. Therefore, we focus on calibrating model parameters that align with the theoretical expected absolute bid-ask spreads to match the corresponding empirical averages of absolute bid-ask spreads.

Table 2. Intraday Turnover of NASDAQ Stocks

This table reports the average turnover over half-hour intervals throughout the trading day, specifically during the time intervals of 9:30 –10:00, 10:00 –10:30, 10:30 –11:00, ..., 15:00 –15:30, and 15:30 –16:00. We calculate the average turnover across stocks and then across dates from January 2000 to December 2021, generating average turnover at these 13 trading points. To avoid the influence of extreme outliers, we trim turnover at 200%.

Time	Turnover	Time	Turnover
9:30 –10:00	25.13%	10:00 –10:30	17.73%
10:30 –11:00	15.17%	11:00 –11:30	13.70%
11:30 –12:00	12.14%	12:00 –12:30	11.18%
12:30 –13:00	10.64%	13:00 –13:30	10.26%
13:30 –14:00	10.05%	14:00 –14:30	10.54%
14:30 –15:00	10.70%	15:00 –15:30	12.17%
15:30 –16:00	18.79%		

find model parameters that can match the theoretical expected absolute bid-ask spreads and cumulative turnover to the corresponding empirical averages of these measures at half-hour intervals throughout the trading day. For this purpose, we set $T = 13$ in our model calibration.³³ To further reduce the number of free parameters, we assume $N_M = 1$ and $N_I = 1.5$, indicating that the population of informed traders is 1.5 times that of the market makers.³⁴ We assume that the risk aversion of uninformed traders, λ^U , is set to 1 for the agency trading market-making case, and to 0 in the principal trading market-making case. We will calibrate the risk aversion of informed traders accordingly.

There remain six parameters in our model, which include the mass of uninformed traders (N_U), the risk aversion of informed traders (λ^I), the precision of information at market opening ($\tau_{\varepsilon,0}$), the precision of information at other periods ($\tau_{\varepsilon,t}$ for $0 < t \leq T$), the precision of liquidity shocks at market closure ($\tau_{\eta,T}$), and the precision of liquidity shocks at other periods ($\tau_{\eta,t}$ for $0 \leq t < T$). As suggested by our model's intuition, the magnitude of the U-shaped intraday pattern of the bid-ask spread is determined by the precision of private information, the precision of liquidity shocks, and the competition level among

³³In our model, without new information or hedging needs, market makers keep their quoted bid and ask prices unchanged. Traders, in these intervals, transact based on the optimal quantity determined by their prevailing information and hedging demand. When new information or hedging needs arise, traders adjust their optimal quantities and market makers update the quoted prices accordingly. The dynamics of trading volume and bid-ask spread when information or liquidity shocks arrive sporadically is illustrated in Section B.5 of the Internet Appendix. Practically speaking, during market opening, there might be more than 14 trading rounds. But, given the infrequent arrival of new signals and changes in hedging needs throughout the day, our calibration assumes $T = 13$ (14 trading rounds).

³⁴As presented in the Internet Appendix B.4, with a different N_I , the model can still generate empirical patterns with varying calibrated parameter values for N_U , λ^I , $\tau_{\varepsilon,0}$, $\tau_{\varepsilon,t}$ for $0 < t \leq T$, $\tau_{\eta,T}$, and $\tau_{\eta,t}$ for $0 \leq t < T$.

Table 3. List of Given and Estimated Parameters

The table reports the given parameter values for $\bar{V}$, τ_V , $\bar{\theta}$, T , λ^U , N_M , and N_I . The bottom six lines report the estimated parameters N_U , λ^I , $\tau_{\varepsilon,0}$, $\tau_{\varepsilon,t}$ for $0 < t \leq T$, $\tau_{\eta,T}$, and $\tau_{\eta,t}$ for $0 \leq t < T$.

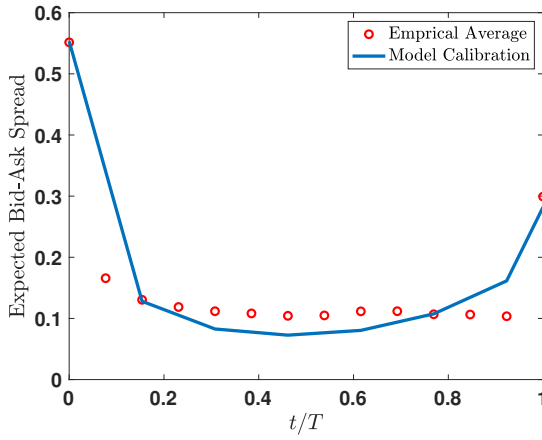
Parameter	Description	Value	
		Agency MM	Principal MM
$\bar{V}$	Expected Payoff	1	1
τ_V	Precision of Payoff V	1	1
$\bar{\theta}$	Total Supply	1	1
T	Trading Periods	13	13
λ^U	Risk Aversion of Uninformed	1	0
N_M	Number of Market Makers	1	1
N_I	Mass of Informed Traders	1.5	1.5
N_U	Mass of Uninformed Traders	283.480	N/A
λ^I	Risk Aversion of Informed	5.373	2.227
$\tau_{\varepsilon,0}$	Precision of Information at Period 0	0.238	0.168
$\tau_{\varepsilon,t}$	Precision of Information at other Periods	0.096	0.106
$\tau_{\eta,T}$	Precision of Hedging at T	15.411	9.070
$\tau_{\eta,t}$	Precision of Hedging at other Periods	111.258	39.552

market makers.

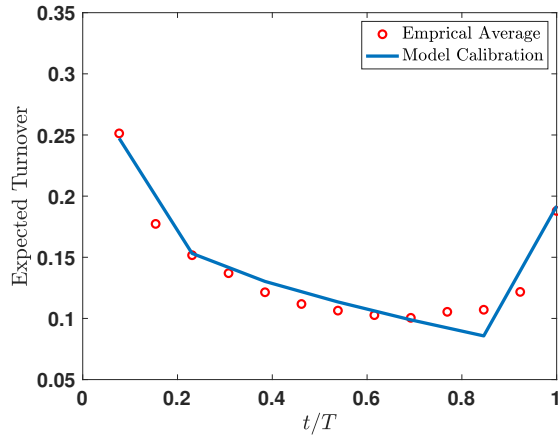
Calibrating model parameters to simultaneously match both the bid-ask spread and trading volume across all trading points is highly challenging due to the need to align 27 data points. Therefore, we calibrate the model parameters to match the empirical averages of absolute bid-ask spreads observed at eight key trading times—9:30, 10:30, 11:30, 12:30, 13:30, 14:30, 15:30, and 16:00—and the empirical cumulative trading volume at seven trading times—10:00, 11:00, 12:00, 13:00, 14:00, 15:00, and 16:00.

Table 3 summarizes the calibrated parameter values. For the case with agency trading market makers, the mass of uninformed traders is calibrated at $N_U = 283.480$, and the risk aversion of informed traders, who trade based on both private information and hedging demand, is calibrated at $\lambda^I = 5.373$. The precision of information at market opening is $\tau_{\varepsilon,0} = 0.238$, and for other periods, it is $\tau_{\varepsilon,t} = 0.096$. The uncertainty of hedging demand at market closure is $\tau_{\eta,T}^{-1/2} = 0.255$, and for other periods, it is $\tau_{\eta,t}^{-1/2} = 0.095$. For the case with principal trading market makers, the precision of information at market opening is $\tau_{\varepsilon,0} = 0.168$, and for other periods, it is $\tau_{\varepsilon,t} = 0.106$. The uncertainty of hedging demand at market closure is $\tau_{\eta,T}^{-1/2} = 0.332$, and for other periods, it is $\tau_{\eta,t}^{-1/2} = 0.159$.

The calibrated parameter values indicate that the precision of private information is higher at market

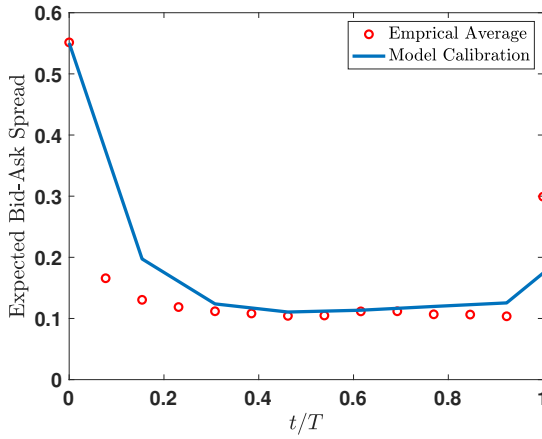


(a) Bid-Ask Spread Calibration

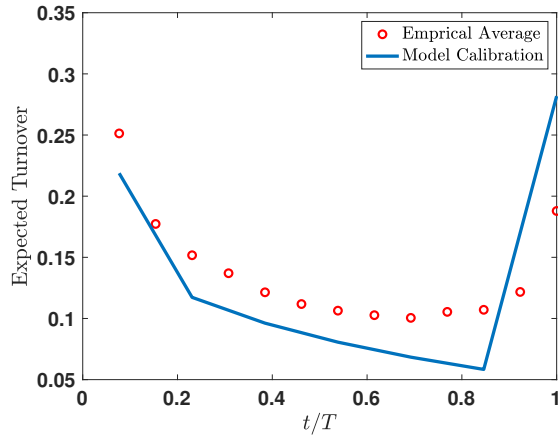


(b) Turnover Calibration

Figure 8. Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for agency trading market makers.



(a) Bid-Ask Spread Calibration



(b) Turnover Calibration

Figure 9. Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for principal trading market makers.

opening compared to other periods, while the uncertainty of hedging needs is greater at market closure. This suggests that trading at market opening is predominantly influenced by private information, while that at market closure is primarily driven by hedging needs. This is also consistent with the model's intuition that traders tend to act quickly on private information, while trading driven by hedging needs is delayed until later periods. As market makers can charge wider bid-ask spreads when trading incentives are greater, bid-ask spreads are wider at market opening and closure.

Figures 8 and 9 depict the expected absolute bid-ask spreads ($E[A_t - B_t]$) and expected turnover from

our model for agency and principal trading market makers, respectively. The left panels show 14 dots corresponding to the empirical averages of absolute bid-ask spreads at trading times $t = 9:30, 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$, as presented in Column (1) of Table 1. The solid curves represent our model's theoretical predictions, which generally align with the empirical values.

In the right panels, we present the expected cumulative turnover for NASDAQ-listed stocks for agency and principal trading market makers. The right panel of Figure 8 is for the case with agency trading market makers, while the right panel of Figure 9 corresponds to the case with principal trading market makers. Each panel displays 13 dots representing the empirical averages of cumulative turnover at trading times $t = 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$. The models' expected cumulative turnover generally aligns with these points, although not perfectly.

4.2. Market Makers' Exogenous Inventory Shocks and Traders' Execution Costs

Although a dynamic setting where risk-averse market makers simultaneously perform both agency and principal market making would not be tractable in our paper, we explore a one-period version of our model to investigate the effects of inventory shocks from risk-averse market makers on other market participants' trading costs. Figure 10 depicts the expected ask price, bid price, and competitive price in two scenarios: one where market makers possess a significantly large positive position (Panel (a)) and another where market makers hold a significantly negative position (Panel (b)).³⁵

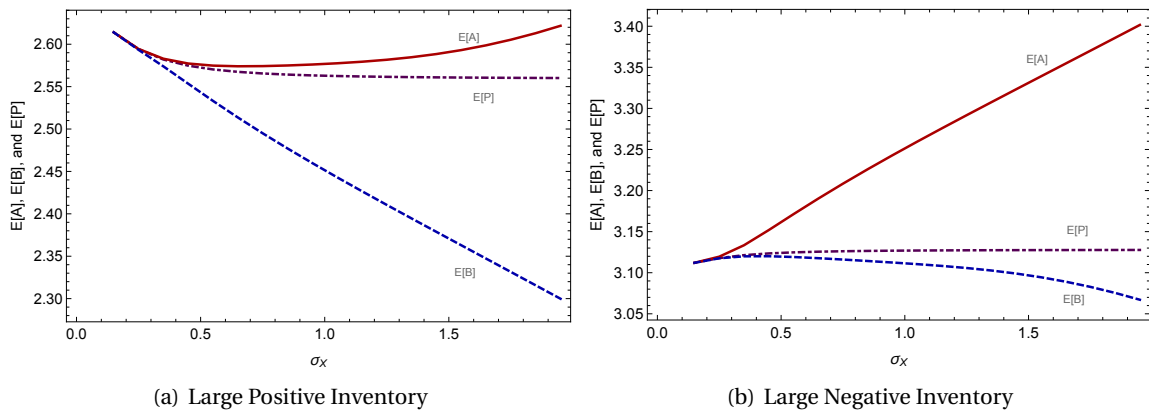


Figure 10. Expected Ask Price (solid curve), Expected Bid Price (dashed curve), and Expected Competitive Price (dot-dashed curve) against σ_X .

³⁵Parameters are $N_M = 1$, $N_I = 10$, $N_U = 100$, $\lambda^I = \lambda^U = 1$, $\tilde{V} = 3$, $\tau_V = 0.4$, $\tau_{\varepsilon,0} = 0.8$, $\bar{\theta}^I = \bar{\theta}^U = 1$, $\bar{\theta}^M = 200$ in Panel (a) and $\bar{\theta}^M = -200$ in Panel (b). Please see detailed analysis in Section B.9 of the Internet Appendix.

In Figure 10, Panel (a) shows that when market makers have a notably positive initial endowment, the expected ask price closely aligns with the competitive price. Meanwhile, Panel (b) illustrates that the expected bid price remains close to the competitive price when the initial endowment is significantly negative. This suggests that with a large stock inventory, market makers offer better prices to buyers, and with a substantial short position, they provide more favorable prices to sellers.

These findings are in line with empirical research by [Bogousslavsky, Collin-Dufresne, and Sağlam \(2021\)](#), who examine the impact of exogenous shocks to the inventory of a designated market maker for a set of NYSE stocks on market liquidity and other market participants' execution costs. They find strong evidence indicating that traders who trade on the “same side” as the market makers consistently experience higher execution costs compared to “opposite side” trades.

5. Conclusion

In this paper, we present a dynamic trading model based on flows of private information and liquidity shocks with imperfectly competitive liquidity suppliers. The market power of market makers leads other traders to smooth out their trading, even though they are not strategic. Traders are motivated to trade on their private information in earlier periods to capitalize on potential profits, as other traders might also act on similar information. However, they delay their trading due to hedging needs to later periods, as the adverse selection effect becomes less severe over time. Consequently, trading volume may exhibit a U-shaped pattern, with higher trading volume observed at the beginning and the end of the trading duration and lower volume in the middle. The market power of liquidity suppliers enables them to widen bid-ask spreads when traders' trading incentives are heightened, resulting in wider bid-ask spreads at market opening and closure. Therefore, our model provides a unified and coherent explanation for the observed U-shaped patterns in both trading volume and bid-ask spread.

Our model explains the empirical finding that liquidity demanders pay most of the bid-ask spread while nondealer liquidity suppliers get much more favorable prices from market makers. Our model predicts that the variation in trading volume tends to be smaller for stocks with less competitive liquidity suppliers during periods of news events. The U-shaped pattern of the bid-ask spread tends to be more pronounced for stocks with less competitive liquidity suppliers (e.g., small stocks) and stocks with higher

order imbalance. The U-shaped pattern of the trading volume tends to be more pronounced for stocks with lower information asymmetry or a higher proportion of liquidity traders. The model's calibration aligns with the empirical intraday patterns of bid-ask spread and trading volume.

Code Availability Statement: The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/N37V4B>.

Appendix

A.1. Proof of Lemma 1

To prove Lemma 1, we first present the following lemma.

Lemma 2. Suppose z_t is an $n \times 1$ vector and it follows the process $z_t = a_{z,t}z_{t-1} + b_{z,t}\zeta_t$, where $\zeta_t \sim \mathcal{N}(0, \Sigma_{\zeta,t})$ is a $k \times 1$ vector, $a_{z,t}$ is an $n \times n$ matrix, and $b_{z,t}$ is an $n \times k$ matrix. The signal s_t is an $m \times 1$ vector, $s_t = a_{s,t}z_t + b_{s,t}\zeta_t$, where the noise of the signal is perfectly correlated with the noise of the process z_t , $a_{s,t}$ is an $m \times n$ matrix, and $b_{s,t}$ is an $m \times k$ matrix. Denote the estimates of the first and second moments as $\hat{z}_t := E_t[z_t]$, $o_t := E[(z_t - \hat{z}_t)(z_t - \hat{z}_t)^\top]$, where $E_t[\cdot] = E[\cdot | \underline{s}_t]$ and $\underline{s}_t := \{s_0, s_1, s_2, \dots, s_t\}$. Then the Kalman filtering formulas for $\hat{z}_t$ and o_t are

$$\hat{z}_t = a_{z,t}\hat{z}_{t-1} + k_t[s_t - E_{t-1}[s_t]], \quad o_t = (I_{n \times n} - k_t a_{s,t})(a_{z,t}o_{t-1}a_{z,t}^\top + b_{z,t}\Sigma_{\zeta,t}b_{z,t}^\top) - k_t b_{s,t}\Sigma_{\zeta,t}b_{z,t}^\top,$$

where k_t is defined below in equation (A-2).

Proof of Lemma 2: Kalman filtering contains two parts: predict and update. The predicted (a priori) state estimate is $\hat{z}_{t|t-1} = E_{t-1}[z_t] = a_{z,t}\hat{z}_{t-1}$, and the innovation (pre-fit residual) is

$$s_t - E_{t-1}[s_t] = a_{s,t}z_t + b_{s,t}\zeta_t - a_{s,t}\hat{z}_{t|t-1} = a_{s,t}a_{z,t}(z_{t-1} - \hat{z}_{t-1}) + (a_{s,t}b_{z,t} + b_{s,t})\zeta_t.$$

The innovation (pre-fit residual) variance and covariance matrix are

$$\begin{aligned} \text{Var}_{t-1}[s_t] &= a_{s,t}a_{z,t}o_{t-1}a_{z,t}^\top a_{s,t}^\top + (a_{s,t}b_{z,t} + b_{s,t})\Sigma_{\zeta,t}(a_{s,t}b_{z,t} + b_{s,t})^\top, \\ \text{Cov}_{t-1}(z_t, s_t) &= a_{z,t}o_{t-1}a_{z,t}^\top a_{s,t}^\top + b_{z,t}\Sigma_{\zeta,t}(a_{s,t}b_{z,t} + b_{s,t})^\top, \end{aligned} \tag{A-1}$$

and the optimal Kalman gain k_t is

$$k_t = \text{Cov}_{t-1}(z_t, s_t)(\text{Var}_{t-1}[s_t])^{-1}, \tag{A-2}$$

where $\text{Cov}_{t-1}(z_t, s_t)$ and $\text{Var}_{t-1}[s_t]$ are as defined in equation (A-1). Therefore, the updated (a posteriori)

state estimate $\hat{z}_t$ and the updated (a posteriori) estimate covariance matrix o_t are

$$\hat{z}_t = a_{z,t} \hat{z}_{t-1} + k_t [s_t - E_{t-1}[s_t]], \quad o_t = E[(z_t - \hat{z}_t)(z_t - \hat{z}_t)^\top],$$

where $z_t - \hat{z}_t = (I_{n \times n} - k_t a_{s,t}) a_{z,t} (z_{t-1} - \hat{z}_{t-1}) + (b_{z,t} - k_t (a_{s,t} b_{z,t} + b_{s,t})) \zeta_t$. From equation (A-2), we have $k_t \text{Var}_{t-1}[s_t] k_t^\top = \text{Cov}_{t-1}(z_t, s_t) k_t^\top$. Using equations (A-1) and (A-2), the updated estimate covariance matrix can be simplified as

$$\begin{aligned} o_t &= (I_{n \times n} - k_t a_{s,t}) a_{z,t} o_{t-1} a_{z,t}^\top (I_{n \times n} - k_t a_{s,t})^\top + (b_{z,t} - k_t (a_{s,t} b_{z,t} + b_{s,t})) \Sigma_{\zeta,t} (b_{z,t} - k_t (a_{s,t} b_{z,t} + b_{s,t}))^\top \\ &= (I_{n \times n} - k_t a_{s,t}) (a_{z,t} o_{t-1} a_{z,t}^\top + b_{z,t} \Sigma_{\zeta,t} b_{z,t}^\top) - k_t b_{s,t} \Sigma_{\zeta,t} b_{z,t}^\top. \end{aligned}$$

Q.E.D.

We now prove Lemma 1 using Lemma 2; the process z_t and the noise $\zeta_t \sim \mathcal{N}(0, \Sigma_{\zeta,t})$ are

$$z_t = \begin{pmatrix} V \\ X_t \end{pmatrix}, \quad \zeta_t = \begin{pmatrix} \varepsilon_t \\ \eta_t \end{pmatrix}, \quad \Sigma_{\zeta,t} = \begin{pmatrix} \tau_{\varepsilon,t}^{-1} & 0 \\ 0 & \tau_{\eta,t}^{-1} \end{pmatrix}.$$

In our model,

$$z_t = a_{z,t} z_{t-1} + b_{z,t} \zeta_t, \quad a_{z,t} = I_{2 \times 2}, \quad b_{z,t} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

State variables X_t and $\hat{V}_t^I$ are Gaussian Markov processes under $\mathcal{F}_t^I$,

$$\begin{aligned} \hat{V}_t^I &= \hat{V}_{t-1}^I + K_{V,t}^I e_t^I, \quad \text{where} \quad e_t^I = v_t - \hat{V}_{t-1}^I = V - \hat{V}_{t-1}^I + \varepsilon_t, \\ e_t^I | \mathcal{F}_{t-1}^I &\sim \mathcal{N}(0, \sigma_t^I), \quad \sigma_t^I = \text{Var}_{t-1}^I(e_t^I) = o_{V,t-1}^I + \tau_{\varepsilon,t}^{-1}, \quad K_{V,t}^I = o_{V,t}^I \tau_{\varepsilon,t}, \end{aligned} \tag{A-3}$$

and the second moment estimation is $o_{V,t}^I = \left(\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s} \right)^{-1}$. Uninformed traders' signal process is $s_t = a_{s,t} z_t + b_{s,t} \zeta_t$, $a_{s,t} = (1 \ 0)$, $b_{s,t} = (1 \ -h_t)$. Uninformed traders' state variables $\hat{V}_t^U$ and $\hat{X}_t^U$ are Gaussian Markov processes under $\mathcal{F}_t^U$:

$$\hat{V}_t^U = \hat{V}_{t-1}^U + K_{V,t}^U e_t^U, \quad \hat{X}_t^U = \hat{X}_{t-1}^U + K_{X,t}^U e_t^U, \quad \text{where} \tag{A-4}$$

$$e_t^U = v_t - h_t \eta_t - \hat{V}_{t-1}^U = V - \hat{V}_{t-1}^U - h_t \eta_t + \varepsilon_t, \quad e_t^U | \mathcal{F}_{t-1}^U \sim \mathcal{N}(0, \Sigma_t^U), \quad (\text{A-5})$$

$$\Sigma_t^U = \text{Var}_{t-1}^U(e_t^U) = o_{V,t-1}^U + \tau_{\varepsilon,t}^{-1} + h_t^2 \tau_{\eta,t}^{-1}, \quad K_{V,t}^U = o_{V,t-1}^U / \Sigma_t^U, \quad K_{X,t}^U = (o_{VX,t-1}^U - h_t \tau_{\eta,t}^{-1}) / \Sigma_t^U,$$

and the second moment estimates are

$$o_{X,t}^U = o_{X,t-1}^U + \tau_{\eta,t}^{-1} - (o_{VX,t-1}^U - h_t \tau_{\eta,t}^{-1})^2 / \Sigma_t^U, \quad o_{VX,t}^U = o_{VX,t-1}^U - (o_{VX,t-1}^U - h_t \tau_{\eta,t}^{-1}) o_{V,t-1}^U / \Sigma_t^U,$$

$$o_{V,t}^U = (\tau_{\varepsilon,t}^{-1} + h_t^2 \tau_{\eta,t}^{-1}) o_{V,t-1}^U / \Sigma_t^U = \left(\tau_V + \sum_{s=0}^t (\tau_{\varepsilon,s}^{-1} + h_s^2 \tau_{\eta,s}^{-1})^{-1} \right)^{-1},$$

where $o_{VX,t}^U := \text{Cov}(V, X_t | \mathcal{F}_t^U)$, the priors are $o_{V,-1}^U = \tau_V^{-1}$, $o_{X,-1}^U = 0$, and $o_{VX,-1}^U = 0$. The coefficients $K_{V,t}^U$ and $K_{X,t}^U$ can be written as

$$K_{V,t}^U = o_{V,t}^U \tau_t, \quad K_{X,t}^U = o_{V,t}^U \tau_t \left(\sum_{s=0}^{t-1} (h_s \tau_{\eta,s}^{-1} - h_t \tau_{\eta,t}^{-1}) \tau_s - \tau_V h_t \tau_{\eta,t}^{-1} \right). \quad (\text{A-6})$$

We next figure out the coefficient h_t in the composite signal $s_t = v_t - h_t \eta_t$. Denote the sufficient statistic of all the composite signals arriving from period 0 to t as $\hat{V}_t^I - \mu_t X_t$, which can be inferred by uninformed traders from prices. It follows that $\hat{V}_t^I - \mu_t X_t = \hat{V}_t^U - \mu_t \hat{X}_t^U$. From equations (A-3) and (A-4), we have

$$\left[K_{V,t}^I - (K_{V,t}^U - \mu_t K_{X,t}^U) \right] (v_t - \hat{V}_{t-1}^U) + \left[(K_{V,t}^U - \mu_t K_{X,t}^U) h_t - \mu_t \right] \eta_t + [(1 - K_{V,t}^I) \mu_{t-1} - \mu_t] (X_{t-1} - \hat{X}_{t-1}^U) = 0. \quad (\text{A-7})$$

Therefore, the coefficients of $v_t - \hat{V}_{t-1}^U$, η_t , and $X_{t-1} - \hat{X}_{t-1}^U$ need to be zero. Setting the coefficients of η_t and $X_{t-1} - \hat{X}_{t-1}^U$ to be zero yields

$$\mu_t / \mu_{t-1} = \left(\tau_V + \sum_{s=0}^{t-1} \tau_{\varepsilon,s} \right) \left(\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s} \right)^{-1}, \quad \mu_t / h_t = \tau_{\varepsilon,t} \left(\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s} \right)^{-1}. \quad (\text{A-8})$$

Informed traders' expected utility at $t = T$ is

$$\mathbb{E}_T^I \left[-e^{-\lambda^I W_{T+1}^I} \right] = -e^{-\lambda^I \left[W_T^I + \theta_T^I (\hat{V}_T^I - P_T^I) + X_T \hat{V}_T^I - \frac{1}{2} \lambda^I o_{V,T}^I (\theta_T^I + X_T)^2 \right]},$$

where P_T^I denotes informed traders' trading price at $t = T$ (it's A_T if informed traders buy and B_T if

informed traders sell), taking the first order condition with respect to θ_T^I yields

$$\theta_T^I = (\hat{V}_T^I - \lambda^I o_{V,T}^I X_T - P_T^I) (\lambda^I o_{V,T}^I)^{-1}.$$

It implies that informed traders' trading reveals the composite signal $\hat{V}_T^I - \mu_T X_T$, where

$$\mu_T = \lambda^I o_{V,T}^I = \lambda^I \left(\tau_V + \sum_{s=0}^T \tau_{\varepsilon,s} \right)^{-1}. \quad (\text{A-9})$$

From equations (A-8) and (A-9), we have

$$\mu_t = \lambda^I o_{V,t}^I = \lambda^I \left(\tau_V + \sum_{s=0}^t \tau_{\varepsilon,s} \right)^{-1}, \quad h_t = \lambda^I / \tau_{\varepsilon,t}. \quad (\text{A-10})$$

We then verify that the coefficient of $v_t - \hat{V}_{t-1}^U$ in equation (A-7) is indeed zero.

A.2. Proof of Proposition 1

Denote P_t^i as trader i 's trading price at t ($i \in \{I, U\}$). We first solve the optimization problems at T .

The expected utility functions of informed and uninformed traders at T are

$$\begin{aligned} \mathbb{E}_T^I \left[-e^{-\lambda^I W_{T+1}^{I,i}} \right] &= -e^{-\lambda^I \left[W_T^{I,i} + \theta_T^{I,i} (\hat{V}_T^I - P_T^I) + X_T (\hat{V}_T^I - \bar{V}) - \frac{1}{2} \lambda^I o_{V,T}^I (\theta_T^{I,i} + X_T)^2 \right]}, \\ \mathbb{E}_T^U \left[-e^{-\lambda^U W_{T+1}^{U,j}} \right] &= -e^{-\lambda^U \left[W_T^{U,j} + \theta_T^{U,j} (\hat{V}_T^U - P_T^U) - \frac{1}{2} \lambda^U o_{V,T}^U (\theta_T^{U,j})^2 \right]}. \end{aligned} \quad (\text{A-11})$$

In the symmetric equilibrium, $\theta_{t-1}^{I,i}$ is the same for any I -trader i and $\theta_{t-1}^{U,j}$ is the same for any U -trader j ; we thus ignore the superscripts i and j in the main text of the paper for notational simplicity. We use $\theta_{t-1}^{I,i}$ and $\theta_{t-1}^{U,j}$ in this proof to make it clear that each individual trader optimizes his expected utility by taking prices as given. Taking the first order condition with respect to $\theta_T^{I,i}$ and $\theta_T^{U,j}$ in traders' expected utility functions yields

$$\begin{aligned} \theta_T^{I,i} - \theta_{T-1}^{I,i} &= (P_T^{IR,i} - P_T^I) / \gamma_T^I, \quad \theta_T^{U,j} - \theta_{T-1}^{U,j} = (P_T^{UR,j} - P_T^U) / \gamma_T^U, \quad \text{where} \\ P_T^{IR,i} &:= \hat{V}_T^I - \mu_T X_T - \gamma_T^I \theta_{T-1}^{I,i}, \quad P_T^{UR,j} := \hat{V}_T^U - \gamma_T^U \theta_{T-1}^{U,j}, \\ \mu_T &= \gamma_T^I = \lambda^I o_{V,T}^I, \quad \gamma_T^U := \lambda^U o_{V,T}^U. \end{aligned} \quad (\text{A-12})$$

The second order conditions ($\gamma_T^I > 0$ and $\gamma_T^U > 0$) are satisfied. The average reservation prices are

$$P_T^{IR} = \hat{V}_T^I - \mu_T X_T - \frac{\gamma_T^I}{N_I} \sum_{i=1}^{N_I} \theta_{T-1}^{I,i}, \quad P_T^{UR} = \hat{V}_T^U - \frac{\gamma_T^U}{N_U} \sum_{j=1}^{N_U} \theta_{T-1}^{U,j}. \quad (\text{A-13})$$

Let $q_t^{I,m}$ and $q_t^{U,m}$ denote the trading quantities of market maker m with I and U traders, respectively.

If $q_t^{i,m}$ is positive (negative), then trader i buys from (sells to) market maker, we have

$$\sum_{m=1}^{N_M} q_t^{I,m} = \sum_{i=1}^{N_I} (\theta_t^{I,i} - \theta_{t-1}^{I,i}), \quad \sum_{m=1}^{N_M} q_t^{U,m} = \sum_{j=1}^{N_U} (\theta_t^{U,j} - \theta_{t-1}^{U,j}). \quad (\text{A-14})$$

At T , market maker m maximizes his profit at trading round T ,

$$\max_{q_T^{I,m} = -q_T^{U,m}} q_T^{I,m} P_T^I + q_T^{U,m} P_T^U. \quad (\text{A-15})$$

Using equations (A-12) and (A-14), we have

$$P_T^I = P_T^{IR} - \frac{\gamma_T^I}{N_I} \sum_{m=1}^{N_M} q_T^{I,m}, \quad P_T^U = P_T^{UR} - \frac{\gamma_T^U}{N_U} \sum_{m=1}^{N_M} q_T^{U,m}. \quad (\text{A-16})$$

Substituting (A-16) into (A-15) and taking first order condition yield,

$$q_T^{I,m} = -q_T^{U,m} = \frac{P_T^{IR} - P_T^{UR}}{(N_M+1)(\gamma_T^I/N_I + \gamma_T^U/N_U)}. \quad (\text{A-17})$$

Substituting equation (A-17) into (A-16) yields traders' trading prices,

$$\begin{aligned} P_T^I &= (1 - w_T^U) P_T^{IR} + w_T^U P_T^{UR} = \hat{V}_T^I - \mu_T X_T + w_T^U \mu_T \hat{X}_T^U - \gamma_T^I \bar{\theta}_{T-1}^I, \\ P_T^U &= w_T^I P_T^{IR} + (1 - w_T^I) P_T^{UR} = \hat{V}_T^U - w_T^I \mu_T \hat{X}_T^I - \gamma_T^U \bar{\theta}_{T-1}^U, \end{aligned} \quad (\text{A-18})$$

implying that trading prices are weighted average of traders' reservation prices, where

$$\begin{aligned} v_T &:= \frac{\gamma_T^U}{\gamma_T^I} = \frac{\lambda^U o_{V,T}^U}{\lambda^I o_{V,T}^I}, \quad w_T^I := \frac{N_M}{N_M+1} \frac{N_I v_T}{N_I v_T + N_U}, \quad w_T^U := \frac{N_M}{N_M+1} \frac{N_U}{N_I v_T + N_U} = \frac{N_U}{N_I v_T} w_T^I, \\ \bar{\theta}_{T-1}^I &:= (1 - w_T^U) \frac{1}{N_I} \sum_{i=1}^{N_I} \theta_{T-1}^{I,i} + w_T^U \frac{v_T}{N_U} \sum_{j=1}^{N_U} \theta_{T-1}^{U,j}, \quad \bar{\theta}_{T-1}^U := w_T^I \frac{1}{N_I v_T} \sum_{i=1}^{N_I} \theta_{T-1}^{I,i} + (1 - w_T^I) \frac{1}{N_U} \sum_{j=1}^{N_U} \theta_{T-1}^{U,j}. \end{aligned}$$

The parameter v_T is the ratio of the product of uninformed traders' risk aversion and the conditional variance for the uninformed to that of the informed and $N_I v_T + N_U$ is the weighted average popula-

tion of non-market-making traders (adjusted for investors' information and risk-aversion). Substituting equation (A-18) into (A-12) yields traders' stock positions,

$$\theta_T^{I,i} = -\frac{N_M \mu_T \hat{X}_T^U}{(N_M+1)(\gamma_T^I + N_I/N_U \gamma_T^U)} + \bar{\theta}_{T-1}^I, \quad \theta_T^{U,j} = \frac{N_M \mu_T \hat{X}_T^U}{(N_M+1)(\gamma_T^U + N_U/N_I \gamma_T^I)} + \bar{\theta}_{T-1}^U. \quad (\text{A-19})$$

Substituting equations (A-18) and (A-19) into traders' expected utility (A-11) yields

$$\begin{aligned} J_T^{I,i} &= -\rho_T^I e^{-\lambda^I \left[W_T^{I,i} + \frac{1}{2} (\Phi_T^I)^\top M_T^I \Phi_T^I + (C_T^I)^\top \Phi_T^I \bar{\theta}_{T-1}^I + \frac{1}{2} m_T^I (\bar{\theta}_{T-1}^I)^2 \right]}, \\ J_T^{U,j} &= -\rho_T^U e^{-\lambda^U \left[W_T^{U,j} + \frac{1}{2} (\Phi_T^U)^\top M_T^U \Phi_T^U + (C_T^U)^\top \Phi_T^U \bar{\theta}_{T-1}^U + \frac{1}{2} m_T^U (\bar{\theta}_{T-1}^U)^2 \right]}, \end{aligned} \quad (\text{A-20})$$

where $\Phi_T^I := [\hat{V}_T^I - \bar{V}, X_T, \hat{X}_T^U, \bar{\theta}]^\top$, $\Phi_T^U := [\hat{V}_T^U - \bar{V}, \hat{X}_T^U, \bar{\theta}]^\top$, parameters $\rho_T^I, \rho_T^U, m_T^I, m_T^U$ are scalars, C_T^I is a 4×1 vector, C_T^U is a 3×1 vector, M_T^I is a 4×4 symmetric matrix, and M_T^U is a 3×3 symmetric matrix and they are given as

$$\begin{aligned} \rho_T^I &= 1, \quad \rho_T^U = 1, \quad C_T^I := [0, 0, -w_T^U \mu_T, 0]^\top, \quad m_T^I := \gamma_T^I, \quad C_T^U := [0, w_T^I \mu_T, 0]^\top, \quad m_T^U := \gamma_T^U, \\ M_T^I &:= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & -\mu_T & 0 & 0 \\ 0 & 0 & (w_T^U \mu_T)^2 / \gamma_T^I & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad M_T^U := \begin{pmatrix} 0 & 0 & 0 \\ 0 & (w_T^I \mu_T)^2 / \gamma_T^U & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{A-21})$$

We next conjecture that traders' value functions at t are given as (A-20) where the subscript T is replaced by t . We also conjecture that traders' reservation prices at t are

$$\begin{aligned} P_t^{IR} &= \bar{V} + \delta_t^I (\hat{V}_t^I - \mu_t X_t - \bar{V}) + g_t^I \hat{X}_t^U - f_t^I \bar{\theta} - \frac{\gamma_t^I}{N_I} \sum_{i=1}^{N_I} \theta_{t-1}^{I,i}, \\ P_t^{UR} &= \bar{V} + \delta_t^U (\hat{V}_t^U - \bar{V}) - g_t^U \hat{X}_t^U - f_t^U \bar{\theta} - \frac{\gamma_t^U}{N_U} \sum_{j=1}^{N_U} \theta_{t-1}^{U,j}. \end{aligned} \quad (\text{A-22})$$

From equation (A-12), the parameter values of $\mu_t, \gamma_t^i, \delta_t^i, g_t^i$, and f_t^i at T are given as

$$\mu_T = \gamma_T^I = \lambda^I o_{V,T}^I, \quad \gamma_T^U = \lambda^U o_{V,T}^U, \quad \delta_T^I = \delta_T^U = 1, \quad g_T^I = g_T^U = 0, \quad f_T^I = f_T^U = 0. \quad (\text{A-23})$$

We then solve the problem backward from t to $t-1$ and verify our conjecture. Denote

$$\Phi_t^I := [\hat{V}_t^I - \bar{V}, X_t, \hat{X}_t^U, \bar{\theta}]^\top, \quad \Phi_t^U := [\hat{V}_t^U - \bar{V}, \hat{X}_t^U, \bar{\theta}]^\top, \quad v_t := \gamma_t^U / \gamma_t^I, \quad (\text{A-24})$$

$$w_t^I := \frac{N_M}{N_M+1} \frac{N_I v_t}{N_I v_t + N_U}, \quad w_t^U := \frac{N_M}{N_M+1} \frac{N_U}{N_I v_t + N_U} = \frac{N_U}{N_I v_t} w_t^I,$$

$$\begin{aligned} \bar{\theta}_{t-1}^I &:= (1 - w_t^U) \frac{1}{N_I} \sum_{i=1}^{N_I} \theta_{t-1}^{I,i} + w_t^U \frac{v_t}{N_U} \sum_{j=1}^{N_U} \theta_{t-1}^{U,j} = \frac{1}{(N_M+1)N_I} \sum_{i=1}^{N_I} \theta_{t-1}^{I,i} + \frac{w_t^I}{N_I} \bar{\theta}, \\ \bar{\theta}_{t-1}^U &:= w_t^I \frac{1}{N_I v_t} \sum_{i=1}^{N_I} \theta_{t-1}^{I,i} + (1 - w_t^I) \frac{1}{N_U} \sum_{j=1}^{N_U} \theta_{t-1}^{U,j} = \frac{1}{(N_M+1)N_U} \sum_{j=1}^{N_U} \theta_{t-1}^{U,j} + \frac{w_t^U}{N_U} \bar{\theta}. \end{aligned} \quad (\text{A-25})$$

Note that $N_I \bar{\theta}_{t-1}^I + N_U \bar{\theta}_{t-1}^U = \bar{\theta}$. Traders' trading prices at t can be written as

$$\begin{aligned} P_t^I &= (1 - w_t^U) P_t^{IR} + w_t^U P_t^{UR} = \bar{V} + (H_{P,t}^I)^\top \Phi_t^I - \gamma_t^I \bar{\theta}_{t-1}^I, \\ P_t^U &= w_t^I P_t^{IR} + (1 - w_t^I) P_t^{UR} = \bar{V} + (H_{P,t}^U)^\top \Phi_t^U - \gamma_t^U \bar{\theta}_{t-1}^U, \end{aligned} \quad (\text{A-26})$$

where $H_{P,t}^I$ is a 4×1 vector and $H_{P,t}^U$ is a 3×1 vector,

$$\begin{aligned} H_{P,t}^I &:= [(1 - w_t^U) \delta_t^I + w_t^U \delta_t^U, -\mu_t((1 - w_t^U) \delta_t^I + w_t^U \delta_t^U), (1 - w_t^U) g_t^I + w_t^U (-g_t^U + \delta_t^U \mu_t), -(1 - w_t^U) f_t^I - w_t^U f_t^U]^\top, \\ H_{P,t}^U &:= [w_t^I \delta_t^I + (1 - w_t^I) \delta_t^U, w_t^I (g_t^I - \delta_t^I \mu_t) - (1 - w_t^I) g_t^U, -w_t^I f_t^I - (1 - w_t^I) f_t^U]^\top. \end{aligned} \quad (\text{A-27})$$

Traders' net trades at t are $\theta_t^{I,i} - \theta_{t-1}^{I,i} = (P_t^{IR,i} - P_t^I) / \gamma_t^I$ and $\theta_t^{U,j} - \theta_{t-1}^{U,j} = (P_t^{UR,j} - P_t^U) / \gamma_t^U$, and they can be written as

$$\theta_t^{I,i} = (H_{\theta,t}^I)^\top \Phi_t^I + \bar{\theta}_{t-1}^I, \quad \theta_t^{U,j} = (H_{\theta,t}^U)^\top \Phi_t^U + \bar{\theta}_{t-1}^U, \quad \text{where} \quad (\text{A-28})$$

$$\begin{aligned} H_{\theta,t}^I &:= \frac{N_M}{(N_M+1)(\gamma_t^I + N_I / N_U \gamma_t^U)} \left[\delta_t^I - \delta_t^U, -\mu_t(\delta_t^I - \delta_t^U), g_t^I + g_t^U - \mu_t \delta_t^U, -(f_t^I - f_t^U) \right]^\top, \\ H_{\theta,t}^U &:= -\frac{N_M}{(N_M+1)(\gamma_t^U + N_U / N_I \gamma_t^I)} \left[\delta_t^I - \delta_t^U, g_t^I + g_t^U - \mu_t \delta_t^I, -(f_t^I - f_t^U) \right]^\top. \end{aligned} \quad (\text{A-29})$$

Next, we solve for traders' optimization problem at $t-1$. We first solve for the informed traders' optimization problem at $t-1$. Informed traders' value function at t can be rewritten as

$$J_t^{I,i} = -\rho_t^I e^{-\lambda^I} \left[w_{t-1}^{I,i} - \theta_{t-1}^{I,i} P_{t-1}^I + \theta_{t-1}^{I,i} P_{t-1}^I + \frac{1}{2} (\Phi_t^I)^\top M_t^I \Phi_t^I + (C_t^I)^\top \Phi_t^I \bar{\theta}_{t-1}^I + \frac{1}{2} m_t^I (\bar{\theta}_{t-1}^I)^2 \right]. \quad (\text{A-30})$$

From equations (A-3)-(A-5), we can write Φ_t^I as a recursive function of Φ_{t-1}^I ,

$$\Phi_t^I := (\hat{V}_t^I - \bar{V}, X_t, \hat{X}_t^U, \bar{\theta})' = H_t^I \Phi_{t-1}^I + F_t^I u_t^I, \quad (\text{A-31})$$

where $\Phi_{t-1}^I := (\hat{V}_{t-1}^I - \bar{V}, X_{t-1}, \hat{X}_{t-1}^U, \bar{\theta})'$, $u_t^I := (e_t^I, \eta_t)'$, and

$$H_t^I := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & K_{X,t}^U \mu_{t-1} & 1 - K_{X,t}^U \mu_{t-1} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad F_t^I := \begin{pmatrix} K_{V,t}^I & 0 \\ 0 & 1 \\ K_{X,t}^U & -K_{X,t}^U h_t \\ 0 & 0 \end{pmatrix}. \quad (\text{A-32})$$

Informed traders' trading price P_t^I in equation (A-26) can thus be rewritten as

$$P_t^I = \bar{V} + (H_{P,t}^I)^\top \Phi_t^I - \gamma_t^I \bar{\theta}_{t-1}^I = \bar{V} + (H_{P,t}^I)^\top H_t^I \Phi_{t-1}^I - \gamma_t^I \bar{\theta}_{t-1}^I + (H_{P,t}^I)^\top F_t^I u_t^I. \quad (\text{A-33})$$

Substituting equations (A-31) and (A-33) into informed traders' value function (A-30) and taking expectation of the value function $J_t^{I,i}$ at period $t-1$, we obtain $E_{t-1}^I[J_t^{I,i}]$ as follows,

$$\begin{aligned} & -\rho_t^I e^{-\lambda^I} \left[W_{t-1}^{I,i} - \theta_{t-1}^{I,i} P_{t-1}^I + \theta_{t-1}^{I,i} (\bar{V} + (H_{P,t}^I)^\top H_t^I \Phi_{t-1}^I - \gamma_t^I \bar{\theta}_{t-1}^I) + (C_t^I)^\top H_t^I \Phi_{t-1}^I \bar{\theta}_{t-1}^I + \frac{1}{2} (H_t^I \Phi_{t-1}^I)^\top M_t^I H_t^I \Phi_{t-1}^I + \frac{1}{2} m_t^I (\bar{\theta}_{t-1}^I)^2 \right] \\ & \times \sqrt{\frac{|\Xi_t^I|}{|\Sigma_t^I|}} e^{\frac{1}{2} (\lambda^I)^2} \left[\theta_{t-1}^{I,i} (H_{P,t}^I)^\top F_t^I + \bar{\theta}_{t-1}^I (C_t^I)^\top F_t^I + (H_t^I \Phi_{t-1}^I)^\top M_t^I F_t^I \right] \Xi_t^I \left[\theta_{t-1}^{I,i} (H_{P,t}^I)^\top F_t^I + \bar{\theta}_{t-1}^I (C_t^I)^\top F_t^I + (H_t^I \Phi_{t-1}^I)^\top M_t^I F_t^I \right]^\top, \end{aligned} \quad (\text{A-34})$$

where

$$\Sigma_t^I = \begin{pmatrix} \sigma_t^I & 0 \\ 0 & \tau_{\eta,t}^{-1} \end{pmatrix}, \quad \Xi_t^I = [(\Sigma_t^I)^{-1} + \lambda^I (F_t^I)^\top M_t^I F_t^I]^{-1}. \quad (\text{A-35})$$

Taking the first order condition with respect to $\theta_{t-1}^{I,i}$ yields

$$-P_{t-1}^I + \bar{V} + (H_{P,t}^I)^\top H_t^I \Phi_{t-1}^I - \gamma_t^I \bar{\theta}_{t-1}^I = \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top [H_{P,t}^I \theta_{t-1}^{I,i} + C_t^I \bar{\theta}_{t-1}^I + M_t^I H_t^I \Phi_{t-1}^I], \quad (\text{A-36})$$

and the second order condition

$$\lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I > 0, \quad (\text{A-37})$$

needs to be satisfied, where $H_{P,t}^I$, F_t^I , and Ξ_t^I are as defined in equations (A-27), (A-32), and (A-35). Similarly, we can rewrite uninformed traders' value function at t as

$$J_t^{U,j} = -\rho_t^U e^{-\lambda^U} \left[W_{t-1}^{U,j} - \theta_{t-1}^{U,j} P_{t-1}^U + \theta_{t-1}^{U,j} P_t^U + \frac{1}{2} (\Phi_t^U)^\top M_t^U \Phi_t^U + (C_t^U)^\top \Phi_t^U \bar{\theta}_{t-1}^U + \frac{1}{2} m_t^U (\bar{\theta}_{t-1}^U)^2 \right]. \quad (\text{A-38})$$

From equation (A-4) in the proof of Lemma 1, we can write Φ_t^U as a recursive function of Φ_{t-1}^U ,

$$\Phi_t^U := (\hat{V}_t^U - \bar{V}, \hat{X}_t^U, \bar{\theta})' = \Phi_{t-1}^U + F_t^U e_t^U, \quad \text{where } F_t^U = (K_{V,t}^U, K_{X,t}^U, 0)'. \quad (\text{A-39})$$

Uninformed traders' trading price P_t^U in equation (A-26) can thus be expressed as

$$P_t^U = \bar{V} + (H_{P,t}^U)^\top \Phi_t^U - \gamma_t^U \bar{\theta}_{t-1}^U = \bar{V} + (H_{P,t}^U)^\top \Phi_{t-1}^U - \gamma_t^U \bar{\theta}_{t-1}^U + (H_{P,t}^U)^\top F_t^U e_t^U. \quad (\text{A-40})$$

Substituting equations (A-39) and (A-40) into uninformed traders' value function (A-38) and taking expectation of the value function $J_t^{U,i}$ at period $t-1$, we obtain $E_{t-1}^U[J_t^{U,j}]$ as follows,

$$\begin{aligned} & -\rho_t^U e^{-\lambda^U} \left[W_{t-1}^{U,j} - \theta_{t-1}^{U,j} P_{t-1}^U + \theta_{t-1}^{U,j} (\bar{V} + (H_{P,t}^U)^\top \Phi_{t-1}^U - \gamma_t^U \bar{\theta}_{t-1}^U) + (C_t^U)^\top \Phi_{t-1}^U \bar{\theta}_{t-1}^U + \frac{1}{2} (\Phi_{t-1}^U)^\top M_t^U \Phi_{t-1}^U + \frac{1}{2} m_t^U (\bar{\theta}_{t-1}^U)^2 \right] \\ & \times \sqrt{\frac{\Xi_t^U}{\Sigma_t^U}} e^{\frac{1}{2} (\lambda^U)^2 \Xi_t^U \left[(H_{P,t}^U)^\top F_t^U \theta_{t-1}^{U,j} + (C_t^U)^\top F_t^U \bar{\theta}_{t-1}^U + (\Phi_{t-1}^U)^\top M_t^U F_t^U \right]^2}, \quad \text{where} \end{aligned} \quad (\text{A-41})$$

$$\Sigma_t^U = \text{Var}_{t-1}^U(e_t^U) = o_{V,t-1}^U + \tau_{\varepsilon,t}^{-1} + h_{\eta,t}^2 \tau_{\eta,t}^{-1}, \quad \text{and } \Xi_t^U = [(\Sigma_t^U)^{-1} + \lambda^U (F_t^U)^\top M_t^U F_t^U]^{-1}. \quad (\text{A-42})$$

Taking the first order condition with respect to $\theta_{t-1}^{U,j}$ yields

$$-P_{t-1}^U + \bar{V} + (H_{P,t}^U)^\top \Phi_{t-1}^U - \gamma_t^U \bar{\theta}_{t-1}^U = \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top \left[H_{P,t}^U \theta_{t-1}^{U,j} + C_t^U \bar{\theta}_{t-1}^U + M_t^U \Phi_{t-1}^U \right], \quad (\text{A-43})$$

and the second order condition ($\lambda^U \Xi_t^U \left[(H_{P,t}^U)^\top F_t^U \right]^2 > 0$) is satisfied. From equations (A-25), (A-36), and (A-43), we show that an informed trader i 's position and an uninformed j 's position are

$$\theta_{t-1}^{I,i} - \theta_{t-2}^{I,i} = (P_{t-1}^{IR,i} - P_{t-1}^I) / \gamma_{t-1}^I, \quad \theta_{t-1}^{U,j} - \theta_{t-2}^{U,j} = (P_{t-1}^{UR,j} - P_{t-1}^U) / \gamma_{t-1}^U, \quad (\text{A-44})$$

where traders' reservation prices at $t-1$ are given as

$$\begin{aligned} P_{t-1}^{IR,i} &:= \bar{V} + \delta_{t-1}^I (\hat{V}_{t-1}^I - \mu_{t-1} X_{t-1} - \bar{V}) + g_{t-1}^I \hat{X}_{t-1}^U - f_{t-1}^I \bar{\theta} - \gamma_{t-1}^I \theta_{t-2}^{I,i}, \\ P_{t-1}^{UR,j} &:= \bar{V} + \delta_{t-1}^U (\hat{V}_{t-1}^U - \bar{V}) - g_{t-1}^U \hat{X}_{t-1}^U - f_{t-1}^U \bar{\theta} - \gamma_{t-1}^U \theta_{t-2}^{U,j}, \end{aligned} \quad (\text{A-45})$$

since $\hat{X}_{t-1}^U = \frac{1}{\mu_{t-1}} (\hat{V}_{t-1}^U - (\hat{V}_{t-1}^I - \mu_{t-1} X_{t-1}))$, traders' reservation prices can thus be rewritten as in equation (14) in Proposition 1. The coefficients δ_{t-1}^I , g_{t-1}^I , f_{t-1}^I , γ_{t-1}^I , ξ_{t-1}^I , and ψ_{t-1}^I in $P_{t-1}^{IR,i}$ (equations (A-45) and (14)) are

$$\begin{aligned}
\delta_{t-1}^I &:= [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,1}, \quad g_{t-1}^I = [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,3}, \\
f_{t-1}^I &:= -[(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,4} + [\gamma_t^I + \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top C_t^I] \frac{w_t^I}{N_I}, \\
\gamma_{t-1}^I &:= \frac{\gamma_t^I}{N_M+1} + \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top \left(H_{P,t}^I + \frac{C_t^I}{N_M+1} \right), \quad \xi_{t-1}^I := \delta_{t-1}^I - g_{t-1}^I / \mu_{t-1}, \quad \psi_{t-1}^I := g_{t-1}^I / \mu_{t-1},
\end{aligned} \tag{A-46}$$

where $H_{P,t}^I$ defined in (A-27) is a 4×1 vector; H_t^I defined in (A-32) is a 4×4 matrix; F_t^I defined in (A-32) is a 4×2 matrix; M_t^I is a 4×4 symmetric matrix, which can be solved recursively from equation (A-54) below; and Ξ_t^I defined in (A-35) is a 2×2 matrix. It follows that the dimension of vector $(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I$ is 1×4 . It is also verified that the coefficient of X_{t-1} in $P_{t-1}^{IR,i}$ (equation (A-45)),

$$-\mu_{t-1} \delta_{t-1}^I = [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,2}.$$

The coefficients δ_{t-1}^U , g_{t-1}^U , f_{t-1}^U , γ_{t-1}^U , ξ_{t-1}^U , and ψ_{t-1}^U in $P_{t-1}^{UR,i}$ (equations (A-45) and (14)) are

$$\begin{aligned}
\delta_{t-1}^U &= [(H_{P,t}^U)^\top - \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top M_t^U]_{1,1}, \quad g_{t-1}^U = -[(H_{P,t}^U)^\top - \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top M_t^U]_{1,2}, \\
f_{t-1}^U &= -[(H_{P,t}^U)^\top - \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top M_t^U]_{1,3} + [\gamma_t^U + \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top C_t^U] \frac{w_t^U}{N_U}, \\
\gamma_{t-1}^U &= \frac{\gamma_t^U}{N_M+1} + \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top \left(H_{P,t}^U + \frac{C_t^U}{N_M+1} \right), \quad \xi_{t-1}^U := g_{t-1}^U / \mu_{t-1}, \quad \psi_{t-1}^U := \delta_{t-1}^U - g_{t-1}^U / \mu_{t-1},
\end{aligned} \tag{A-47}$$

where $H_{P,t}^U$ defined in (A-27) is a 3×1 vector; F_t^U defined in (A-39) is a 3×1 vector; M_t^U is a 3×3 symmetric matrix, which can be solved recursively from equation (A-55) below; and Ξ_t^U defined in (A-42) is a scalar. The dimension of vector $(H_{P,t}^U)^\top - \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top M_t^U$ is 1×3 .

At $t-1$, market maker m maximizes his profit at trading round $t-1$,

$$\max_{q_{t-1}^{I,m} = -q_{t-1}^{U,m}} q_{t-1}^{I,m} P_{t-1}^I + q_{t-1}^{U,m} P_{t-1}^U. \tag{A-48}$$

Using equations (A-44) and (A-14), we have

$$P_{t-1}^I = P_{t-1}^{IR} - \frac{\gamma_{t-1}^I}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m}, \quad P_{t-1}^U = P_{t-1}^{UR} - \frac{\gamma_{t-1}^U}{N_U} \sum_{m=1}^{N_M} q_{t-1}^{U,m}. \tag{A-49}$$

Substituting (A-49) into (A-48) and taking first order condition yield,

$$q_{t-1}^{I,m} = -q_{t-1}^{U,m} = \frac{P_{t-1}^{IR} - P_{t-1}^{UR}}{(N_M+1)(\gamma_{t-1}^I / N_I + \gamma_{t-1}^U / N_U)}. \tag{A-50}$$

Substituting equation (A-50) into (A-49) yields traders' trading prices,

$$P_{t-1}^I = P_{t-1}^{IR} + w_{t-1}^U (P_{t-1}^{UR} - P_{t-1}^{IR}), \quad P_{t-1}^U = P_{t-1}^{UR} + w_{t-1}^I (P_{t-1}^{IR} - P_{t-1}^{UR}), \quad (\text{A-51})$$

where w_{t-1}^I and w_{t-1}^U are as defined in equation (A-24) where t is replaced by $t-1$. We then define $H_{P,t-1}^I$, $H_{P,t-1}^U$, $H_{\theta,t-1}^I$, and $H_{\theta,t-1}^U$ similar to equations (A-27) and (A-29) where t is replaced by $t-1$. Substituting (A-45) and (A-51) into (A-44), traders' optimal positions at $t-1$ can be written as

$$\theta_{t-1}^{I,i} = (H_{\theta,t-1}^I)^\top \Phi_{t-1}^I + \bar{\theta}_{t-2}^I, \quad \theta_{t-1}^{U,j} = (H_{\theta,t-1}^U)^\top \Phi_{t-1}^U + \bar{\theta}_{t-2}^U. \quad (\text{A-52})$$

Using the definition of $\bar{\theta}_{t-1}^I$ and $\bar{\theta}_{t-1}^U$ in equation (A-25), it can be shown that

$$\begin{aligned} \bar{\theta}_{t-1}^I &= (\hat{H}_{\theta,t-1}^I)^\top \Phi_{t-1}^I + \frac{\bar{\theta}_{t-2}^I}{N_M+1}, \quad \bar{\theta}_{t-1}^U = (\hat{H}_{\theta,t-1}^U)^\top \Phi_{t-1}^U + \frac{\bar{\theta}_{t-2}^U}{N_M+1}, \quad \text{where} \\ \hat{H}_{\theta,t-1}^I &:= \frac{H_{\theta,t-1}^I}{N_M+1} + [0, 0, 0, \frac{w_t^I}{N_I}]^\top, \quad \hat{H}_{\theta,t-1}^U := \frac{H_{\theta,t-1}^U}{N_M+1} + [0, 0, \frac{w_t^U}{N_U}]^\top. \end{aligned} \quad (\text{A-53})$$

Finally, using equations (A-36), (A-43), (A-52), and (A-53), we verify that traders' value functions at period $t-1$ (equations (A-34) and (A-41)) have the conjectured forms as follows

$$\begin{aligned} J_{t-1}^{I,i} &= -\rho_{t-1}^I e^{-\lambda^I \left[W_{t-1}^{I,i} + \frac{1}{2} (\Phi_{t-1}^I)^\top M_{t-1}^I \Phi_{t-1}^I + (C_{t-1}^I)^\top \Phi_{t-1}^I \bar{\theta}_{t-2}^I + \frac{1}{2} m_{t-1}^I (\bar{\theta}_{t-2}^I)^2 \right]}, \\ J_{t-1}^{U,j} &= -\rho_{t-1}^U e^{-\lambda^U \left[W_{t-1}^{U,j} + \frac{1}{2} (\Phi_{t-1}^U)^\top M_{t-1}^U \Phi_{t-1}^U + (C_{t-1}^U)^\top \Phi_{t-1}^U \bar{\theta}_{t-2}^U + \frac{1}{2} m_{t-1}^U (\bar{\theta}_{t-2}^U)^2 \right]}, \end{aligned}$$

the coefficients M_{t-1}^I , C_{t-1}^I , m_{t-1}^I , and ρ_{t-1}^I , are

$$\begin{aligned} M_{t-1}^I &= (H_t^I)^\top M_t^I H_t^I + m_t^I \hat{H}_{\theta,t-1}^I (\hat{H}_{\theta,t-1}^I)^\top + \lambda^I H_{\theta,t-1}^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I (H_{\theta,t-1}^I)^\top + \hat{H}_{\theta,t-1}^I (C_t^I)^\top H_t^I \\ &\quad + (H_t^I)^\top C_t^I (\hat{H}_{\theta,t-1}^I)^\top - \lambda^I [C_t^I (\hat{H}_{\theta,t-1}^I)^\top + M_t^I H_t^I]^\top F_t^I \Xi_t^I (F_t^I)^\top [C_t^I (\hat{H}_{\theta,t-1}^I)^\top + M_t^I H_t^I], \\ C_{t-1}^I &= \frac{1}{N_M+1} ((H_t^I)^\top C_t^I + m_t^I \hat{H}_{\theta,t-1}^I) + \lambda^I H_{\theta,t-1}^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I \\ &\quad - \frac{\lambda^I}{N_M+1} [C_t^I (\hat{H}_{\theta,t-1}^I)^\top + M_t^I H_t^I]^\top F_t^I \Xi_t^I (F_t^I)^\top C_t^I, \\ m_{t-1}^I &= \frac{m_t^I}{(N_M+1)^2} + \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I - \frac{\lambda^I}{(N_M+1)^2} (C_t^I)^\top F_t^I \Xi_t^I (F_t^I)^\top C_t^I, \quad \rho_{t-1}^I = \rho_t^I \sqrt{|\Xi_t^I|/|\Sigma_t^I|}, \end{aligned} \quad (\text{A-54})$$

and the coefficients M_{t-1}^U , C_{t-1}^U , m_{t-1}^U , and ρ_{t-1}^U are

$$\begin{aligned}
M_{t-1}^U &= M_t^U + m_t^U \hat{H}_{\theta,t-1}^U (\hat{H}_{\theta,t-1}^U)^\top + \lambda^U \Xi_t^U H_{\theta,t-1}^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top H_{P,t}^U (H_{\theta,t-1}^U)^\top + \hat{H}_{\theta,t-1}^U (C_t^U)^\top \\
&\quad + C_t^U (\hat{H}_{\theta,t-1}^U)^\top - \lambda^U \Xi_t^U [(C_t^U)^\top F_t^U (\hat{H}_{\theta,t-1}^U)^\top + (F_t^U)^\top M_t^U]^\top [(C_t^U)^\top F_t^U (\hat{H}_{\theta,t-1}^U)^\top + (F_t^U)^\top M_t^U], \\
C_{t-1}^U &= \frac{C_t^U}{N_M+1} + \frac{m_t^U}{N_M+1} \hat{H}_{\theta,t-1}^U + \lambda^U \Xi_t^U H_{\theta,t-1}^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top H_{P,t}^U \\
&\quad - \frac{\lambda^U \Xi_t^U}{N_M+1} [(C_t^U)^\top F_t^U (\hat{H}_{\theta,t-1}^U)^\top + (F_t^U)^\top M_t^U]^\top (F_t^U)^\top C_t^U, \\
m_{t-1}^U &= \frac{m_t^U}{(N_M+1)^2} + \lambda^U \Xi_t^U (H_{P,t}^U)^\top F_t^U (F_t^U)^\top H_{P,t}^U - \frac{\lambda^U}{(N_M+1)^2} \Xi_t^U (C_t^U)^\top F_t^U (F_t^U)^\top C_t^U, \quad \rho_{t-1}^U = \rho_t^U \sqrt{\Xi_t^U / \Sigma_t^U}.
\end{aligned} \tag{A-55}$$

Therefore, by backward induction, we conclude that traders' reservation prices are linear functions of key state variables as conjectured in equation (A-22). In the symmetric equilibrium, $\theta_{t-1}^{I,i}$ is the same for all I traders and $\theta_{t-1}^{U,j}$ is the same for all U traders, equation (A-22) can be rewritten as (14) in Proposition 1. All coefficients can be solved recursively. The recursive formulas for coefficients ξ_t^I , ξ_t^U , ψ_t^I , ψ_t^U , f_t^I , f_t^U , γ_t^I , and γ_t^U are presented in equations (A-46)–(A-47). Their values at $t = T$ are $\xi_T^I = 1$, $\xi_T^U = 0$, $\psi_T^I = 0$, $\psi_T^U = 1$, and for $i \in \{I, U\}$, $f_T^i = 0$, $\gamma_T^i = \lambda^i o_{V,T}^i$. The coefficients in traders' value functions ρ_t^I , M_t^I , C_t^I , m_t^I , ρ_t^U , M_t^U , C_t^U , and m_t^U can be computed recursively by equations (A-54) and (A-55). Their values at $t = T$ are given by equation (A-21).

A.3. Proof of Proposition 2

We assume that market makers do not carry inventory over time. For $j = 1, 2, \dots, N_M$, the designated market maker M_j 's problem is equivalent to solving

$$\max_{\alpha_t^j = \beta_t^j} \mathbb{E} \left[\alpha_t^j A_t - \beta_t^j B_t \mid \mathcal{F}_t^M \right]. \tag{A-56}$$

This implies that market makers who trade in both the “ask” market and the “bid” market optimally make offsetting trades with other traders to maximize the expected profit from each round of trading conditional on market makers' information set at period t , $\mathcal{F}_t^M$. We prove formally in Section B.8 of the Internet Appendix that market makers optimally do not carry inventory over time when they are extremely risk averse.

We prove the case when $\Delta_t > 0$. The other case when $\Delta_t < 0$ can be proved similarly. If $\Delta_t > 0$, I investors buy at the ask price and U investors sell at the bid price, implying $P_t^{IR} > A_t > B_t > P_t^{UR}$.

Given A_t and B_t , the optimal demand of trader I and U is

$$\theta_t^I - \theta_{t-1}^I = \frac{P_t^{IR} - A_t}{\gamma_t^I}, \quad \theta_t^U - \theta_{t-1}^U = \frac{P_t^{UR} - B_t}{\gamma_t^U}, \tag{A-57}$$

where the reservation price of trader i , P_t^{iR} presented in Proposition 1 is a linear function of key state variables $\hat{V}_t^I - \mu_t X_t$, $\hat{V}_t^U$, and $\hat{X}_t^U$. The parameter γ_t^i calculated in the proof of Proposition 1 depends on trader i 's risk aversion and conditional variance and covariance of key state variables.

Substituting (A-57) into the market clearing condition (6), we get

$$A_t = P_t^{IR} - \frac{\gamma_t^I}{N_I} \sum_{j=1}^{N_M} \alpha_t^j, \quad B_t = P_t^{UR} + \frac{\gamma_t^U}{N_U} \sum_{j=1}^{N_M} \beta_t^j. \quad (\text{A-58})$$

Substituting (A-58) into the objective function (A-56), and taking the first order conditions with respect to α_t^j and β_t^j yield

$$\alpha_t^j = \beta_t^j = \frac{1}{N_M + 1} \frac{\Delta_t}{\gamma_t^I / N_I + \gamma_t^U / N_U}. \quad (\text{A-59})$$

Substituting Equation (A-59) into (A-58), we have

$$A_t = P_t^{IR} - \frac{N_M}{N_M + 1} \frac{\gamma_t^I / N_I}{\gamma_t^I / N_I + \gamma_t^U / N_U} \Delta_t, \quad B_t = P_t^{UR} + \frac{N_M}{N_M + 1} \frac{\gamma_t^U / N_U}{\gamma_t^I / N_I + \gamma_t^U / N_U} \Delta_t.$$

A.4. Proof of Propositions 3 and 4

We first solve the problem at T . We denote P_T^I as informed traders' trading price, informed trader i 's optimal demand at T is

$$\theta_T^{I,i} - \theta_{T-1}^{I,i} = \frac{P_T^{IR,i} - P_T^I}{\gamma_T^I}, \text{ where } P_T^{IR,i} = \hat{V}_T^I - \mu_T X_T - \gamma_T^I \theta_{T-1}^{I,i}, \quad \gamma_T^I = \mu_T = \lambda^I o_{V,T}^I. \quad (\text{A-60})$$

Uninformed traders are risk neutral ($\lambda^U = 0$), they maximize their expected terminal wealth:

$$\max_{\theta_T^{U,j}} E_T^U[W_{T+1}^{U,j}] = W_T^{U,j} + \theta_T^{U,j} (\hat{V}_T^U - P_T^U).$$

Therefore, if $P_T^U > \hat{V}_T^U$, uninformed traders have infinitely negative demand, and if $P_T^U < \hat{V}_T^U$, uninformed traders have infinitely positive demand. For $t \leq T$, denote $q_t^{I,m}$ and $q_t^{U,m}$ as the trading quantities of market maker m with informed traders and uninformed traders, respectively. If $q_t^{i,m}$ is positive (negative), then trader i buys from (sells to) market maker, we have

$$\sum_{m=1}^{N_M} q_t^{I,m} = \sum_{i=1}^{N_I} (\theta_t^{I,i} - \theta_{t-1}^{I,i}), \quad \sum_{m=1}^{N_M} q_t^{U,m} = \sum_{j=1}^{N_U} (\theta_t^{U,j} - \theta_{t-1}^{U,j}). \quad (\text{A-61})$$

At T , a risk-neutral market maker m maximizes his expected terminal wealth, $E_T^M[W_{T+1}^{M,m}]$, i.e.,

$$\begin{aligned} \max_{q_T^{I,m}, q_T^{U,m}} W_T^{M,m} + q_T^{I,m} P_T^I + q_T^{U,m} P_T^U + (\theta_{T-1}^{M,m} - q_T^{I,m} - q_T^{U,m}) \hat{V}_T^U \\ = W_T^{M,m} + \theta_{T-1}^{M,m} \hat{V}_T^U + q_T^{I,m} (P_T^I - \hat{V}_T^U) + q_T^{U,m} (P_T^U - \hat{V}_T^U). \end{aligned} \quad (\text{A-62})$$

If $P_T^U > \hat{V}_T^U$, then uninformed traders sell ($q_T^{U,m} < 0$), and if $P_T^U < \hat{V}_T^U$, then uninformed traders buy ($q_T^{U,m} > 0$).

Therefore, in equilibrium, $P_T^U = \hat{V}_T^U$ and $q_T^{U,m} = 0$. Using equations (A-60) and (A-61), we have

$$P_T^I = P_T^{IR} - \frac{\gamma_T^I}{N_I} \sum_{m=1}^{N_M} q_T^{I,m}, \text{ where } P_T^{IR} = \hat{V}_T^I - \mu_T X_T - \gamma_T^I \theta_{T-1}^I, \theta_{T-1}^I := \frac{1}{N_I} \sum_{i=1}^{N_I} \theta_{T-1}^{I,i}. \quad (\text{A-63})$$

Substituting equation (A-63) into (A-62) and taking first order condition yield,

$$q_T^{I,m} = \frac{P_T^{IR} - P_T^{MR}}{(N_M+1)\gamma_T^I/N_I}, \text{ where } P_T^{MR} = \hat{V}_T^U.$$

Therefore, P_T^I can be written as $P_T^I = (1 - \omega_T^M) P_T^{IR} + \omega_T^M P_T^{MR}$, where $\omega_T^M = N_M/(N_M+1)$. It can be shown that the value function of informed traders and market makers at T can be written as

$$\begin{aligned} J_T^{I,i} &= E_T^I[-e^{-\lambda^I W_{T+1}^{I,i}}] = -\rho_T^I e^{-\lambda^I [W_T^{I,i} + \frac{1}{2}(\Phi_T^I)^\top M_T^I \Phi_T^I + (C_T^I)^\top \Phi_T^I \theta_{T-1}^I + \frac{1}{2} m_T^I (\theta_{T-1}^I)^2]}, \\ J_T^{M,m} &= E_T^M[W_{T+1}^{M,m}] = W_T^{M,m} + \theta_{T-1}^{M,m} \hat{V}_T^U + (C_T^M)^\top \Phi_T^M \theta_{T-1}^I + \frac{1}{2} m_T^M (\theta_{T-1}^I)^2 + \frac{1}{2} (\Phi_T^M)^\top M_T^M \Phi_T^M + \rho_T^M, \end{aligned} \quad (\text{A-64})$$

where $\Phi_T^I := [\hat{V}_T^I - \bar{V}, X_T, \hat{X}_T^U]^\top$, $\Phi_T^M := [\hat{V}_T^U - \bar{V}, \hat{X}_T^U]^\top$, and the coefficients are

$$\begin{aligned} \rho_T^I &= 1, \quad m_T^I = \frac{\gamma_T^I}{(N_M+1)^2}, \quad \rho_T^M = 0, \quad m_T^M = \frac{2N_I\gamma_T^I}{(N_M+1)^2}, \quad C_T^I = \frac{N_M\gamma_T^I}{(N_M+1)^2} (0, 0, -1)', \\ C_T^M &= \left[0, \frac{2N_I\gamma_T^I}{(N_M+1)^2}\right]^\top, \quad M_T^I = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -\gamma_T^I & 0 \\ 0 & 0 & \frac{N_M^2\gamma_T^I}{(N_M+1)^2} \end{pmatrix}, \quad M_T^M = \begin{pmatrix} 0 & 0 \\ 0 & \frac{2N_I\gamma_T^I}{(N_M+1)^2} \end{pmatrix}. \end{aligned} \quad (\text{A-65})$$

We conjecture that traders' value function at t is given as (A-64) where the subscript T is replaced by t . We also conjecture that traders' average reservation prices at t are

$$P_t^{IR} = \bar{V} + \delta_t^I (\hat{V}_t^I - \mu_t X_t - \bar{V}) + g_t^I \hat{X}_t^U - \gamma_t^I \theta_{t-1}^I, \quad P_t^{MR} = \bar{V} + \delta_t^M (\hat{V}_t^U - \bar{V}) - g_t^M \hat{X}_t^U - \gamma_t^M \theta_{t-1}^I.$$

We then solve the problem backward from t to $t-1$ and verify our conjecture.

Denote $\Phi_t^I := [\hat{V}_t^I - \bar{V}, X_t, \hat{X}_t^U]^\top$ and $\Phi_t^M := [\hat{V}_t^U - \bar{V}, \hat{X}_t^U]^\top$. Using Lemma 1, the vectors can be expressed recur-

sively as

$$\Phi_t^I = H_t^I \Phi_{t-1}^I + F_t^I u_t^I, \quad \Phi_t^M = \Phi_{t-1}^M + F_t^M e_t^U, \quad (\text{A-66})$$

where $u_t^I = [e_t^I, \eta_t]^T$, $F_t^M = [K_{V,t}^U, K_{X,t}^U]^T$, and

$$H_t^I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & K_{X,t}^U \mu_{t-1} & 1 - K_{X,t}^U \mu_{t-1} \end{pmatrix}, \quad F_t^I = \begin{pmatrix} K_{V,t}^I & 0 \\ 0 & 1 \\ K_{X,t}^U & -K_{X,t}^U h_t \end{pmatrix}. \quad (\text{A-67})$$

We conjecture that informed traders' trading price at t can be written as

$$P_t^I = (1 - \omega_t^M) P_t^{IR} + \omega_t^M P_t^{MR} = \bar{V} + (H_{P,t}^I)^T \Phi_t^I - f_t^I \theta_{t-1}^I, \quad \text{where}$$

$$\begin{aligned} H_{P,t}^I &:= [(1 - \omega_t^M) \delta_t^I + \omega_t^M \delta_t^M, -\mu_t ((1 - \omega_t^M) \delta_t^I + \omega_t^M \delta_t^M), (1 - \omega_t^M) g_t^I + \omega_t^M (\mu_t \delta_t^M - g_t^M)]^T, \\ f_t^I &:= (1 - \omega_t^M) \gamma_t^I + \omega_t^M \gamma_t^M. \end{aligned} \quad (\text{A-68})$$

Note that $[H_{P,t}^I]_{1,2} = -\mu_t [H_{P,t}^I]_{1,1}$. From traders' reservation prices at T , we obtain

$$\delta_T^I = \delta_T^M = 1, \quad g_T^I = g_T^M = 0, \quad \gamma_T^I = \lambda^I o_{V,T}^I, \quad \mu_T = \lambda^I o_{V,T}^I, \quad \gamma_T^M = 0, \quad \omega_T^M = N_M / (N_M + 1). \quad (\text{A-69})$$

Then informed traders' value function at t can be written as

$$\begin{aligned} J_t^{I,i} &= -\rho_t^I e^{-\lambda^I [W_{t-1}^{I,i} - \theta_{t-1}^{I,i} P_{t-1}^I + \theta_{t-1}^{I,i} (\bar{V} + (H_{P,t}^I)^T H_t^I \Phi_{t-1}^I - f_t^I \theta_{t-1}^I) + (C_t^I)^T H_t^I \Phi_{t-1}^I \theta_{t-1}^I + \frac{1}{2} (H_t^I \Phi_{t-1}^I)^T M_t^I H_t^I \Phi_{t-1}^I]} \\ &\quad \times e^{-\lambda^I [\frac{1}{2} m_t^I (\theta_{t-1}^I)^2 + \frac{1}{2} (F_t^I u_t^I)^T M_t^I F_t^I u_t^I + (\theta_{t-1}^{I,i} (H_{P,t}^I)^T F_t^I + \theta_{t-1}^I (C_t^I)^T F_t^I + (H_t^I \Phi_{t-1}^I)^T M_t^I F_t^I) u_t^I]}. \end{aligned}$$

The value function at $t-1$, $J_{t-1}^{I,i} := E_{t-1}^I [J_t^{I,i}]$ is

$$\begin{aligned} J_{t-1}^{I,i} &= -\rho_t^I e^{-\lambda^I [W_{t-1}^{I,i} - \theta_{t-1}^{I,i} P_{t-1}^I + \theta_{t-1}^{I,i} (\bar{V} + (H_{P,t}^I)^T H_t^I \Phi_{t-1}^I - f_t^I \theta_{t-1}^I) + (C_t^I)^T H_t^I \Phi_{t-1}^I \theta_{t-1}^I + \frac{1}{2} (H_t^I \Phi_{t-1}^I)^T M_t^I H_t^I \Phi_{t-1}^I + \frac{1}{2} m_t^I (\theta_{t-1}^I)^2]} \\ &\quad \times \sqrt{\frac{|\Xi_t^I|}{|\Sigma_t^I|}} e^{\frac{1}{2} (\lambda^I)^2 [\theta_{t-1}^{I,i} (H_{P,t}^I)^T + \theta_{t-1}^I (C_t^I)^T + (H_t^I \Phi_{t-1}^I)^T M_t^I] F_t^I \Xi_t^I (F_t^I)^T [\theta_{t-1}^{I,i} (H_{P,t}^I)^T + \theta_{t-1}^I (C_t^I)^T + (H_t^I \Phi_{t-1}^I)^T M_t^I]^T}, \end{aligned} \quad (\text{A-70})$$

where

$$\Sigma_t^I = \begin{pmatrix} \sigma_t^I & 0 \\ 0 & \tau_{\eta,t}^{-1} \end{pmatrix}, \quad \Xi_t^I = [(\Sigma_t^I)^{-1} + \lambda^I (F_t^I)^T M_t^I F_t^I]^{-1}.$$

Taking the first order condition with respect to $\theta_{t-1}^{I,i}$ yields

$$-P_{t-1}^I + \bar{V} + (H_{P,t}^I)^\top H_t^I \Phi_{t-1}^I - f_t^I \theta_{t-1}^I = \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top (H_{P,t}^I \theta_{t-1}^{I,i} + C_t^I \theta_{t-1}^I + M_t^I H_t^I \Phi_{t-1}^I),$$

and the second order condition requires

$$\lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I > 0. \quad (\text{A-71})$$

In the symmetric equilibrium, $\theta_{t-1}^{I,i} = \theta_{t-1}^I$. After reorganizing, we can express informed trader i 's demand at $t-1$ as

$$\begin{aligned} \theta_{t-1}^{I,i} - \theta_{t-2}^{I,i} &= (P_{t-1}^{IR,i} - P_{t-1}^I) / \gamma_{t-1}^I, \quad \text{where} \\ P_{t-1}^{IR,i} &:= \bar{V} + [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I] \Phi_{t-1}^I - \gamma_{t-1}^I \theta_{t-2}^{I,i}, \end{aligned} \quad (\text{A-72})$$

the informed traders' reservation price at $t-1$ can be written as

$$P_{t-1}^{IR} = \bar{V} + \delta_{t-1}^I (\hat{V}_{t-1}^I - \mu_{t-1} X_{t-1} - \bar{V}) + g_{t-1}^I \hat{X}_{t-1}^U - \gamma_{t-1}^I \theta_{t-2}^I,$$

since $\hat{X}_{t-1}^U = \frac{1}{\mu_{t-1}} (\hat{V}_{t-1}^U - (\hat{V}_{t-1}^I - \mu_{t-1} X_{t-1}))$, traders' reservation prices can thus be rewritten as in equation (26) in Proposition 3. The coefficients $\delta_{t-1}^I, g_{t-1}^I, \gamma_{t-1}^I, \xi_{t-1}^I$, and ψ_{t-1}^I are given as

$$\begin{aligned} \delta_{t-1}^I &:= [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,1}, \quad g_{t-1}^I := [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,3}, \\ \gamma_{t-1}^I &:= f_t^I + \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top (H_{P,t}^I + C_t^I), \quad \xi_{t-1}^I := \delta_{t-1}^I - g_{t-1}^I / \mu_{t-1}, \quad \psi_{t-1}^I := g_{t-1}^I / \mu_{t-1}, \end{aligned} \quad (\text{A-73})$$

and we analytically verify that

$$[(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,2} = -\mu_{t-1} [(H_{P,t}^I)^\top H_t^I - \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top M_t^I H_t^I]_{1,1}.$$

Uninformed traders' value function at $t-1$ is

$$\max_{\theta_{t-1}^{U,j}} E_{t-1}^U [W_t^{U,j}] = W_{t-1}^{U,j} + \theta_{t-1}^{U,j} (\hat{V}_{t-1}^U - P_{t-1}^U).$$

Substituting T by t in equation (A-64), market makers' value function at t can be written as,

$$J_t^{M,m} = W_{t-1}^{M,m} + q_{t-1}^{I,m}(P_{t-1}^I - \hat{V}_{t-1}^U) + q_{t-1}^{U,m}(P_{t-1}^U - \hat{V}_{t-1}^U) + \theta_{t-2}^{M,m} \hat{V}_{t-1}^U + (C_t^M)^\top \Phi_t^M \left(\frac{1}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m} + \theta_{t-2}^I \right) \\ + \frac{1}{2} m_t^M \left(\frac{1}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m} + \theta_{t-2}^I \right)^2 + \frac{1}{2} (\Phi_t^M)^\top M_t^M \Phi_t^M + \rho_t^M.$$

If $P_{t-1}^U > \hat{V}_{t-1}^U$, then uninformed traders sell infinite amount ($q_{t-1}^{U,m} < 0$), and if $P_{t-1}^U < \hat{V}_{t-1}^U$, then uninformed traders buy infinite amount ($q_{t-1}^{U,m} > 0$). Therefore, in equilibrium, $P_{t-1}^U = \hat{V}_{t-1}^U$, $q_{t-1}^{U,m} = 0$. Using equation (A-72) and the market-clearing condition, informed traders' trading price at time $t-1$ can be written as

$$P_{t-1}^I = P_{t-1}^{IR} - \frac{\gamma_{t-1}^I}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m}. \quad (\text{A-74})$$

Substituting equation (A-74) and $q_{t-1}^{U,m} = 0$ into market makers' value function and we get

$$J_{t-1}^{M,m} := E_{t-1}^M[J_t^{M,m}] = W_{t-1}^{M,m} + q_{t-1}^{I,m} \left(P_{t-1}^{IR} - \frac{\gamma_{t-1}^I}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m} \right) + (C_t^M)^\top \Phi_{t-1}^M \left(\frac{1}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m} + \theta_{t-2}^I \right) + \rho_t^M \\ + \left(\theta_{t-2}^{M,m} - q_{t-1}^{I,m} \right) \hat{V}_{t-1}^U + \frac{1}{2} m_t^M \left(\frac{1}{N_I} \sum_{m=1}^{N_M} q_{t-1}^{I,m} + \theta_{t-2}^I \right)^2 + \frac{1}{2} (\Phi_{t-1}^M)^\top M_t^M \Phi_{t-1}^M + \frac{1}{2} (F_t^M)^\top M_t^M F_t^M \Sigma_t^U,$$

where Σ_t^U is as given in equation (A-5). Taking the first order condition with respect to $q_{t-1}^{I,m}$ yields

$$\left(\frac{\gamma_{t-1}^I}{N_I} (N_M + 1) - \frac{m_t^M}{N_I^2} N_M \right) q_{t-1}^{I,m} = P_{t-1}^{IR} - \hat{V}_{t-1}^U + \frac{1}{N_I} (C_t^M)^\top \Phi_{t-1}^M + \frac{m_t^M}{N_I} \theta_{t-2}^I, \quad (\text{A-75})$$

and the second order condition

$$\frac{\gamma_{t-1}^I}{N_I} (N_M + 1) - \frac{m_t^M}{N_I^2} N_M > 0, \quad (\text{A-76})$$

needs to be satisfied. Using equation (A-75), we have

$$q_{t-1}^{I,m} = (P_{t-1}^{IR} - P_{t-1}^{MR}) / \left(\frac{\gamma_{t-1}^I}{N_I} (N_M + 1) - \frac{m_t^M}{N_I^2} N_M \right), \quad (\text{A-77})$$

and market makers' reservation price at time $t-1$ is defined as

$$P_{t-1}^{MR} := \bar{V} + \delta_{t-1}^M (\hat{V}_{t-1}^U - \bar{V}) - g_{t-1}^M \hat{X}_{t-1}^U - \gamma_{t-1}^M \theta_{t-2}^I, \quad (\text{A-78})$$

since $\hat{X}_{t-1}^U = \frac{1}{\mu_{t-1}} (\hat{V}_{t-1}^U - (\hat{V}_{t-1}^I - \mu_{t-1} X_{t-1}))$, P_{t-1}^{MR} can thus be rewritten as in equation (26) in Proposition 3. The

coefficients $\delta_{t-1}^M, g_{t-1}^M, \gamma_{t-1}^M, \xi_{t-1}^M$, and ψ_{t-1}^M are

$$\delta_{t-1}^M := 1 - \frac{1}{N_I} [C_t^M]_{1,1}, \quad g_{t-1}^M := \frac{1}{N_I} [C_t^M]_{2,1}, \quad \gamma_{t-1}^M := \frac{m_t^M}{N_I}, \quad \xi_{t-1}^M := g_{t-1}^M / \mu_{t-1}, \quad \psi_{t-1}^M := \delta_{t-1}^M - g_{t-1}^M / \mu_{t-1}. \quad (\text{A-79})$$

Substituting equation (A-77) into (A-74) yields $P_{t-1}^I = (1 - \omega_{t-1}^M) P_{t-1}^{IR} + \omega_{t-1}^M P_{t-1}^{MR}$, where the coefficient

$$\omega_{t-1}^M := \gamma_{t-1}^I \left(\gamma_{t-1}^I (1 + 1/N_M) - m_t^M / N_I \right)^{-1} = \frac{N_M}{N_M + 1} \frac{\gamma_{t-1}^I}{\gamma_{t-1}^I - \frac{N_M}{N_M + 1} \gamma_{t-1}^M} > 0, \quad (\text{A-80})$$

$\omega_{t-1}^M > 0$ is from the second order condition (A-76). This verifies that the equilibrium trading prices of informed traders can be written as equation (27) in Proposition 4. From equations (A-77) and (A-80), we verify that the trading quantities can be written as equation (28) in Proposition 4. The trading volume is equal to the total trading quantities of informed traders.

Informed traders' trading price can thus be written as

$$P_{t-1}^I = \bar{V} + (H_{P,t-1}^I)^\top \Phi_{t-1}^I - f_{t-1}^I \theta_{t-2}^I, \quad (\text{A-81})$$

where $H_{P,t-1}^I$ and f_{t-1}^I are as defined in equation (A-68) with subscript t replaced by $t-1$, *i.e.*,

$$\begin{aligned} H_{P,t-1}^I &:= \left[(1 - \omega_{t-1}^M) \delta_{t-1}^I + \omega_{t-1}^M \delta_{t-1}^M, -\mu_{t-1} \left((1 - \omega_{t-1}^M) \delta_{t-1}^I + \omega_{t-1}^M \delta_{t-1}^M \right), (1 - \omega_{t-1}^M) g_{t-1}^I + \omega_{t-1}^M (\mu_{t-1} \delta_{t-1}^M - g_{t-1}^M) \right]^\top, \\ f_{t-1}^I &:= (1 - \omega_{t-1}^M) \gamma_{t-1}^I + \omega_{t-1}^M \gamma_{t-1}^M. \end{aligned} \quad (\text{A-82})$$

The equilibrium trading quantity of informed traders at $t-1$ is

$$q_{t-1}^{I,m} = \left(H_{q,t-1}^I \right)^\top \Phi_{t-1}^I - f_{t-1}^M \theta_{t-2}^I = \left(H_{q,t-1}^M \right)^\top \Phi_{t-1}^M - f_{t-1}^M \theta_{t-2}^I, \quad \text{where} \quad (\text{A-83})$$

$$\begin{aligned} H_{q,t-1}^I &:= \frac{N_I \omega_{t-1}^M}{N_M \gamma_{t-1}^I} \left[\delta_{t-1}^I - \delta_{t-1}^M, -\mu_{t-1} (\delta_{t-1}^I - \delta_{t-1}^M), g_{t-1}^I + g_{t-1}^M - \mu_{t-1} \delta_{t-1}^M \right]^\top, \\ H_{q,t-1}^M &:= \frac{N_I \omega_{t-1}^M}{N_M \gamma_{t-1}^I} \left[\delta_{t-1}^I - \delta_{t-1}^M, g_{t-1}^I + g_{t-1}^M - \mu_{t-1} \delta_{t-1}^I \right]^\top, \quad f_{t-1}^M := \frac{N_I \omega_{t-1}^M}{N_M \gamma_{t-1}^I} (\gamma_{t-1}^I - \gamma_{t-1}^M). \end{aligned} \quad (\text{A-84})$$

Define

$$H_{\theta,t-1}^I := \frac{N_M}{N_I} H_{q,t-1}^I, \quad f_{\theta,t-1}^I := 1 - \frac{N_M}{N_I} f_{t-1}^M, \quad (\text{A-85})$$

then informed trader i 's inventory at $t-1$ is

$$\theta_{t-1}^{I,i} = \theta_{t-1}^I = \left(H_{\theta,t-1}^I \right)^\top \Phi_{t-1}^I + f_{\theta,t-1}^I \theta_{t-2}^I. \quad (\text{A-86})$$

Using equations (A-70) and (A-71), the value function at $t-1$ can be written as

$$J_{t-1}^{I,i} = -\rho_t^I \sqrt{\frac{|\Xi_t^I|}{|\Sigma_t^I|}} e^{-\lambda^I [W_{t-1}^{I,i} + (C_t^I)^\top H_t^I \Phi_{t-1}^I \theta_{t-1}^I + \frac{1}{2} (H_t^I \Phi_{t-1}^I)^\top M_t^I H_t^I \Phi_{t-1}^I + \frac{1}{2} m_t^I (\theta_{t-1}^I)^2]} \times e^{\frac{1}{2} (\lambda^I)^2 \left[(\theta_{t-1}^I (C_t^I)^\top + (H_t^I \Phi_{t-1}^I)^\top M_t^I) F_t^I \Xi_t^I (F_t^I)^\top [\theta_{t-1}^I (C_t^I)^\top + (H_t^I \Phi_{t-1}^I)^\top M_t^I]^\top - (\theta_{t-1}^I)^2 (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I \right]}. \quad (\text{A-87})$$

Substituting (A-86) into (A-87) yields

$$J_{t-1}^{I,i} = -\rho_{t-1}^I e^{-\lambda^I \left[W_{t-1}^{I,i} + \frac{1}{2} (\Phi_{t-1}^I)^\top M_{t-1}^I \Phi_{t-1}^I + (C_{t-1}^I)^\top \Phi_{t-1}^I \theta_{t-2}^I + \frac{1}{2} m_{t-1}^I (\theta_{t-2}^I)^2 \right]},$$

where M_{t-1}^I is a 3×3 symmetric matrix, C_{t-1}^I is a 3×1 vector, m_{t-1}^I is a scalar, and they are given as

$$\begin{aligned} C_{t-1}^I &:= f_{\theta,t-1}^I \left[(H_t^I)^\top C_t^I + m_t^I H_{\theta,t-1}^I + \lambda^I H_{\theta,t-1}^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I \right] \\ &\quad - \lambda^I f_{\theta,t-1}^I \left[H_{\theta,t-1}^I (C_t^I)^\top + (H_t^I)^\top M_t^I \right] F_t^I \Xi_t^I (F_t^I)^\top C_t^I, \\ m_{t-1}^I &:= (f_{\theta,t-1}^I)^2 \left[m_t^I + \lambda^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I - \lambda^I (C_t^I)^\top F_t^I \Xi_t^I (F_t^I)^\top C_t^I \right], \quad \rho_{t-1}^I := \rho_t^I \sqrt{\frac{|\Xi_t^I|}{|\Sigma_t^I|}}, \\ M_{t-1}^I &:= (H_t^I)^\top M_t^I H_t^I + m_t^I H_{\theta,t-1}^I (H_{\theta,t-1}^I)^\top + \lambda^I H_{\theta,t-1}^I (H_{P,t}^I)^\top F_t^I \Xi_t^I (F_t^I)^\top H_{P,t}^I (H_{\theta,t-1}^I)^\top + H_{\theta,t-1}^I (C_t^I)^\top H_t^I \\ &\quad + (H_t^I)^\top C_t^I (H_{\theta,t-1}^I)^\top - \lambda^I \left[C_t^I (H_{\theta,t-1}^I)^\top + (M_t^I)^\top H_t^I \right]^\top F_t^I \Xi_t^I (F_t^I)^\top \left[C_t^I (H_{\theta,t-1}^I)^\top + (M_t^I)^\top H_t^I \right]. \end{aligned} \quad (\text{A-88})$$

Similarly, the value function of market maker m at $t-1$ is

$$J_{t-1}^{M,m} = W_{t-1}^{M,m} + \theta_{t-2}^{M,m} \hat{V}_{t-1}^U + (C_{t-1}^M)^\top \Phi_{t-1}^M \theta_{t-2}^M + \frac{1}{2} m_{t-1}^M (\theta_{t-2}^M)^2 + \frac{1}{2} (\Phi_{t-1}^M)^\top M_{t-1}^M \Phi_{t-1}^M + \rho_{t-1}^M,$$

where C_{t-1}^M is a 2×1 vector, m_{t-1}^M is a scalar, M_{t-1}^M is a 2×2 symmetric matrix, and they are given as

$$\begin{aligned} C_{t-1}^M &:= \left(m_t^M \frac{N_M-1}{N_I} - m_t^M f_{t-1}^M \frac{N_M(N_M-2)}{N_I^2} - 2 \frac{f_{t-1}^M \gamma_{t-1}^I}{N_I} \right) H_{q,t-1}^M + \left(1 - \frac{N_M-1}{N_I} f_{t-1}^M \right) C_t^M, \\ m_{t-1}^M &:= m_t^M \left(1 - 2 f_{t-1}^M \frac{N_M-1}{N_I} + \frac{N_M(N_M-2)}{N_I^2} (f_{t-1}^M)^2 \right) + 2 \frac{\gamma_{t-1}^I}{N_I} (f_{t-1}^M)^2, \quad \rho_{t-1}^M := \rho_t^M + \frac{1}{2} (F_t^M)^\top M_t^M F_t^M \Sigma_t^U, \\ M_{t-1}^M &:= M_t^M + 2 \left(\frac{\gamma_{t-1}^I}{N_I} + \frac{1}{2} m_t^M \frac{N_M(N_M-2)}{N_I^2} \right) H_{q,t-1}^M (H_{q,t-1}^M)^\top + \frac{N_M-1}{N_I} \left(H_{q,t-1}^M (C_t^M)^\top + C_t^M (H_{q,t-1}^M)^\top \right). \end{aligned} \quad (\text{A-89})$$

Parameters $H_{P,t-1}^I$, $H_{q,t-1}^I$, and $H_{\theta,t-1}^I$ are 3×1 vectors. Parameter $H_{q,t-1}^M$ is a 2×1 vector. Parameters f_{t-1}^I , f_{t-1}^M , and $f_{\theta,t-1}^I$ are scalars. The recursive formulas are presented in equations (A-82), (A-84), and (A-85). The recursive formulas of scalars ξ_{t-1}^I , ψ_{t-1}^I , γ_{t-1}^I , δ_{t-1}^I , g_{t-1}^I , ξ_{t-1}^M , ψ_{t-1}^M , γ_{t-1}^M , δ_{t-1}^M , g_{t-1}^M , and ω_{t-1}^M are presented in equations (A-73), (A-79), and (A-80). The recursive formulas of parameters in informed traders' value function, C_{t-1}^I , m_{t-1}^I , M_{t-1}^I , and ρ_{t-1}^I are given in equation (A-88). The recursive formulas of parameters in market makers' value function,

$C_{t-1}^M, m_{t-1}^M, M_{t-1}^M$ and ρ_{t-1}^M are given in equation (A-89). Parameter μ_{t-1} is as given in equation (A-10), *i.e.*, $\mu_{t-1} = \lambda^I o_{t-1}^I$. The parameter values at period T are $\xi_T^I = 1, \xi_T^M = 0, \psi_T^I = 0, \psi_T^M = 1, \gamma_T^I = \lambda^I o_{V,T}^I$, and $\gamma_T^M = 0$. The parameters in value functions at T , $\rho_T^I, m_T^I, M_T^I, C_T^I, \rho_T^M, m_T^M, C_T^M$, and M_T^M are given in equation (A-65). We can thus solve the equilibrium recursively.

A.5. Proof of Propositions 5

For bid and ask prices, at $t < T$, there are four possible cases in equilibrium: Informed traders buy or sell and uninformed traders are indifferent between selling or no-trading, informed traders sell or buy and uninformed traders are indifferent between buying or no-trading.

More specifically, (1) If $P_t^{IR} \geq P_t^I$, then informed traders buy from market makers. There are two subcases: (i) If $P_t^U \leq P_t^I$, then $A_t = P_t^I$ and $B_t = P_t^U$; (ii) If $P_t^U > P_t^I$, then P_t^U and P_t^I are both ask prices, any bid price B_t that satisfies $B_t \leq A_t^I := P_t^I < A_t^U := P_t^U$ can be an equilibrium. We thus set $B_t = P_t^I$ and the ask price is the average of the two ask prices, *i.e.*, $A_t = \frac{1}{2} (P_t^U + P_t^I)$.

(2) If $P_t^{IR} < P_t^I$, then informed traders sell to market makers. There are two subcases: (i) If $P_t^U \geq P_t^I$, then $A_t = P_t^U$ and $B_t = P_t^I$; (ii) If $P_t^U < P_t^I$, then P_t^U and P_t^I are both bid prices, any ask price A_t that satisfies $A_t \geq B_t^I := P_t^I > B_t^U := P_t^U$ can be an equilibrium. We thus set $A_t = P_t^I$ and the bid price is the average of the two bid prices, *i.e.*, $B_t = \frac{1}{2} (P_t^U + P_t^I)$. Therefore, bid-ask spread at t can be written as in equation (30).

References

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Internet Appendix. Additional Results for “Dynamic Market Making with Asymmetric Information and Market Power”

Wen Chen and Yajun Wang

This Internet Appendix is structured as follows. We present the formulas for expected trading volume and bid-ask spread for both agency and principal trading market making cases in Section B.1. Section B.2 decomposes trading volume into trading due to information and trading due to hedging demand. Section B.3 demonstrates the parameter value dynamics of $\xi_t^I - \xi_t^U$, $\xi_t^I - \xi_t^M$, $\psi_t^U - \psi_t^I$, and $\psi_t^M - \psi_t^I$. Section B.4 presents model calibration with a different number of informed traders. Section B.5 presents the dynamics of trading volume and bid-ask spread when information or liquidity shocks arrive sporadically. Additional figures depicting the dynamics of trading volume and bid-ask spread for the case with principal trading market making are provided in Section B.6. In Section B.7, we present analytical results for two straightforward cases. Section B.8 establishes that extremely risk-averse market makers refrain from holding inventory. The equilibrium bid and ask prices in the presence of risk-averse market makers with substantial positions are discussed in Section B.9.

B.1. Expected Trading Volume and Bid-Ask Spreads

B.1.1. Expected Trading Volume and Spreads with Agency Trading Market Makers

From equation (A-22) in the proof of Proposition 1, it can be shown that the equilibrium quantity at t is

$$\begin{aligned} \frac{N_M}{N_M+1} \frac{\Delta_t}{\gamma_t^I/N_I + \gamma_t^U/N_U} = & - \frac{w_t^I}{\gamma_t^U/N_U} \left[(\delta_t^U - \delta_t^I)(\hat{V}_t^U - \bar{V}) + (\mu_t \delta_t^I - g_t^U - g_t^I) \hat{X}_t^U - (f_t^U - f_t^I) \bar{\theta} \right] \\ & + \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} \left\{ \frac{w_s^I}{\gamma_s^U/N_U} \left[(\delta_s^U - \delta_s^I)(\hat{V}_s^U - \bar{V}) + (\mu_s \delta_s^I - g_s^U - g_s^I) \hat{X}_s^U - (f_s^U - f_s^I) \bar{\theta} \right] + w_s^U \bar{\theta} \right\} \\ & - w_t^U \bar{\theta} + \frac{N_M}{(N_M+1)^t} N_U \bar{\theta}^U \sim \mathcal{N}(\mu_t^\Delta, (\sigma_t^\Delta)^2). \end{aligned} \quad (\text{B-1})$$

Therefore, the average trading volume at period t is

$$\mathbb{E}[\text{Vol}_t] = \frac{N_M}{N_M+1} \left[\sqrt{2/\pi} \sigma_t^\Delta e^{-\frac{1}{2}(\mu_t^\Delta/\sigma_t^\Delta)^2} + \mu_t^\Delta (1 - 2\Phi(-\mu_t^\Delta/\sigma_t^\Delta)) \right], \quad (\text{B-2})$$

where Φ is the normal cumulative distribution function, μ_t^Δ is given by equation (B-3), and σ_t^Δ is given by equation (B-7) below. It can be shown that

$$\mu_t^\Delta = \frac{N_M}{(N_M+1)^t} N_U \bar{\theta}^U + \left[\frac{w_t^I}{\gamma_t^U/N_U} (f_t^U - f_t^I) - w_t^U \right] \bar{\theta} - \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} \left[\frac{w_s^I}{\gamma_s^U/N_U} (f_s^U - f_s^I) - w_s^U \right] \bar{\theta}. \quad (\text{B-3})$$

Since $\hat{X}_t^U = \mu_t^{-1}(\hat{V}_t^U - \hat{V}_t^I) + X_t$, and

$$\hat{V}_t^I = \frac{\tau_V \bar{V} + V \sum_{n=0}^t \tau_{\varepsilon,n} + \sum_{n=0}^t \tau_{\varepsilon,n} \varepsilon_n}{\tau_V + \sum_{n=0}^t \tau_{\varepsilon,n}}, \quad \hat{V}_t^U = \frac{\tau_V \bar{V} + V \sum_{n=0}^t \tau_n + \sum_{n=0}^t \tau_n \varepsilon_n - \sum_{n=0}^t \tau_n h_n \eta_n}{\tau_V + \sum_{n=0}^t \tau_n},$$

where $\tau_t = \left(\tau_{\varepsilon,t}^{-1} + h_t^2 \tau_{\eta,t}^{-1} \right)^{-1}$. Define

$$\begin{aligned} d_t^V &:= \frac{w_t^I}{\gamma_t^U / N_U} \left[(\delta_t^U - (g_t^U + g_t^I) / \mu_t) o_{V,t}^U \sum_{n=0}^t \tau_n - (\delta_t^I - (g_t^U + g_t^I) / \mu_t) o_{V,t}^I \sum_{n=0}^t \tau_{\varepsilon,n} \right], \\ d_{t,n}^\varepsilon &:= \frac{w_t^I}{\gamma_t^U / N_U} \left[(\delta_t^U - (g_t^U + g_t^I) / \mu_t) o_{V,t}^U \tau_n - (\delta_t^I - (g_t^U + g_t^I) / \mu_t) o_{V,t}^I \tau_{\varepsilon,n} \right], \\ d_{t,n}^\eta &:= \frac{w_t^I}{\gamma_t^U / N_U} \left[(\delta_t^I - (g_t^U + g_t^I) / \mu_t) \mu_t - (\delta_t^U - (g_t^U + g_t^I) / \mu_t) o_{V,t}^U \tau_n h_n \right], \end{aligned} \quad (\text{B-4})$$

we can rewrite

$$\frac{w_t^I}{\gamma_t^U / N_U} \left[(\delta_t^U - \delta_t^I) (\hat{V}_t^U - \bar{V}) + (\mu_t \delta_t^I - g_t^U - g_t^I) \hat{X}_t^U \right] = d_t^V (V - \bar{V}) + \sum_{n=0}^t d_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t d_{t,n}^\eta \eta_n. \quad (\text{B-5})$$

The standard deviation σ_t^Δ depends on the variance of independent variables V , ε_t , and η_t . Define

$$D_t^V := -d_t^V + \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_s^V, \quad D_{t,n}^\varepsilon := -d_{t,n}^\varepsilon + \sum_{s=n}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_{s,n}^\varepsilon, \quad D_{t,n}^\eta := -d_{t,n}^\eta + \sum_{s=n}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_{s,n}^\eta,$$

using equations (B-1), (B-3), and (B-5), we show that the equilibrium quantity demanded at t ,

$$\frac{N_M}{N_M+1} \frac{\Delta_t}{\gamma_t^I / N_I + \gamma_t^U / N_U} = \mu_t^\Delta + D_t^V (V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t D_{t,n}^\eta \eta_n. \quad (\text{B-6})$$

It follows that the variance $(\sigma_t^\Delta)^2$ is

$$(\sigma_t^\Delta)^2 = (D_t^V)^2 / \tau_V + \sum_{n=0}^t (D_{t,n}^\varepsilon)^2 / \tau_{\varepsilon,n} + \sum_{n=0}^t (D_{t,n}^\eta)^2 / \tau_{\eta,n}. \quad (\text{B-7})$$

We then can calculate the expected trading volume using equation (B-2). From Proposition 2, the average bid-ask spread is $E[A_t - B_t] = \frac{1}{N_M+1} E|\Delta_t| = (\gamma_t^I / N_I + \gamma_t^U / N_U) \frac{E[\text{Vol}_t]}{N_M}$. Thus, the expected bid-ask spread

$$E[A_t - B_t] = \frac{1}{N_M} (\gamma_t^I / N_I + \gamma_t^U / N_U) \left(\sqrt{2/\pi} \sigma_t^\Delta e^{-\frac{1}{2}(\mu_t^\Delta / \sigma_t^\Delta)^2} + \mu_t^\Delta (1 - 2\Phi(-\mu_t^\Delta / \sigma_t^\Delta)) \right),$$

where μ_t^Δ is given by equation (B-3) and σ_t^Δ is given by equation (B-7).

B.1.2. Expected Trading Volume and Spreads with Principal Trading Market Makers

From the proof of Proposition 4, the trading volume at time t , $\text{Vol}_t = |N_M q_t^{I,m}| = \frac{N_t \omega_t^M}{\gamma_t^I} |P_t^{IR} - P_t^{MR}|$. From equation (28) and the proof of Proposition 4, trader I 's inventory at time $t-1$ can be written as $\theta_{t-1}^I = \frac{N_M}{N_I} (H_{q,t-1}^M)^\top \Phi_{t-1}^M + f_{\theta,t-1}^I \theta_{t-2}^I$. From equation (A-83), the equilibrium trading quantity of informed traders at t is

$$\begin{aligned} N_M q_t^{I,m} &= N_M \left(H_{q,t}^M \right)^\top \Phi_t^M - N_M f_t^M \theta_{t-1}^I \\ &= N_M (H_{q,t}^M)^\top \Phi_t^M - \frac{N_M^2}{N_I} f_t^M \left[(H_{q,t-1}^M)^\top \Phi_{t-1}^M + \sum_{i=2}^t \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) (H_{q,t-i}^M)^\top \Phi_{t-i}^M \right] - N_M f_t^M \left(\prod_{i=1}^t f_{\theta,t-i}^I \right) \theta_{t-1}^I, \end{aligned} \quad (\text{B-8})$$

where $H_{q,t}^M$ and f_t^M are as defined in equation (A-84), with the replacement of $t-1$ by t . Note that

$$\hat{V}_t^I = \frac{\tau_V \bar{V} + V \sum_{n=0}^t \tau_{\varepsilon,n} + \sum_{n=0}^t \tau_{\varepsilon,n} \varepsilon_n}{\tau_V + \sum_{n=0}^t \tau_{\varepsilon,n}}, \quad \hat{V}_t^U = \frac{\tau_V \bar{V} + V \sum_{n=0}^t \tau_n + \sum_{n=0}^t \tau_n \varepsilon_n - \sum_{n=0}^t \tau_n h_n \eta_n}{\tau_V + \sum_{n=0}^t \tau_n},$$

where $\tau_t = (\tau_{\varepsilon,t}^{-1} + h_t^2 \tau_{\eta,t}^{-1})^{-1}$. Define

$$\begin{aligned} d_t^V &:= N_M \left(g_t^M / \mu_t o_{V,t}^U \sum_{n=0}^t \tau_n - (\delta_t^M + g_t^M / \mu_t) o_{V,t}^I \sum_{n=0}^t \tau_{\varepsilon,n} \right), \\ d_{t,n}^\varepsilon &:= N_M \left(g_t^M / \mu_t o_{V,t}^U \tau_n - (\delta_t^M + g_t^M / \mu_t) o_{V,t}^I \tau_{\varepsilon,n} \right), \\ d_{t,n}^\eta &:= N_M \left(\delta_t^M \mu_t + g_t^M - g_t^M / \mu_t o_{V,t}^U \tau_n h_n \right), \end{aligned} \quad (\text{B-9})$$

we can rewrite

$$(H_{q,t}^M)^\top \Phi_t^M = \delta_t^M \bar{V} + g_t^M / \mu_t \hat{V}_t^U - (\delta_t^M + g_t^M / \mu_t) (\hat{V}_t^I - \mu_t X_t) = \frac{1}{N_M} \left(d_t^V (V - \bar{V}) + \sum_{n=0}^t d_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t d_{t,n}^\eta \eta_n \right). \quad (\text{B-10})$$

It follows that $(H_{q,t}^M)^\top \Phi_t^M$ is normally distributed with zero mean. Therefore, $N_M q_t^{I,m} \sim \mathcal{N}(\mu_t^q, (\sigma_t^q)^2)$, where

$$\mu_t^q = -N_M f_t^M \left(\prod_{i=1}^t f_{\theta,t-i}^I \right) \theta_{t-1}^I \quad \text{and} \quad (\sigma_t^q)^2 = \frac{(D_t^V)^2}{\tau_V} + \sum_{n=0}^t \frac{(D_{t,n}^\varepsilon)^2}{\tau_{\varepsilon,n}} + \sum_{n=0}^t \frac{(D_{t,n}^\eta)^2}{\tau_{\eta,n}}, \quad \text{where} \quad (\text{B-11})$$

$$\begin{aligned} D_t^V &:= d_t^V - \frac{N_M}{N_I} f_t^M \left[d_{t-1}^V + \sum_{i=2}^t \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i}^V \right], \quad D_{t,n}^\varepsilon := d_{t,n}^\varepsilon - \frac{N_M}{N_I} f_t^M \left[d_{t-1,n}^\varepsilon + \sum_{i=2}^{t-n} \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i,n}^\varepsilon \right], \\ D_{t,n}^\eta &:= d_{t,n}^\eta - \frac{N_M}{N_I} f_t^M \left[d_{t-1,n}^\eta + \sum_{i=2}^{t-n} \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i,n}^\eta \right]. \end{aligned} \quad (\text{B-12})$$

Therefore, trading quantity at time t is

$$N_M q_t^{I,m} = \mu_t^q + D_t^V (V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t D_{t,n}^\eta \eta_n, \quad (\text{B-13})$$

and the expected trading volume is

$$\mathbb{E}[\text{Vol}_t] = N_M \mathbb{E} |q_t^{I,m}| = \sqrt{\frac{2}{\pi}} \sigma_t^q e^{-\frac{1}{2}(\mu_t^q/\sigma_t^q)^2} + \mu_t^q \left(1 - 2\Psi(-\mu_t^q/\sigma_t^q)\right). \quad (\text{B-14})$$

Similarly, we can compute $P_t^I - P_t^U = (H_{P,t}^M)^\top \Phi_t^M - f_t^I \theta_{t-1}^I$, where $H_{P,t}^M = (\delta_t^I - 1, g_t^I - \delta_t^I \mu_t)^\top$. Define

$$\begin{aligned} k_t^V &:= -(1 - g_t^I/\mu_t) o_{V,t}^U \sum_{n=0}^t \tau_n + (\delta_t^I - g_t^I/\mu_t) o_{V,t}^I \sum_{n=0}^t \tau_{\varepsilon,n}, \\ k_{t,n}^\varepsilon &:= -(1 - g_t^I/\mu_t) o_{V,t}^U \tau_n + (\delta_t^I - g_t^I/\mu_t) o_{V,t}^I \tau_{\varepsilon,n}, \quad k_{t,n}^\eta := -(\delta_t^I \mu_t - g_t^I) + (1 - g_t^I/\mu_t) o_{V,t}^U \tau_n h_n. \end{aligned} \quad (\text{B-15})$$

We can rewrite

$$(H_{P,t}^M)^\top \Phi_t^M = (\delta_t^I - g_t^I/\mu_t) (\hat{V}_t^I - \mu_t X_t - \bar{V}) - (1 - g_t^I/\mu_t) (\hat{V}_t^U - \bar{V}) = k_t^V (V - \bar{V}) + \sum_{n=0}^t k_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t k_{t,n}^\eta \eta_n.$$

It follows that $(H_{P,t}^M)^\top \Phi_t^M$ is normally distributed with zero mean. We rewrite the difference in trading prices

$$P_t^I - P_t^U = (H_{P,t}^M)^\top \Phi_t^M - \frac{N_M}{N_I} f_t^I \left[(H_{q,t-1}^M)^\top \Phi_{t-1}^M + \sum_{i=2}^t \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) (H_{q,t-i}^M)^\top \Phi_{t-i}^M \right] - f_t^I \left(\prod_{i=1}^t f_{\theta,t-i}^I \right) \theta_{-1}^I. \quad (\text{B-16})$$

Therefore, $P_t^I - P_t^U \sim \mathcal{N}(\mu_t^D, (\sigma_t^D)^2)$, the mean μ_t^D and the variance $(\sigma_t^D)^2$ are given as

$$\begin{aligned} \mu_t^D &= -f_t^I \left(\prod_{i=1}^t f_{\theta,t-i}^I \right) \theta_{-1}^I \quad \text{and} \quad (\sigma_t^D)^2 = \frac{(K_t^V)^2}{\tau_V} + \sum_{n=0}^t \frac{(K_{t,n}^\varepsilon)^2}{\tau_{\varepsilon,n}} + \sum_{n=0}^t \frac{(K_{t,n}^\eta)^2}{\tau_{\eta,n}}, \quad \text{where} \\ K_t^V &:= k_t^V - \frac{N_M}{N_I} f_t^I \left[d_{t-1}^V + \sum_{i=2}^{t-1} \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i}^V \right], \quad K_{t,n}^\varepsilon := k_{t,n}^\varepsilon - \frac{N_M}{N_I} f_t^I \left[d_{t-1,n}^\varepsilon + \sum_{i=2}^{t-n} \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i,n}^\varepsilon \right], \\ K_{t,n}^\eta &:= k_{t,n}^\eta - \frac{N_M}{N_I} f_t^I \left[d_{t-1,n}^\eta + \sum_{i=2}^{t-n} \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i,n}^\eta \right]. \end{aligned} \quad (\text{B-17})$$

Therefore, the expected difference in trading prices

$$\mathbb{E} |P_t^I - P_t^U| = \sqrt{2/\pi} \sigma_t^D e^{-\frac{1}{2}(\mu_t^D/\sigma_t^D)^2} + \mu_t^D (1 - 2\Psi(-\mu_t^D/\sigma_t^D)). \quad (\text{B-18})$$

B.2. Decomposition of Trading Volume

B.2.1. Decomposition of Trading Volume with Agency Trading Market Makers

The trading quantity at time t is as shown in equation (B-6). To analyze the breakdown of trading volume attributed to information and hedging demand, we classify the term involving ε_n for $n = 0, 1, \dots, t$ as trading driven by information, and the term involving η_n for $n = 0, 1, \dots, t$ as trading driven by hedging demand.

Regarding the term $V - \bar{V}$ in the trading quantity (equation (B-6)), it includes contributions from both information and hedging demand. We decompose the coefficient on $V - \bar{V}$, $D_t^V = -d_t^V + \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_s^V$ into two distinct components, where d_t^V and d_s^V are as defined in equation (B-4). The component related to information is represented by terms involving $o_{V,s}^I \sum_{n=0}^s \tau_{\varepsilon,n}$ for $s = 0, 1, \dots, t$, whereas the component associated with hedging demand is represented by terms involving $o_{V,s}^U \sum_{n=0}^s \tau_n$ for $s = 0, 1, \dots, t$. Specifically,

$$\begin{aligned} d_s^{V,\text{in}} &= -\frac{w_s^I}{\gamma_s^U / N_U} \left[(\delta_s^I - (g_s^U + g_s^I) / \mu_s) o_{V,s}^I \sum_{n=0}^s \tau_{\varepsilon,n} \right] = -c_s^{V,\text{in}} \sum_{n=0}^s \tau_{\varepsilon,n}, \\ d_s^{V,\text{hd}} &= \frac{w_s^I}{\gamma_s^U / N_U} \left[(\delta_s^U - (g_s^U + g_s^I) / \mu_s) o_{V,s}^U \sum_{n=0}^s \tau_n \right] = c_s^{V,\text{hd}} \sum_{n=0}^s \tau_n, \end{aligned} \quad (\text{B-19})$$

for $s = 0, 1, \dots, t$, and $D_t^{V,\text{in}} := -d_t^{V,\text{in}} + \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_s^{V,\text{in}}$, $D_t^{V,\text{hd}} := -d_t^{V,\text{hd}} + \sum_{s=0}^{t-1} \frac{N_M}{(N_M+1)^{t-s}} d_s^{V,\text{hd}}$. Thus, the trading quantity is split into information driven part and hedging driven part as following

$$\text{Trading Quantity}_t = \underbrace{D_t^{V,\text{in}}(V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n + \mu_t^\Delta}_{\text{information}} + \underbrace{D_t^{V,\text{hd}}(V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\eta \eta_n}_{\text{hedging}}. \quad (\text{B-20})$$

Given that the trading quantity follows a normal distribution, it is common practice to measure the trading volume using the volatility of the trading quantity. Specifically, if $x \sim \mathcal{N}(0, \sigma_x^2)$, then the expected absolute value of x can be calculated as $E |x| = \sqrt{\frac{2}{\pi}} \sigma_x$. However, it is important to note that we cannot decompose the trading volume measured by the volatility of the trading quantity because $\sigma_{x+y} \neq \sigma_x + \sigma_y$ when $y \sim \mathcal{N}(0, \sigma_y^2)$ and x and y are independent. Instead, we decompose the variance of the trading quantity into two distinct components: the part driven by information and the part driven by hedging. This decomposition is expressed as follows:

$$\begin{aligned} \text{Var}[\text{Trading Quantity}_t] &= \text{Var}[D_t^{V,\text{in}}(V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n] + \text{Var}[\mu_t^\Delta + D_t^{V,\text{hd}}(V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\eta \eta_n] \\ &\quad + 2\text{Cov}[D_t^{V,\text{in}}(V - \bar{V}), D_t^{V,\text{hd}}(V - \bar{V})] \\ &= \underbrace{\frac{(D_t^{V,\text{in}})^2}{\tau_V} + \sum_{n=0}^t \frac{(D_{t,n}^\varepsilon)^2}{\tau_{\varepsilon,n}} + \frac{D_t^{V,\text{in}} D_t^{V,\text{hd}}}{\tau_V}}_{\text{information}} + \underbrace{\frac{(D_t^{V,\text{hd}})^2}{\tau_V} + \sum_{n=0}^t \frac{(D_{t,n}^\eta)^2}{\tau_{\eta,n}} + \frac{D_t^{V,\text{in}} D_t^{V,\text{hd}}}{\tau_V}}_{\text{hedging}}. \end{aligned} \quad (\text{B-21})$$

Given that trading driven by information and trading driven by hedging demand are interrelated, their interaction is reflected in the covariance between $D_t^{V,\text{in}}(V - \bar{V})$ and $D_t^{V,\text{hd}}(V - \bar{V})$. Achieving a clear separation between trading due to information and trading due to hedging demand can be challenging due to this interdependence.

To address this issue, we decompose the covariance term $2D_t^{V,\text{in}} D_t^{V,\text{hd}} / \tau_V$ into information and hedging components. It is important to note that the covariance term can have negative values, as the trading direction resulting from information might be opposite to the trading direction driven by hedging demand. Consequently, when

performing the decomposition, it is possible for the resulting components to also become negative.

Therefore, in the illustrated decomposition examples in Section B.2.3, we carefully select parameter values to guarantee that the expected total trading quantity is greater than both the expected trading quantity attributed to information and the expected trading quantity resulting from hedging demand (i.e., ensuring that both components are positive).

B.2.2. Decomposition of Trading Volume with Principal Trading Market Makers

In the case with principal trading market makers, the trading quantity at time t is as given in equation (B-13),

$$N_M q_t^{I,m} = \mu_t^q + D_t^V (V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n + \sum_{n=0}^t D_{t,n}^\eta \eta_n, \quad (\text{B-22})$$

where D_t^V , $D_{t,n}^\varepsilon$, and $D_{t,n}^\eta$ are as defined in equation (B-12). Similar to the case with agency trading market makers, in the case with principal trading market makers, the trading quantity can be split into information driven part and hedging driven part as following,

$$\text{Trading Quantity}_t = \underbrace{D_t^{V,\text{in}} (V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\varepsilon \varepsilon_n}_{\text{information}} + \underbrace{\mu_t^q + D_t^{V,\text{hd}} (V - \bar{V}) + \sum_{n=0}^t D_{t,n}^\eta \eta_n}_{\text{hedging}}, \quad (\text{B-23})$$

$$D_t^{V,\text{in}} := d_t^{V,\text{in}} - \frac{N_M}{N_I} f_t^M \left[d_{t-1}^{V,\text{in}} + \sum_{i=2}^t \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i}^{V,\text{in}} \right], \quad D_t^{V,\text{hd}} := d_t^{V,\text{hd}} - \frac{N_M}{N_I} f_t^M \left[d_{t-1}^{V,\text{hd}} + \sum_{i=2}^t \left(\prod_{s=1}^{i-1} f_{\theta,t-s}^I \right) d_{t-i}^{V,\text{hd}} \right], \quad (\text{B-24})$$

and $d_s^{V,\text{in}}$, $d_s^{V,\text{hd}}$, for $s = 0, 1, \dots, t$ are defined as follows,

$$d_s^{V,\text{in}} = -N_M \left(\delta_s^M + \frac{g_s^M}{\mu_s} \right) o_{V,s}^I \sum_{n=0}^s \tau_{\varepsilon,n} = -c_s^{V,\text{in}} \sum_{n=0}^s \tau_{\varepsilon,n}, \quad d_s^{V,\text{hd}} = N_M \frac{g_s^M}{\mu_s} o_{V,s}^U \sum_{n=0}^s \tau_n = c_s^{V,\text{hd}} \sum_{n=0}^s \tau_n. \quad (\text{B-25})$$

We then decompose the variance of the trading quantity into two distinct components: the part driven by information and the part driven by hedging. Similar to the case with agency trading market makers, this decomposition is expressed as follows:

$$\text{Var}[\text{Trading Quantity}_t] = \underbrace{\frac{(D_t^{V,\text{in}})^2}{\tau_V} + \sum_{n=0}^t \frac{(D_{t,n}^\varepsilon)^2}{\tau_{\varepsilon,n}} + \frac{D_t^{V,\text{in}} D_t^{V,\text{hd}}}{\tau_V}}_{\text{information}} + \underbrace{\frac{(D_t^{V,\text{hd}})^2}{\tau_V} + \sum_{n=0}^t \frac{(D_{t,n}^\eta)^2}{\tau_{\eta,n}} + \frac{D_t^{V,\text{in}} D_t^{V,\text{hd}}}{\tau_V}}_{\text{hedging}}. \quad (\text{B-26})$$

B.2.3. Examples of Trading Volume Decomposition Due to Information and Hedging Demand

As discussed previously, when performing the decomposition, it is possible for the resulting components to be negative. Therefore, in the illustrated decomposition examples below, we carefully select parameter values so that both components are positive.

Figure B-1 illustrates the decomposition of the variance of trading quantity using equations (B-21) and (B-26). This decomposition presents the case where new information and hedging demand arrive gradually throughout the trading day. As shown in Figure B-1, the results of the decomposition, as observed in both panels, align with the model's intuition. Informed traders tend to act promptly upon receiving private information, as there is a possibility of other traders trading on the same information. Consequently, trading volume due to information diminishes over time.

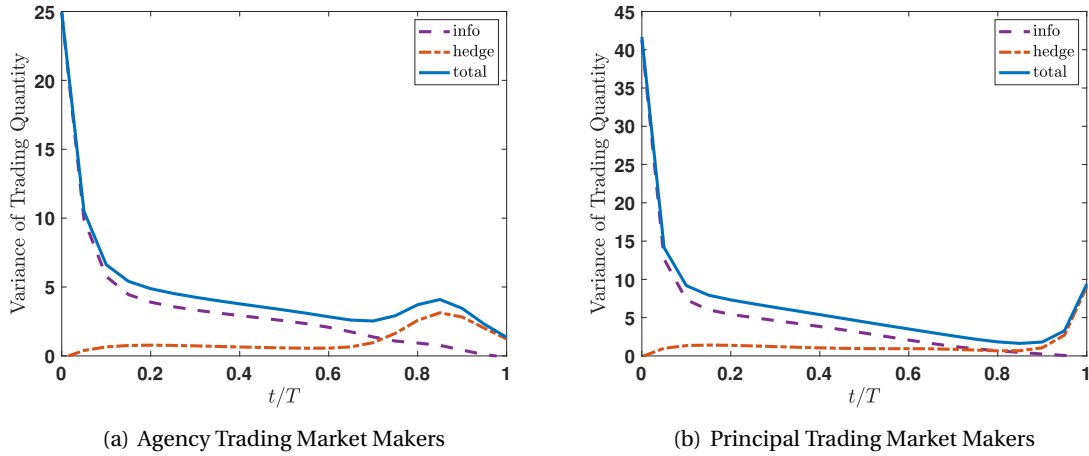


Figure B-1. Decomposition of Trading Quantity into Information (dashed) and Hedging (dot-dashed) Components for Agency (left) and Principal (right) Trading Market Makers. Parameters are $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\hat{\theta} = 1$, $\tau_V = 4$, $\tau_{\eta,t} = 10$, and $\tau_{\varepsilon,t} = 0.25$, for $t \neq 0$, $\tau_{\varepsilon,0} = 1.5$. In the agency market making case, $N_M = 1$, $\lambda^U = 1$, and $\tau_{\eta,0} = 5$, while in the case of principal market making, $N_M = 2$, $\lambda^U = 0$, and $\tau_{\eta,0} = 4.5$.

As more information is revealed through trading, the adverse selection effect becomes less severe. Consequently, traders tend to postpone their trading for hedging purposes to later periods. As a result, trading volume due to hedging demand increases over time.

For the case where market makers primarily offset trading and do not hold inventory (left panel), there is a noticeable bump in trading volume before market closure. This is because informed traders optimally hedge a portion of their risk resulting from liquidity shocks during the earlier periods, prior to market closure.

Figure B-2 illustrates the decomposition of the variance of trading quantity in information-driven markets, while Figure B-3 presents the decomposition of the variance of trading quantity in liquidity-driven markets. As

depicted in both Figures B-2 and B-3, trading volume attributed to information diminishes over time, while the trading volume in later periods is primarily driven by hedging demand.

It is worth noting that in an information-driven market, where no new liquidity shocks occur after period 0, informed traders' private information observed after period 0 becomes public information due to the absence of new noise. Consequently, in Figure B-2, we observe a small spike in trading volume due to liquidity shocks precisely at trading period $t/T = 1/20 = 0.05$. This spike occurs because adverse selection diminishes significantly after period 0, as information received thereafter becomes public knowledge.

In summary, the findings depicted in Figures B-1, B-2, and B-3 align with the main results of our paper. It is evident that trading volume in later periods is predominantly driven by hedging demand, while trading volume attributed to information gradually diminishes over time. These results reinforce our key findings and provide further support for the dynamics of trading volume in our paper.

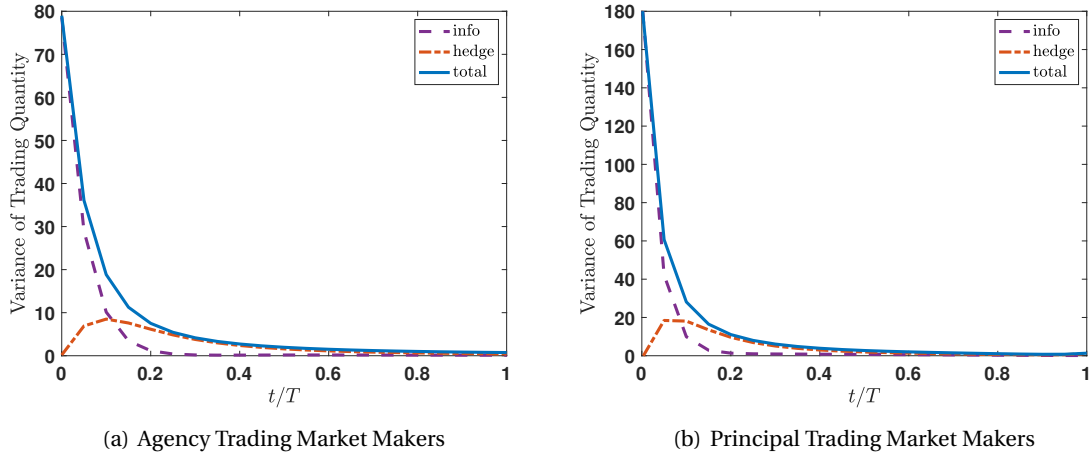


Figure B-2. Decomposition of Trading Quantity into Information (dashed) and Hedging (dot-dashed) Components for Agency (left) and Principal (right) Trading Market Makers. Parameters are $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\hat{\theta} = 1$, $\tau_V = 1$. In the agency market making case, $N_M = 1$, $\lambda^U = 1$, $\tau_{\varepsilon,t} = 0.55$, and $\tau_{\eta,t} = 10^6$, expect for $\tau_{\varepsilon,0} = 3$ and $\tau_{\eta,0} = 0.096$. In the principal market making case, $N_M = 2$, $\lambda^U = 0$, $\tau_{\varepsilon,t} = 0.65$, and $\tau_{\eta,t} = 10^6$, expect for $\tau_{\varepsilon,0} = 3$ and $\tau_{\eta,0} = 0.097$.

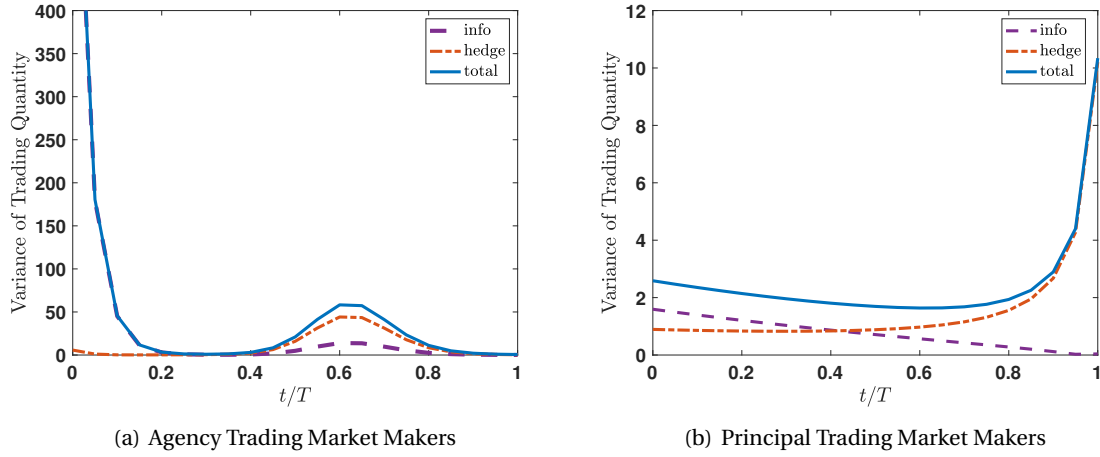


Figure B-3. Decomposition of Trading Quantity into Information (dashed) and Hedging (dot-dashed) Components for Agency (left) and Principal (right) Trading Market Makers. Parameters are $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$. In the agency market making case, $N_M = 1$, $\lambda^U = 1$, $\tau_{\varepsilon,t} = 10^{-6}$, and $\tau_{\eta,t} = 75$, expect for $\tau_{\varepsilon,0} = 5$ and $\tau_{\eta,0} = 0.9$. In the principal market making case, $N_M = 2$, $\lambda^U = 0$, $\tau_{\varepsilon,t} = 10^{-12}$, and $\tau_{\eta,t} = 10$, expect for $\tau_{\varepsilon,0} = 1.8$ and $\tau_{\eta,0} = 0.2$.

B.3. Parameter Values of $\xi_t^I - \xi_t^U$, $\xi_t^I - \xi_t^M$, $\psi_t^U - \psi_t^I$ and $\psi_t^M - \psi_t^I$

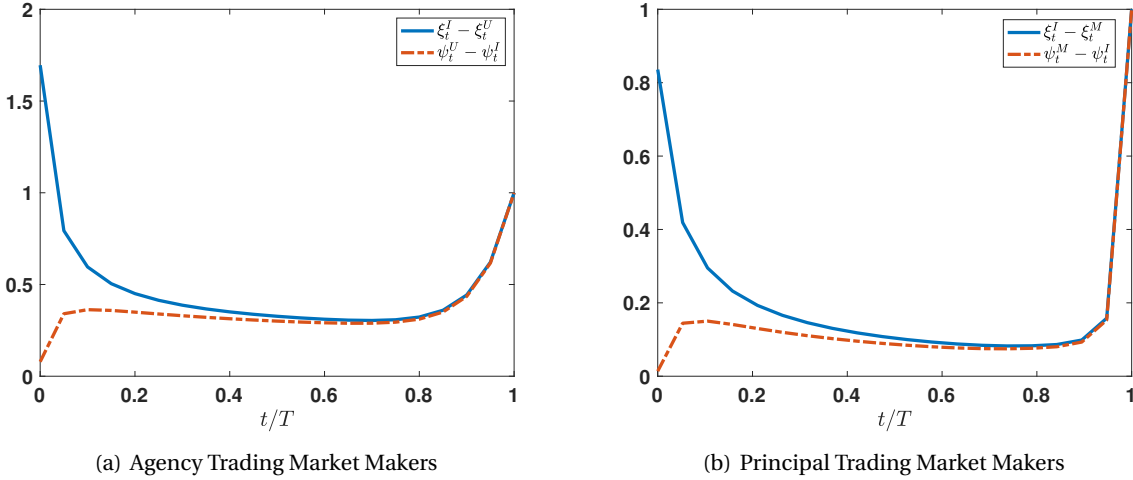


Figure B-4. Parameter Values of $\xi_t^I - \xi_t^U$, $\xi_t^I - \xi_t^M$, $\psi_t^U - \psi_t^I$ and $\psi_t^M - \psi_t^I$. Panel (a) (and Panel (b)) presents the dynamics of parameters for the case with agency (and principal) trading market makers. The exogenous parameter values are $N_I = 10$, $N_U = 100$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\bar{V} = 1$, $\tau_V = 1$, $\tau_{\eta,t} = 10$, and $\tau_{\varepsilon,t} = 1$. In Panel (a), $\lambda^U = 1$ and $N_M = 1$. In Panel (b), $\lambda^U = 0$ and $N_M = 2$.

Given that both the long-term stock position and short-term speculative position of each trader are based on his own estimate of the asset's payoff, and only the short-term speculative position is influenced by the other trader's estimates, the coefficient on $\hat{V}_t^I - \mu_t X_t$ in the informed traders' reservation price is typically greater than

that in the uninformed traders' reservation price. This implies that $\xi_t^I > \xi_t^U$ for $t < T$. Similarly, the coefficient on $\hat{V}_t^U$ in the uninformed traders' reservation price tends to be larger than that in the informed traders' reservation price, suggesting $\psi_t^U > \psi_t^I$ for $t < T$.

Similarly, the parameter $\xi_t^I > \xi_t^M$ and $\psi_t^M > \psi_t^I$ for the case with principal trading market makers. We perform thorough numerical analysis using a wide range of parameters. In our analysis, we observe that the parameter values consistently demonstrate $\xi_t^I - \xi_t^U > 0$, $\xi_t^I - \xi_t^M > 0$, $\psi_t^U - \psi_t^I > 0$, and $\psi_t^M - \psi_t^I > 0$, for $t < T$, as illustrated in Figure B-4.

B.4. Model Calibration with a Different Number of Informed Traders

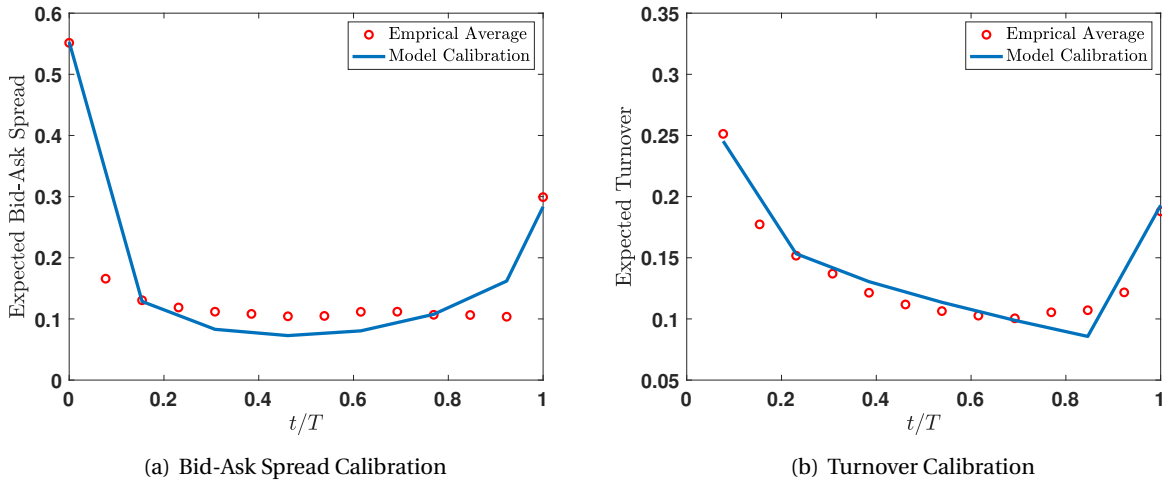


Figure B-5. Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for agency trading market makers. The fixed parameters are $N_M = 1$, $N_I = 2$, $\lambda_U = 1$, $\tau_V = 1$, $\bar{V} = 1$, and $\bar{\theta} = 1$. The calibrated parameters are $\lambda_I = 7.098$, $N_U = 97.904$, $\tau_{\varepsilon, t} = 0.095$, $\tau_{\eta, t}^{-1/2} = 0.071$, except for $\tau_{\varepsilon, 0} = 0.234$ and $\tau_{\eta, T}^{-1/2} = 0.193$.

In this subsection, we present the model calibration with a different number of informed traders. As discussed in Section 4 of the paper, the calibrated parameter values indicate that the precision of private information is higher at market opening compared to other periods, while the uncertainty of hedging needs is greater at market closure. This suggests that trading at market opening is predominantly influenced by private information, whereas trading at market closure is primarily driven by hedging needs.

Figures B-5 and B-6 depict the expected absolute bid-ask spreads ($E[A_t - B_t]$) and expected turnover from our model for agency and principal trading market makers, respectively. The left panels show 14 dots corresponding to the empirical averages of absolute bid-ask spreads at trading times $t = 9:30, 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$, as presented in Column (1) of Table 1. The solid curves represent our model's theoretical

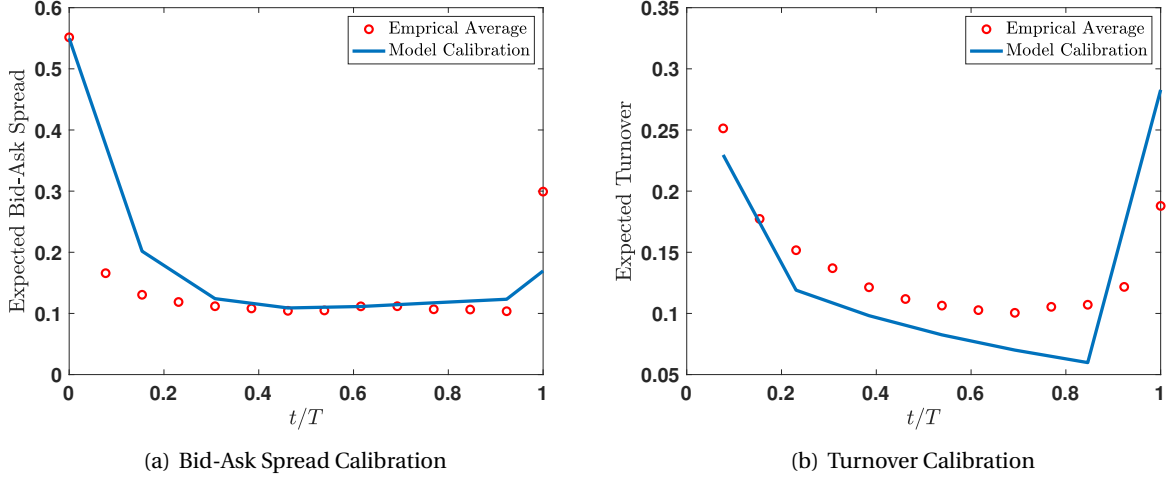


Figure B-6. Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for principal trading market makers. The fixed parameters are $N_M = 1$, $N_I = 2$, $\lambda_U = 0$, $\tau_V = 1$, $\bar{V} = 1$, and $\bar{\theta} = 1$. The calibrated parameters are $\lambda_I = 2.855$, $\tau_{\varepsilon,t} = 0.105$, $\tau_{\eta,t}^{-1/2} = 0.123$, except for $\tau_{\varepsilon,0} = 0.164$ and $\tau_{\eta,T}^{-1/2} = 0.251$.

predictions, which approximate these empirical values. In the right panels, we present the expected cumulative turnover for NASDAQ-listed stocks for agency and principal trading market makers. Each panel displays 13 dots representing the empirical averages of cumulative turnover at trading times $t = 10:00, 10:30, 11:00$, and so on until $15:00, 15:30$, and $16:00$. The models' expected cumulative turnover roughly traces these points, although not perfectly.

B.5. Dynamics of Trading Volume and Bid-Ask Spread When Information or Liquidity Shocks Arrive Sporadically

In this subsection, we examine the dynamics of trading volume and bid-ask spread when information or liquidity shocks only arrive sporadically. Figures B-7a and B-7b depict the dynamics of trading volume for the case where information (Panel (a)) or liquidity shocks (Panel (b)) arrive at certain points in time ($t = 0, 7, 14$) for the oligopolistic case (solid curve) and the competitive case (dashed curve).

When market makers possess market power, the arrival of new information or liquidity shocks creates a ripple effect on the average trading volume. In such cases, traders don't immediately adjust to their target levels. Instead, they spread their trades over several periods after receiving a fresh piece of information or new liquidity shocks. This is in stark contrast to scenarios with perfectly competitive market makers, where traders instantly adapt to their target levels and then stop trading until the arrival of new information or liquidity shocks.

In our model, it's noteworthy that even though both informed and uninformed traders aren't strategic, they

still make optimal decisions regarding when to trade and how much to trade at any given period. When market makers are less competitive (i.e., N_M is smaller), traders spread out their trading over an extended time frame. This is because the monopoly power of market makers makes it too costly for other traders to trade aggressively. Our model implies that when market makers have significant market power, other traders trade gradually over time even though they are not strategic.

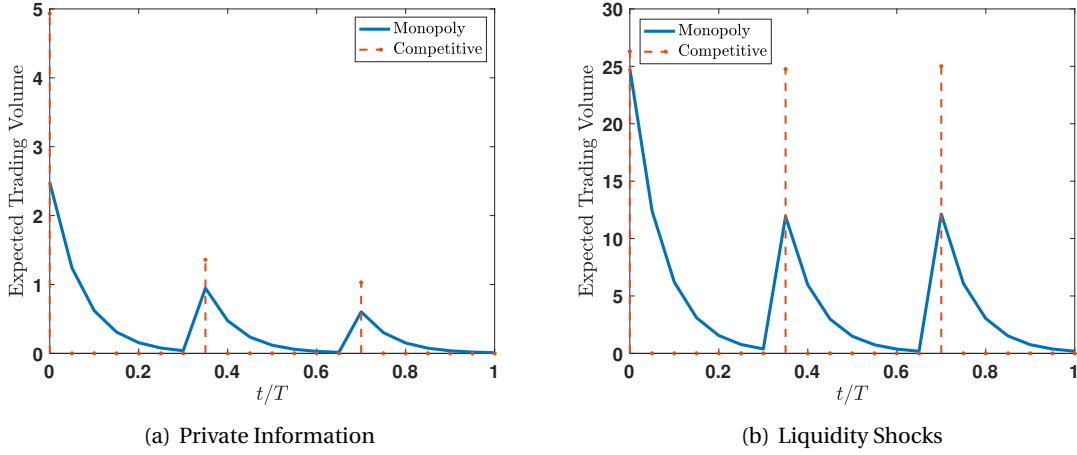


Figure B-7a. Agency Trading Market Markers. The Dynamics of Expected Trading Volume with Sporadically Arrived Information (left) or Liquidity Shocks (right). Parameter values are $T = 20$, $N_M = 1$, $N_I = 10$, $N_U = 100$, $\lambda^I = \lambda^U = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,0} = 1$, and $\tau_{\varepsilon,0} = 0.5$. For $t = 1, 2, \dots, T$, in Panel (a), $\tau_{\eta,t} = 10^6$ and $\tau_{\varepsilon,t} = 10^{-6}$ except for $\tau_{\varepsilon,7} = \tau_{\varepsilon,14} = 0.5$; in Panel (b), $\tau_{\varepsilon,t} = 10^{-6}$ and $\tau_{\eta,t} = 10^6$ except for $\tau_{\eta,7} = \tau_{\eta,14} = 1$.

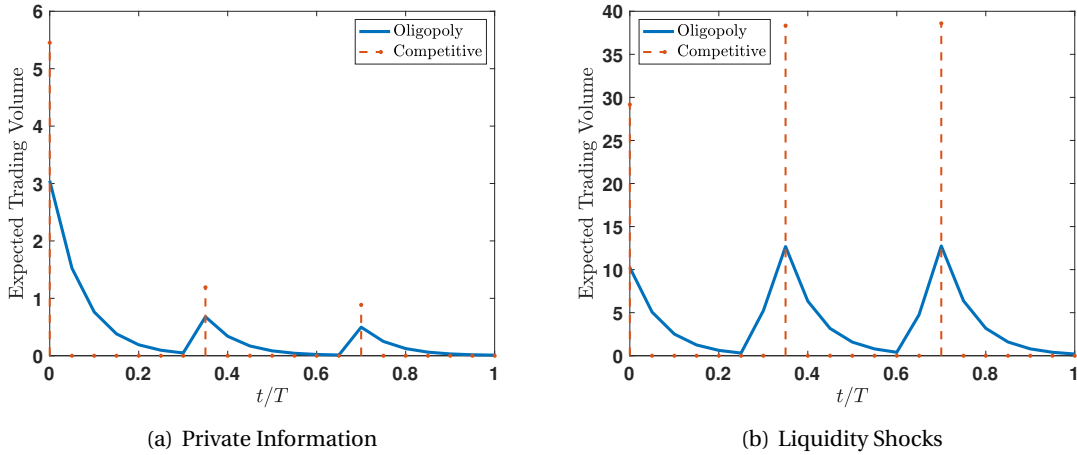


Figure B-7b. Principal Trading Market Markers. The Dynamics of Expected Trading Volume with Sporadically Arrived Information (left) or Liquidity Shocks (right). Parameter values are $T = 20$, $N_M = 3$, $N_I = 10$, $N_U = 100$, $\lambda^I = 1$, $\lambda^U = 0$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,0} = 1$, and $\tau_{\varepsilon,0} = 0.5$. For $t = 1, 2, \dots, T$, in Panel (a), $\tau_{\eta,t} = 10^6$ and $\tau_{\varepsilon,t} = 10^{-6}$ except for $\tau_{\varepsilon,7} = \tau_{\varepsilon,14} = 0.5$; in Panel (b), $\tau_{\varepsilon,t} = 10^{-6}$ and $\tau_{\eta,t} = 10^6$ except for $\tau_{\eta,7} = \tau_{\eta,14} = 1$.

Figures B-7a and B-7b also imply that the monopoly power of market makers tends to dampen trading spikes due to either information or liquidity shocks: the solid curve (oligopolistic case) is smoother than the dashed curve (competitive case) over the whole trading period. An empirically testable implication is that the variation in trading volume tends to be smaller for stocks with less competitive market makers during periods of news events.

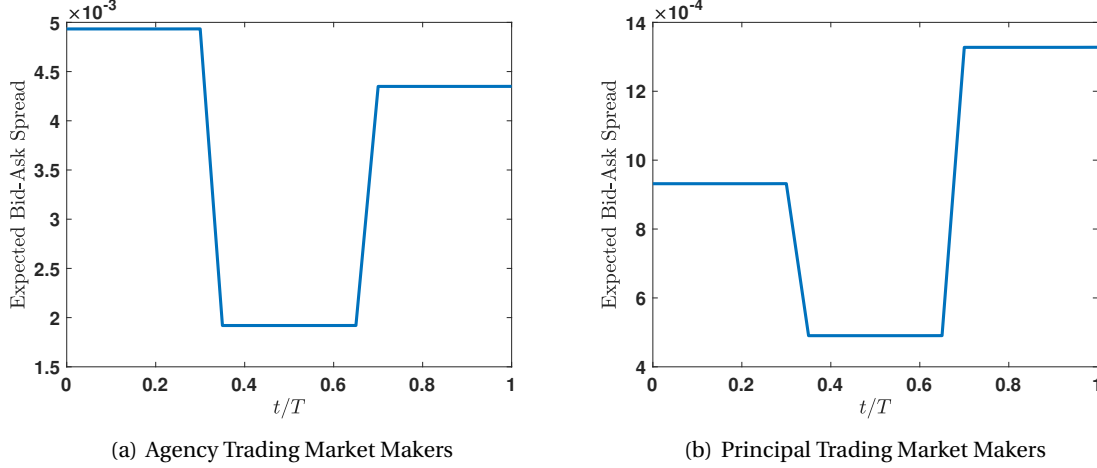


Figure B-8. The Dynamics of Expected Bid-Ask Spread with Sporadically Arrived Information and Liquidity Shocks. Parameter values are $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\varepsilon,t} = 10^{-6}$, and $\tau_{\eta,t} = 10^6$ except for $\tau_{\varepsilon,0} = \tau_{\varepsilon,7} = \tau_{\varepsilon,14} = 0.5$ and $\tau_{\eta,0} = \tau_{\eta,7} = \tau_{\eta,14} = 1$. In Panel (a), $N_M = 1$, $N_I = 10$, $\lambda^U = 1$, and $N_U = 100$. In Panel (b), $N_M = 3$, $\lambda^U = 0$, and $N_I = 10$.

Figure B-8 depicts the dynamics of expected bid-ask spread with sporadically arrived information and liquidity shocks. As evident in Figures B-7a, B-7b, and B-8, without new information or hedging needs, market makers keep their quoted bid and ask prices unchanged. Traders, in these intervals, transact based on the optimal quantity determined by their prevailing information and hedging demand. When new information or hedging needs arise, traders adjust their optimal quantities and market makers update the quoted prices accordingly.

B.6. Additional Figures of the Dynamics of Trading Volume and Bid-Ask Spread for the case with Principal Market Making

As discussed in the paper, the expected trading volume tends to be smoother for the case with $N_M \geq 2$, due to greater competition among market makers compared to the monopolistic principal trading market maker case. This observation is supported by the analytical results presented in Proposition B.2 in Appendix B.7.

In this subsection, we present the the expected trading volume and spread for the principal market making case with $N_M = 2$ for information-driven market (Figure B-9) and liquidity-driven market (Figure B-10) cases and $N_M = 1$ for the case with both Information and liquidity shocks arriving at constant rates (Figure B-11).

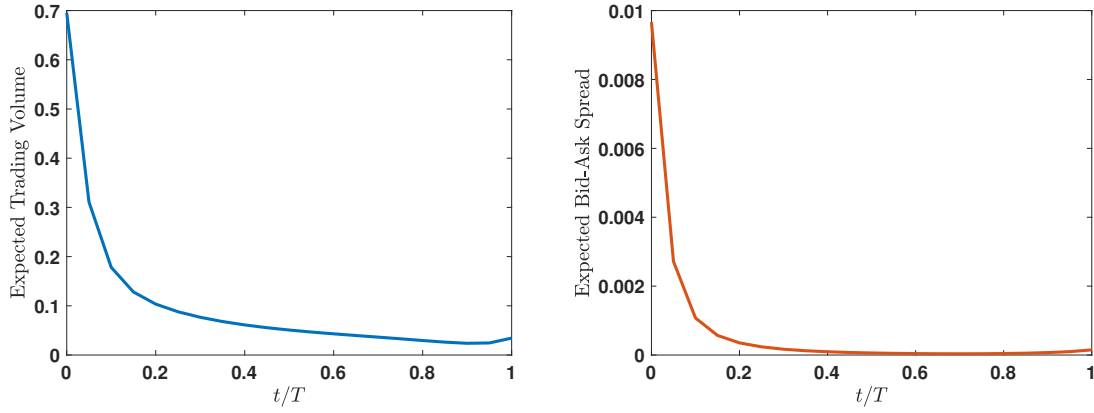


Figure B-9. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Information-Driven Markets. Parameters are $N_M = 2$, $N_I = 10$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\varepsilon,t} = 0.5$, and $\tau_{\eta,t} = 10^6$ ($\tau_{\eta,t}^{-1} \approx 0$) except for $\tau_{\eta,0} = 10$.

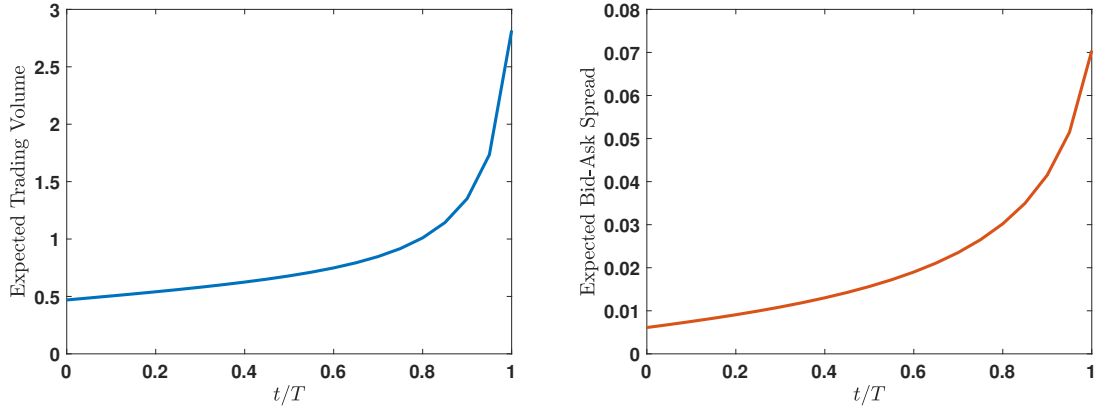


Figure B-10. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration in Liquidity-Driven Markets. Parameters are $N_M = 2$, $N_I = 10$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,t} = 10$, $\tau_{\varepsilon,t} = 10^{-6} \approx 0$ except for $\tau_{\varepsilon,0} = 0.1$.

B.7. A Simple Case with Analytical Proofs

In this subsection, we provide analytical proofs for the case where information and liquidity shocks only arrive at date 0. The reservation prices of informed and uninformed traders at date 0 are defined as

$$\begin{aligned}
 P_0^{IR} &:= E[V|v_0] - \lambda^I \text{Var}[V|v_0](X + \bar{\theta}^I), \\
 P_0^{UR} &:= E[V|s_0] - \lambda^U \text{Var}[V|s_0]\bar{\theta}^U, \\
 \Delta_0 &:= P_0^{IR} - P_0^{UR}.
 \end{aligned}$$

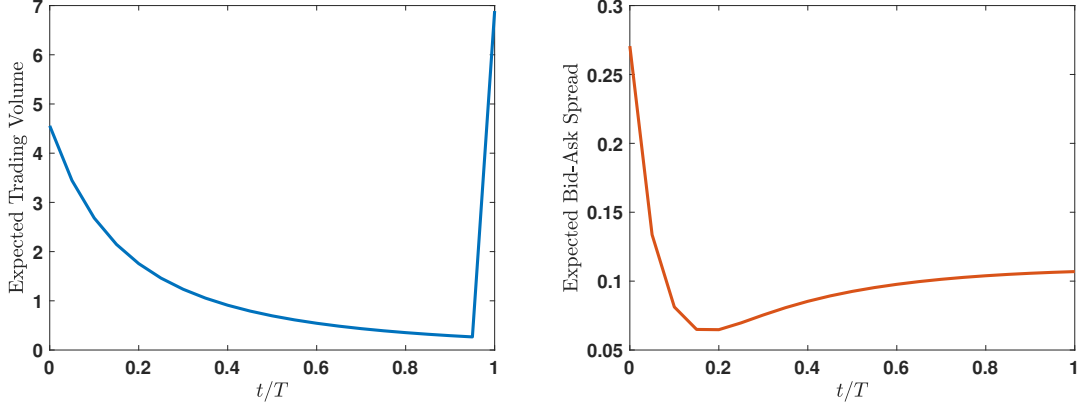


Figure B-11. Principal Trading Market Making. The Dynamics of Expected Trading Volume and Spread against Trading Duration with Both Information and Liquidity Shocks Arriving at Constant Rates. Parameters are $N_M = 1$, $N_I = 10$, $T = 20$, $\lambda^I = 1$, $\bar{\theta} = 1$, $\tau_V = 1$, $\tau_{\eta,t} = 10$, and $\tau_{\varepsilon,t} = 0.5$.

Proposition B.1. (*The equilibrium with agency trading market makers*)

1. Bid-ask spreads decrease with the number of trading rounds. The bid-ask spread at t is

$$A_t - B_t = \frac{|\Delta_0|}{(N_M + 1)^{t+1}}. \quad (\text{B-27})$$

2. The total trading volume increases with the number of trading rounds. The equilibrium quantities demanded at t are

$$N_I(\theta_t^I - \theta_{t-1}^I) = -N_U(\theta_t^U - \theta_{t-1}^U) = \frac{N_M}{(N_M + 1)^{t+1}} \frac{\Delta_0}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U}.$$

3. The equilibrium ask and bid depths at t are

$$\alpha_t = \beta_t = \frac{1}{(N_M + 1)^{t+1}} \frac{|\Delta_0|}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U}. \quad (\text{B-28})$$

Proposition B.1 shows analytically that more trading rounds reduce market makers' power and thus decrease bid-ask spread and increase total trading volume. If the market is competitive, both informed and uninformed traders will trade to their optimal target positions immediately on date 0. Due to market makers' market power, other market participants trade gradually over time. The total net trades of informed traders over trading rounds T are

$$\sum_{t=0}^T N_I(\theta_t^I - \theta_{t-1}^I) = \left(1 - \frac{1}{(N_M + 1)^{T+1}}\right) \frac{\Delta_0}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U}.$$

In the limit when $T \rightarrow \infty$, the total net trade of informed traders over time converges to

$$\lim_{T \rightarrow \infty} \sum_{t=0}^T N_I(\theta_t^I - \theta_{t-1}^I) = \frac{\Delta_0}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U},$$

which is informed traders' optimal target position in a competitive market where informed and uninformed traders can trade directly with each other without going through oligopolistic market makers.

Proof of Proposition B.1: We prove the case when $\Delta_0 > 0$. The other case when $\Delta_0 < 0$ can be proved similarly. In this special case where no new information or liquidity shocks arrive after date 0, from optimization of the CARA utility function for traders, the optimal positions of informed and uninformed traders at period T are

$$\theta_T^I = \frac{E[V|v_0] - A_T}{\lambda^I \text{Var}[V|v_0]} - X_0, \quad \theta_T^U = \frac{E[V|s_0] - B_T}{\lambda^U \text{Var}[V|s_0]}, \quad (\text{B-29})$$

where X_0 is the total number of shares of a nontradable risky asset on date 0. Equation (B-29) yields

$$\begin{aligned} A_T(\alpha_{jT}) &= E[V|v_0] - \lambda^I \text{Var}[V|v_0] (\theta_{T-1}^I + X_{-1} + \frac{1}{N_I} \sum_{j=1}^{N_M} \alpha_T^j) = P_T^{IR} - \lambda^I \text{Var}[V|v_0] \frac{1}{N_I} \sum_{j=1}^{N_M} \alpha_T^j, \\ B_T(\beta_{jT}) &= E[V|s_0] - \lambda^U \text{Var}[V|s_0] (\theta_{T-1}^U - \frac{1}{N_U} \sum_{j=1}^{N_M} \beta_T^j) = P_T^{UR} + \lambda^U \text{Var}[V|s_0] \frac{1}{N_U} \sum_{j=1}^{N_M} \beta_T^j. \end{aligned} \quad (\text{B-30})$$

Market makers with limited risk-bearing capacity maximize the round-trip profit from the bid-ask spread by off-setting trades. Then, for $j = 1, 2, \dots, N_M$, the designated market maker M_j 's problem is equivalent to solving

$$\max_{\alpha_T^j = \beta_T^j} E \left[\alpha_T^j A_T - \beta_T^j B_T \mid s_0 \right].$$

The first-order condition yields

$$\begin{aligned} \alpha_T = \beta_T &= \frac{P_T^{IR} - P_T^{UR}}{(N_M + 1) (\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U)}, \\ N_I(\theta_T^I - \theta_{T-1}^I) &= -N_U(\theta_T^U - \theta_{T-1}^U) = \frac{N_M}{N_M + 1} \frac{P_T^{IR} - P_T^{UR}}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U}. \end{aligned} \quad (\text{B-31})$$

Substituting Equation (B-31) into (B-30), we obtain

$$\begin{aligned} A_T &= P_T^{IR} - \frac{N_M}{N_M + 1} \frac{\lambda^I \text{Var}[V|v_0]/N_I}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U} (P_T^{IR} - P_T^{UR}), \\ B_T &= P_T^{UR} + \frac{N_M}{N_M + 1} \frac{\lambda^U \text{Var}[V|s_0]/N_U}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U} (P_T^{IR} - P_T^{UR}). \end{aligned}$$

In this special case where no new information or liquidity shocks arrive after date 0, when we solve informed

and uninformed traders optimization at period $T-1$, for a finite solution to exist, we show that $A_{T-1} = A_T$, and $B_{T-1} = B_T$. Then, for $j = 1, 2, \dots, N_M$, at period $T-1$, the designated market maker M_j 's problem is to solve

$$\max_{\alpha_{T-1}^j = \beta_{T-1}^j} \mathbb{E} \left[\alpha_{T-1}^j A_{T-1} - \beta_{T-1}^j B_{T-1} \mid s_0 \right].$$

The first-order condition yields

$$\alpha_{T-1} = \beta_{T-1} = \frac{P_{T-1}^{IR} - P_{T-1}^{UR}}{(N_M + 1)(\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U)}. \quad (\text{B-32})$$

It can be shown that

$$\begin{aligned} P_T^{IR} - P_T^{UR} &= P_{T-1}^{IR} - \lambda^I \text{Var}[V|v_0] \frac{1}{N_I} \sum_{j=1}^{N_M} \alpha_{T-1}^j - P_{T-1}^{UR} - \lambda^U \text{Var}[V|s_0] \frac{1}{N_U} \sum_{j=1}^{N_M} \beta_{T-1}^j \\ &= \frac{P_{T-1}^{IR} - P_{T-1}^{UR}}{N_M + 1}. \end{aligned}$$

Similarly, we can prove that at period t , $P_t^{IR} - P_t^{UR} = \frac{P_{t-1}^{IR} - P_{t-1}^{UR}}{N_M + 1} = \frac{P_0^{IR} - P_0^{UR}}{(N_M + 1)^t}$.

$$\begin{aligned} \alpha_t = \beta_t &= \frac{P_0^{IR} - P_0^{UR}}{(N_M + 1)^{t+1}(\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U)}, \\ A_t &= P_0^{IR} - \left(1 - \frac{1}{(N_M + 1)^{T+1}}\right) \frac{\lambda^I \text{Var}[V|v_0]/N_I}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U} (P_0^{IR} - P_0^{UR}), \\ B_t &= P_0^{UR} + \left(1 - \frac{1}{(N_M + 1)^{T+1}}\right) \frac{\lambda^U \text{Var}[V|s_0]/N_U}{\lambda^I \text{Var}[V|v_0]/N_I + \lambda^U \text{Var}[V|s_0]/N_U} (P_0^{IR} - P_0^{UR}). \end{aligned}$$

Proposition B.2. *(The equilibrium with principal trading market makers and risk-neutral uninformed traders)*

1. *Informed and risk neutral uninformed traders' trading prices P_t^I and P_t^U are*

$$P_t^I = \frac{1}{N_M^2 + 1} P_0^{IR} + \frac{N_M^2}{N_M^2 + 1} P_0^{UR}, \quad P_t^U = P_0^{UR} = \mathbb{E}[V|s_0]. \quad (\text{B-33})$$

2. *The equilibrium trading quantities are*

$$\begin{aligned} \alpha_T &= \frac{N_I}{N_M^2 + 1} \frac{(P_0^{IR} - P_0^{UR})^+}{\lambda^I \text{Var}[V|v_0]}, & \beta_T &= \frac{N_I}{N_M^2 + 1} \frac{(P_0^{IR} - P_0^{UR})^-}{\lambda^I \text{Var}[V|v_0]}, \\ \sum_{t=0}^{T-1} \alpha_t &= \frac{(N_M - 1)N_I}{N_M^2 + 1} \frac{(P_0^{IR} - P_0^{UR})^+}{\lambda^I \text{Var}[V|v_0]}, & \sum_{t=0}^{T-1} \beta_t &= \frac{(N_M - 1)N_I}{N_M^2 + 1} \frac{(P_0^{IR} - P_0^{UR})^-}{\lambda^I \text{Var}[V|v_0]}. \end{aligned} \quad (\text{B-34})$$

Proposition B.2 implies that principal trading market makers wield more market power than agency trading market makers. In this simple case where information and liquidity shocks only arrive at date 0, risk neutral market

makers' profits are not affected by the number of trading rounds since trading prices stay the same over time. In more general case with new arrival of information or liquidity shocks, market makers are still worse off with more trading rounds.

Equation (B-34) suggests that if there is only one principal market maker ($N_M = 1$), then he optimally quotes prices so trading only occurs once at market closure. For the case with multiple oligopolistic market makers, traders still spread out trading over time.

Proof of Proposition B.2: Similar to the proof of Propositions 3, we first solve the problem at T . Informed trading quantity and trading price are

$$q_T^{I,m} = \frac{P_T^{IR} - \hat{V}_T^U}{(N_M+1)\gamma_T^I/N_I}, \quad P_T^I = \frac{1}{N_M+1}P_T^{IR} + \frac{N_M}{N_M+1}\hat{V}_T^U, \quad \text{where} \quad (\text{B-35})$$

$$P_T^{IR} = \hat{V}_0^I - \lambda^I \text{Var}[V|v_0]X_0 - \lambda^I \text{Var}[V|v_0] \left(\bar{\theta}^I + \frac{1}{N_I} \sum_{m=1}^{N_M} \sum_{t=0}^{T-1} q_t^{I,m} \right). \quad (\text{B-36})$$

Uninformed traders are risk neutral ($\lambda^U = 0$), $P_T^U = \hat{V}_T^U = \hat{V}_0^U$ and $q_T^{U,m} = 0$. For the case where information and liquidity shocks only arrive at date 0, it is straightforward to show that, in equilibrium, informed traders' trading prices stay the same over time, $P_t^I = P_T^I$, for $t = 0, 1, \dots, T-1$. Then, a risk-neutral market maker m maximizes,

$$\max_{q_t^{I,m}} \bar{\theta}^M \hat{V}_0^U + (q_0^{I,m} + \dots + q_{T-1}^{I,m} + q_T^{I,m})(P_T^I - \hat{V}_0^U). \quad (\text{B-37})$$

First order condition with respect to $\sum_{t=0}^{T-1} q_t^{I,m}$ yields

$$\sum_{t=0}^{T-1} q_t^{I,m} = \frac{(N_M-1)N_I}{N_M^2+1} \frac{(P_0^{IR} - P_0^{UR})}{\lambda^I \text{Var}[V|v_0]}. \quad (\text{B-38})$$

Substituting equations (B-38) and (B-36) into equation (B-35) yields,

$$q_T^{I,m} = \frac{N_I}{N_M^2+1} \frac{(P_0^{IR} - P_0^{UR})}{\lambda^I \text{Var}[V|v_0]}, \quad P_t^I = P_T^I = \frac{1}{N_M^2+1}P_0^{IR} + \frac{N_M^2}{N_M^2+1}P_0^{UR}. \quad (\text{B-39})$$

B.8. Extremely Risk-Averse Market Makers Do Not Hold Inventory

For notational simplicity, in this proof we set $N_M = 1$. The proof stays the same for $N_M \geq 2$. We assume that the market maker has zero initial endowment, i.e., $\bar{\theta}^M = 0$. The market maker's final wealth at $T+1$ is $W_{T+1}^M = W_T^M + \theta_T^M V$. At period t ($t = 0, 1, \dots, T$),

$$W_t^M = W_{t-1}^M + \alpha_t A_t - \beta_t B_t, \quad \theta_t^M = \theta_{t-1}^M + \beta_t - \alpha_t.$$

The objective function of the market maker with risk aversion λ_M at $t = T$ is

$$\max_{\alpha_T, \beta_T} E_T^M [-e^{-\lambda^M W_{T+1}^M}].$$

The market maker's wealth at period $T + 1$ depends on his previous inventory θ_{T-1}^M , as $\lambda^M \rightarrow \infty$; the eight cases described in [Liu and Wang \(2016\)](#) become:

- (1) I traders buy and U traders sell if $-\frac{N_U}{2\gamma_T^U} \Delta_T < \theta_{T-1}^M < \frac{N_I}{2\gamma_T^I} \Delta_T$;
- (2) I traders buy and U traders do not trade if $\frac{N_I}{2\gamma_T^I} \Delta_T \leq \theta_{T-1}^M \leq \frac{N_I}{\gamma_T^I} \Delta_T$;
- (3) Both I and U buy if $\theta_{T-1}^M \geq \max\{-\frac{N_U}{\gamma_T^U} \Delta_T, \frac{N_I}{\gamma_T^I} \Delta_T\}$;
- (4) I investors do not trade and U traders buy if $-\frac{N_U}{2\gamma_T^U} \Delta_T \leq \theta_{T-1}^M \leq -\frac{N_U}{\gamma_T^U} \Delta_T$;
- (5) I traders sell and U traders buy if $\frac{N_I}{2\gamma_T^I} \Delta_T < \theta_{T-1}^M < -\frac{N_U}{2\gamma_T^U} \Delta_T$;
- (6) I traders sell and U traders do not trade if $\frac{N_I}{\gamma_T^I} \Delta_T \leq \theta_{T-1}^M \leq \frac{N_I}{2\gamma_T^I} \Delta_T$;
- (7) Both I and U traders sell if $\theta_{T-1}^M \leq \min\{-\frac{N_U}{\gamma_T^U} \Delta_T, \frac{N_I}{\gamma_T^I} \Delta_T\}$;
- (8) I traders do not trade and U traders sell if $-\frac{N_U}{\gamma_T^U} \Delta_T \leq \theta_{T-1}^M \leq -\frac{N_U}{2\gamma_T^U} \Delta_T$.

Since Δ_T is normally distributed, the probability of each case is the cumulative density function of normal variable depending on θ_{T-1}^M . We denote the probability of case j as $\text{Prob}_T^j(\theta_{T-1}^M)$, where $j = 1, 2, \dots, 8$. If $\theta_{T-1}^M > 0$, then the probabilities of cases (1), (2), (3), (4), and (5) are not zero. If $\theta_{T-1}^M < 0$, then the probabilities of cases (1), (5), (6), (7), and (8) are not zero. If $\theta_{T-1}^M = 0$, then the probabilities of case (2), (3), (4), (6), (7), and (8) are all zero.

We next show that $\theta_{T-1}^M = 0$, otherwise infinite risk-averse market makers' utility will explode. If $\theta_{T-1}^M \neq 0$, we prove the case when $\theta_{T-1}^M > 0$; the case when $\theta_{T-1}^M < 0$ can be proved similarly. Note that if $\theta_{T-1}^M > 0$, then the probabilities of cases (1), (2), (3), (4), and (5) are not zero. In case (2), informed traders buy and uninformed traders do not trade at $t = T$, implying that $\beta_T = 0$ and the expected utility of the market maker is

$$E_T^M [-e^{-\lambda^M W_{T+1}^M}] = -e^{-\lambda^M [W_{T-1}^M + \alpha_T A_T + (\theta_{T-1}^M - \alpha_T) \hat{V}_T^U - \frac{1}{2} \lambda^M (\theta_{T-1}^M - \alpha_T)^2 o_{V,T}^U]},$$

where $A_T = P_T^{IR} - \frac{\gamma_T^I}{N_I} \alpha_T$. Taking the first order condition with respect to α_T yields

$$\alpha_T = \frac{P_T^{IR} - \hat{V}_T^U + \lambda^M o_{V,T}^U \theta_{T-1}^M}{\lambda^M o_{V,T}^U + 2\gamma_T^I / N_I}.$$

We can express the above equations in terms of $1/\lambda^M$ as follows

$$\alpha_T = \theta_{T-1}^M + o\left(\frac{1}{\lambda^M}\right), \quad A_T = P_T^{IR} - \frac{\gamma_T^I}{N_I} \theta_{T-1}^M + o\left(\frac{1}{\lambda^M}\right),$$

$$J_T^M = -e^{-\lambda^M \left[W_{T-1}^M + \theta_{T-1}^M (P_T^{IR} - \frac{\gamma_T^I}{N_I} \theta_{T-1}^M) + o\left(\frac{1}{\lambda^M}\right) \right]},$$

where $o(\frac{1}{\lambda^M})$ means higher orders of $1/\lambda^M$. As $\lambda^M \rightarrow \infty$, $o(\frac{1}{\lambda^M}) \rightarrow 0$. At period $T-1$, the market maker maximizes $E_{T-1}^M[J_T^M]$ by choosing the optimal number of shares to sell α_{T-1} and the optimal number of shares to buy β_{T-1} and $\beta_{T-1} - \alpha_{T-1} = \theta_{T-1}^M - \theta_{T-2}^M$; then the conditional expected utility of case 2 is

$$\begin{aligned} \Psi_{T-1}^2(\theta_{T-1}^M) &\equiv E_{T-1}^M[J_T^M \mid \frac{N_I}{2\gamma_T^I} \Delta_T \leq \theta_{T-1}^M \leq \frac{N_I}{\gamma_T^I} \Delta_T] \\ &= -\exp \left\{ -\lambda^M \left[o\left(\frac{1}{\lambda^M}\right) + W_{T-2}^M + A_{T-1} \alpha_{T-1} - B_{T-1} (\theta_{T-1}^M + \alpha_{T-1} - \theta_{T-2}^M) \right] \right\} \\ &\quad \times \exp \left\{ -\lambda^M \left[\theta_{T-1}^M \left(E_{T-1}^M[P_T^{IR} \mid \text{case}(2)] - \frac{\gamma_T^I}{N_I} \theta_{T-1}^M \right) \right] \right\} \\ &\quad \times \exp \left\{ -\lambda^M \left[-\frac{1}{2} \lambda^M \text{Var}_{T-1}^M[P_T^{IR} \mid \text{case}(2)] (\theta_{T-1}^M)^2 \right] \right\}. \end{aligned}$$

Similarly, we can also compute the conditional expected utility of cases (1), (3), (4), (5). Denote the conditional expected utility of case k as $\Psi_{T-1}^k(\theta_{T-1}^M)$, where $k = 1, 2, \dots, 5$. Therefore, the expected utility at $t = T-1$ is

$$E_{T-1}^M[J_T^M] = \sum_{k=1}^5 \text{Prob}_{T-1}^k(\theta_{T-1}^M) \Psi_{T-1}^k(\theta_{T-1}^M).$$

If the probability of case (2) is not zero, then the expected utility contains the conditional expected utility of the second case, which includes the term

$$-\exp \left\{ -\lambda^M \left[\theta_{T-1}^M \left(E_{T-1}^M[P_T^{IR} \mid \text{case}(2)] - \frac{\gamma_T^I}{N_I} \theta_{T-1}^M \right) - \frac{1}{2} \lambda^M \text{Var}_{T-1}^M[P_T^{IR} \mid \text{case}(2)] (\theta_{T-1}^M)^2 \right] \right\}.$$

This implies that when $\lambda^M \rightarrow \infty$, the expected utility goes to negative infinity. Since $\Delta_T \in \mathcal{F}_T$ is a normal random variable at $T-1$, the probability of case (2) is positive except when $\theta_{T-1}^M = 0$. The above argument can be applied to any period t and we can then conclude that when market makers are extremely risk averse, they optimally hold zero inventory at any period t , $\theta_t^M \rightarrow 0$.

B.9. Equilibrium Bid and Ask Prices When Risk-Averse Market Makers Have Large Positions

For simplicity, we assume $\delta_I = \delta_U = \delta$ and focus on the limiting case where $\lambda^M \rightarrow \infty$. Denote $v := \text{Var}^U[V]/\text{Var}^I[V]$.

If $\bar{\theta}^M > 0$, market makers need to offload their large position and there are five cases:

- (1) Informed traders buy and uninformed traders sell if $\Delta > \frac{2\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M$, and

$$A = ((2N_I v + N_U)P^{IR} + N_U P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)),$$

$$B = (N_I v P^{IR} + (N_I v + 2N_U)P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)).$$

- (2) Informed traders sell and uninformed traders buy if $\Delta < -\frac{2\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M$,

$$A = (N_I v P^{IR} + (N_I v + 2N_U)P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)),$$

$$B = ((2N_I v + N_U)P^{IR} + N_U P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)).$$

- (3) Informed traders buy and uninformed traders do not trade if $\frac{\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M < \Delta \leq \frac{2\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M$,

$$A = P^{IR} - \delta\text{Var}^U[V]\bar{\theta}^M/(N_I v), \quad B \leq P^{UR}.$$

- (4) Both informed and uninformed investors buy if $-\frac{\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M \leq \Delta \leq \frac{\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M$,

$$A = (N_I v P^{IR} + N_U P^{UR} - \delta\text{Var}^U[V]\bar{\theta}^M)/(N_I v + N_U), \quad B \leq A.$$

- (5) Informed traders do not trade and uninformed traders buy if $-\frac{2\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M \leq \Delta < -\frac{\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M$,

$$A = P^{UR} - \delta\text{Var}^U[V]\bar{\theta}^M/N_U, \quad B \leq P^{IR}.$$

If $\bar{\theta}^M < 0$, market makers need to buy to close existing short position and there are five cases:

- (1) Informed traders buy and uninformed traders sell if $\Delta > -\frac{2\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M$, and

$$A = ((2N_I v + N_U)P^{IR} + N_U P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)),$$

$$B = (N_I v P^{IR} + (N_I v + 2N_U)P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)).$$

(2) Informed traders sell and uninformed traders buy if $\Delta < \frac{2\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M$,

$$A = (N_I v P^{IR} + (N_I v + 2N_U)P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)),$$

$$B = ((2N_I v + N_U)P^{IR} + N_U P^{UR} - 2\delta\text{Var}^U[V]\bar{\theta}^M)/(2(N_I v + N_U)).$$

(3) Informed traders sell and uninformed traders do not trade if $\frac{2\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M < \Delta \leq \frac{\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M$,

$$B = P^{IR} - \delta\text{Var}^U[V]\bar{\theta}^M/(N_I v), \quad A \geq P^{UR}.$$

(4) Both informed and uninformed investors sell if $\frac{\delta\text{Var}^I[V]}{N_I}\bar{\theta}^M \leq \Delta \leq -\frac{\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M$,

$$B = (N_I v P^{IR} + N_U P^{UR} - \delta\text{Var}^U[V]\bar{\theta}^M)/(N_I v + N_U), \quad A \geq B.$$

(5) Informed traders do not trade and uninformed traders sell if $-\frac{\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M \leq \Delta < -\frac{2\delta\text{Var}^U[V]}{N_U}\bar{\theta}^M$,

$$B = P^{UR} - \delta\text{Var}^U[V]\bar{\theta}^M/N_U, \quad A \geq P^{IR}.$$

The competitive price where nobody has market power is

$$P = (N_I v P^{IR} + N_U P^{UR} - \delta\text{Var}^U[V]\bar{\theta}^M)/(N_I v + N_U).$$